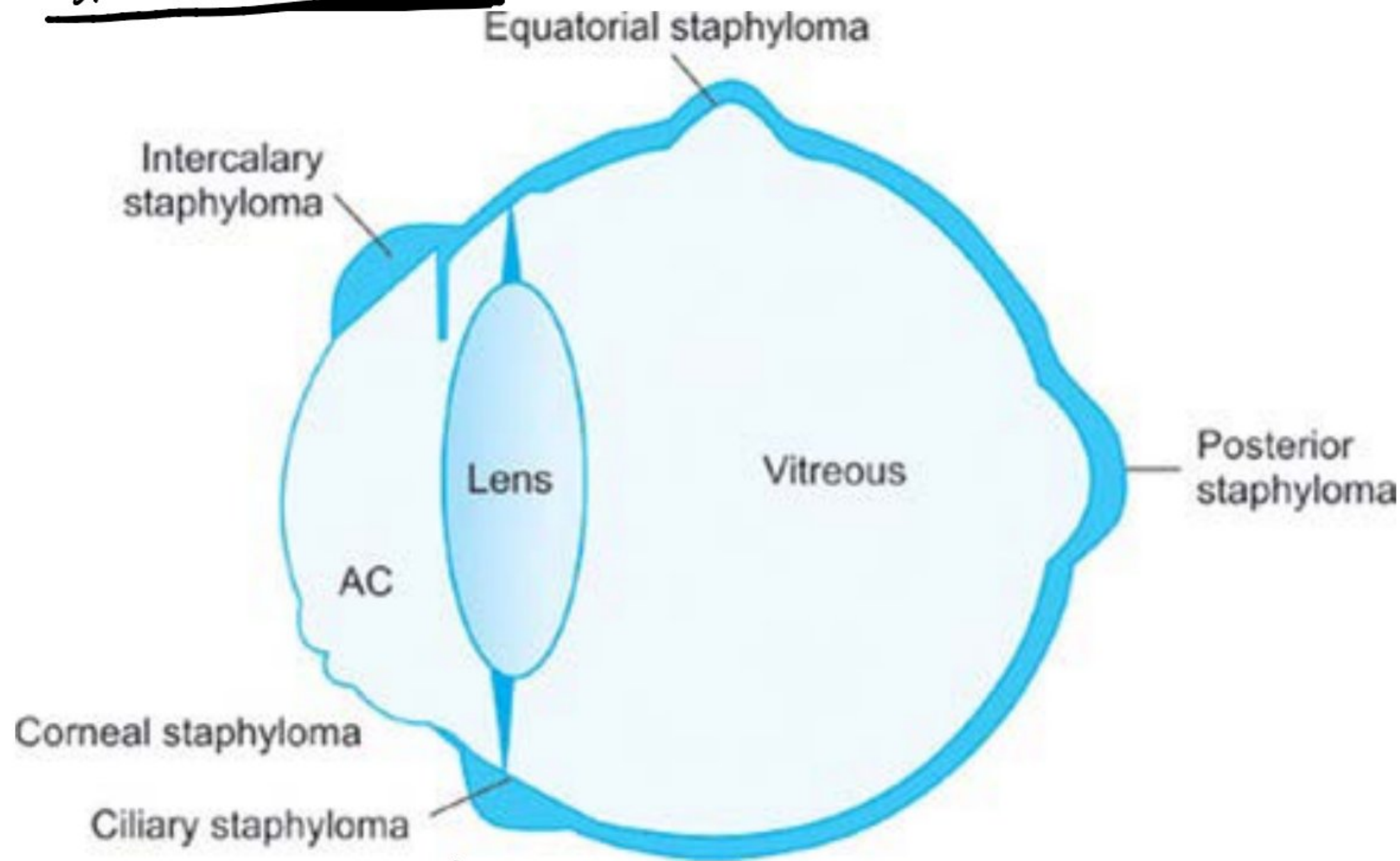


STAPHYLOMA: {Bunch of grapes}

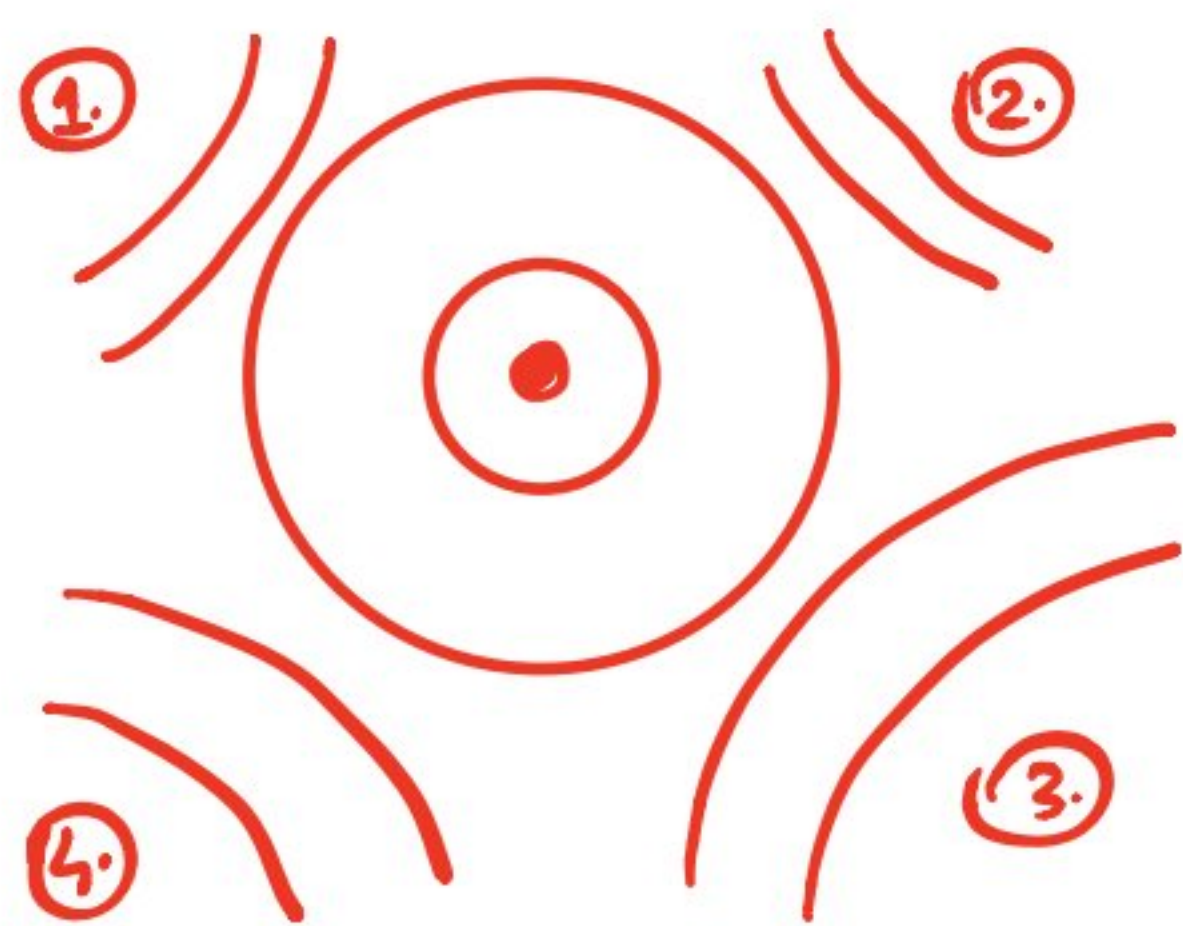
- Ectasia & Bulging of the outer coats {cornea or sclera or both} of the eye
- Incarceration of uveal tissue {iris, ciliary body, choroid} in to the bulge.

Types of Staphyloma:



ADD NOTE:

Equator of Eyeball:



⇒ Encircled by 4 vortex vein

SCLERITIS: Inflammation of sclera
cause: Rheumatoid arthritis & MCC

Types

Anterior

- Redness
- Pain
- Tenderness

Posterior

- Pain
- ↓sed vision d/t fluid @
↳ Sub tenons space 'T' sign

Anterior

Nodular

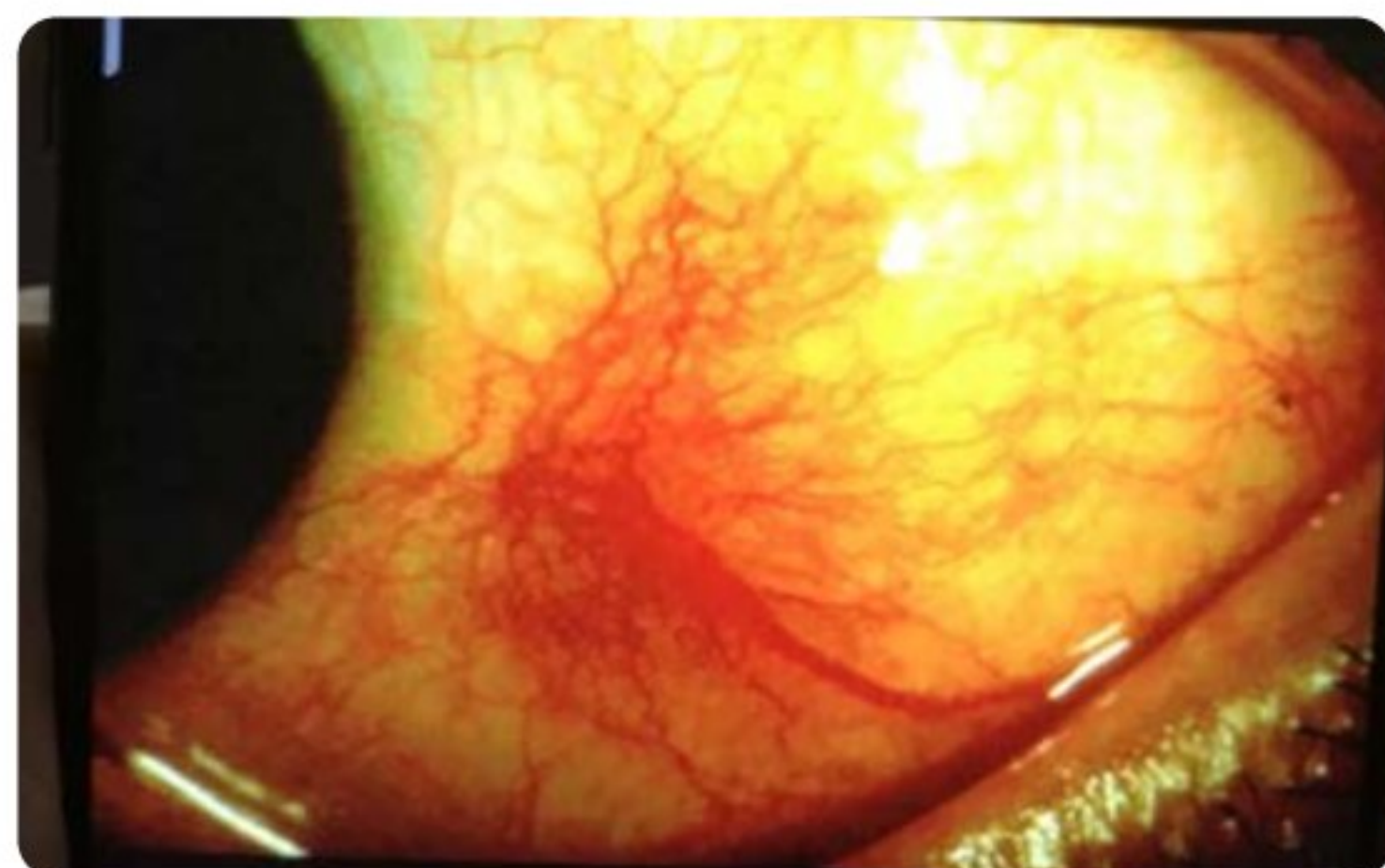
Diffuse

Necrotizing

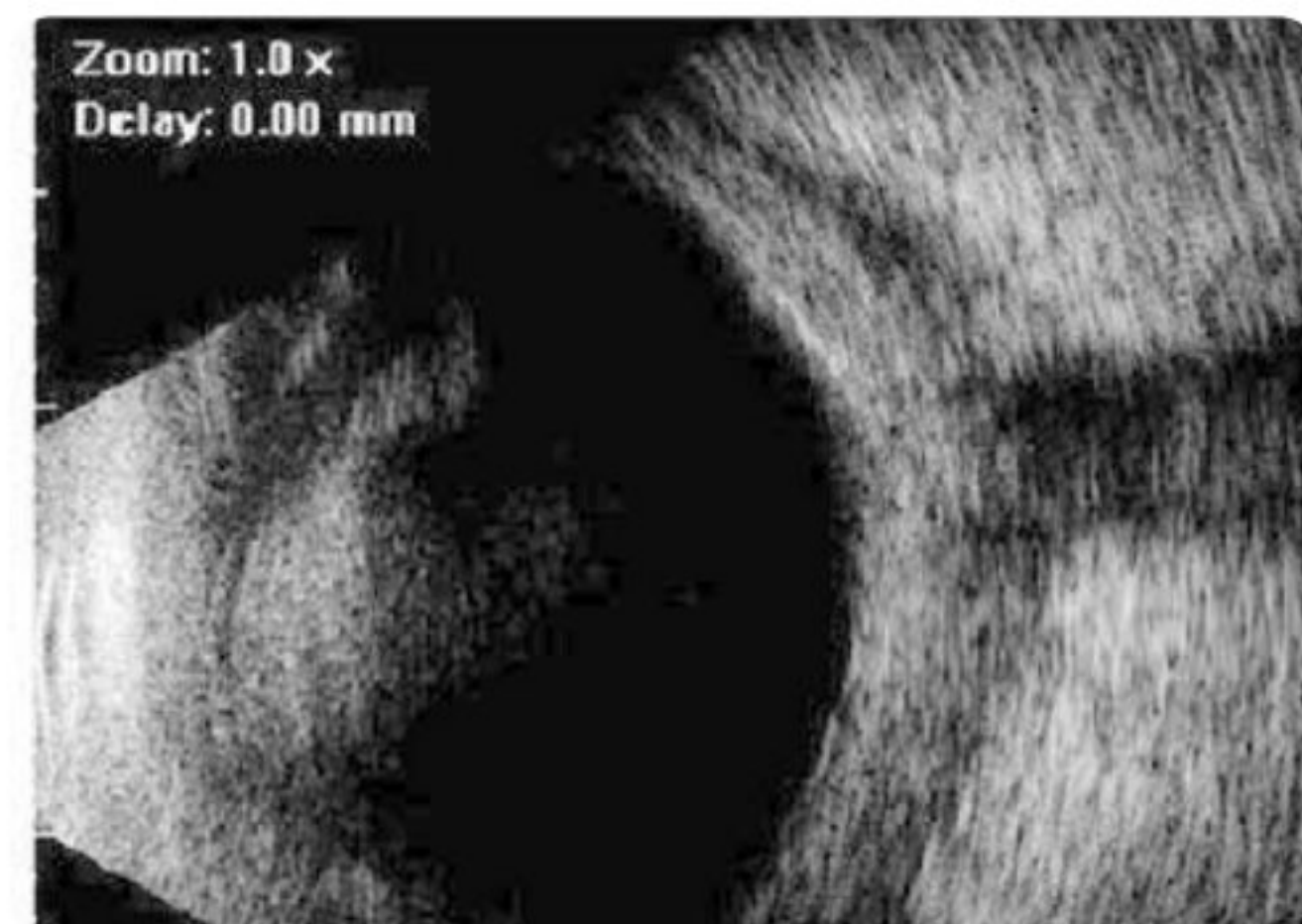
↑ Inflammation

↑ out
inflammation

↓
SCLEROMALAKIA



Anterior nodular scleritis



SCLEROMALACIA PERFORANS:

• Ischemic necrosis of sclera

causes: Rheumatoid Arthritis & elderly women?

C/F:

- No pain
- No redness



Scleromalacia Perforans

Goldenhar Syndrome {oculoauriculovertebral spectrum}

Systemic features:

- Hypoplasia of the malar, maxillary and mandibular regions.
- Macrostomia & microtia
- Preauricular facial skin tags
- Hemivertebrae {cervical}
- Mental Retardation
- Cardiac, renal & CNS anomalies

Ocular feature:

- Dermoids
- Upper lid notching or Coloboma
- Microphthalmos
- Disc coloboma

GOLDENHAR SYNDROME



BLUE SCLERA:

Cause: **PO4E MAN - BC**

- 1) Pseudo xanthoma Elasticum
- 2) Osteogenesis Imperfecta → (MCC)
- 3) Glaucoma → Congenital
- 4) Ehler's Danlos syndrome Type VI
- 5) Marfan's Syndrome
- 6) Nevus of Ota
- 7) Big eye → High myopia
- 8) Scleritis



Nevus of Ota: Proliferation of melanocytes & Lamin fusca?

- Hamartoma → proliferation of native tissue
- Dit tyndal effect → blue colour of sclera

Osteogenesis Imperfecta : • **MCC of Blue sclera**
• Defect in synthesis & structures of **Type 1 collagen**

Type 1 : Autosomal Dominant

Features

- i) Few features with little or no deformity
- ii) Wormian bone
- iii) Hyperextensible joint
- iv) Dental hypoplasia
- v) Deafness
- vi) Easy bruising
- vii) **Ocular feature** →
 - Blue sclera
 - Megalocornea
 - Corneal Ulcer

Type II A ⇒ Sporadic or Inherited ⇒ AD

Features

- i) Deafness
- ii) Dental hypoplasia
- iii) **Multiple fractures**
- iv) Short limbs
- v) **Death in infancy is d/t respiratory infection**
- vi) **Ocular manifestation** →
 - Blue sclera
 - Shallow orbits

Trauma to sclera by blow out

Rupture occurs at:

- at the **insertion of rectus muscle**
- at the equator
- In an area parallel to the limbus opposite from the site of impact

Cornea:

Function

- i) Act as a clear & transparent window for transmission of light.
- ii) Refraction of light (2/3rd)

Refractive Power of eye = +60D

(max) Refractive Power of cornea = +44D (2/3rd)

" " of lens = +16D (1/3rd)

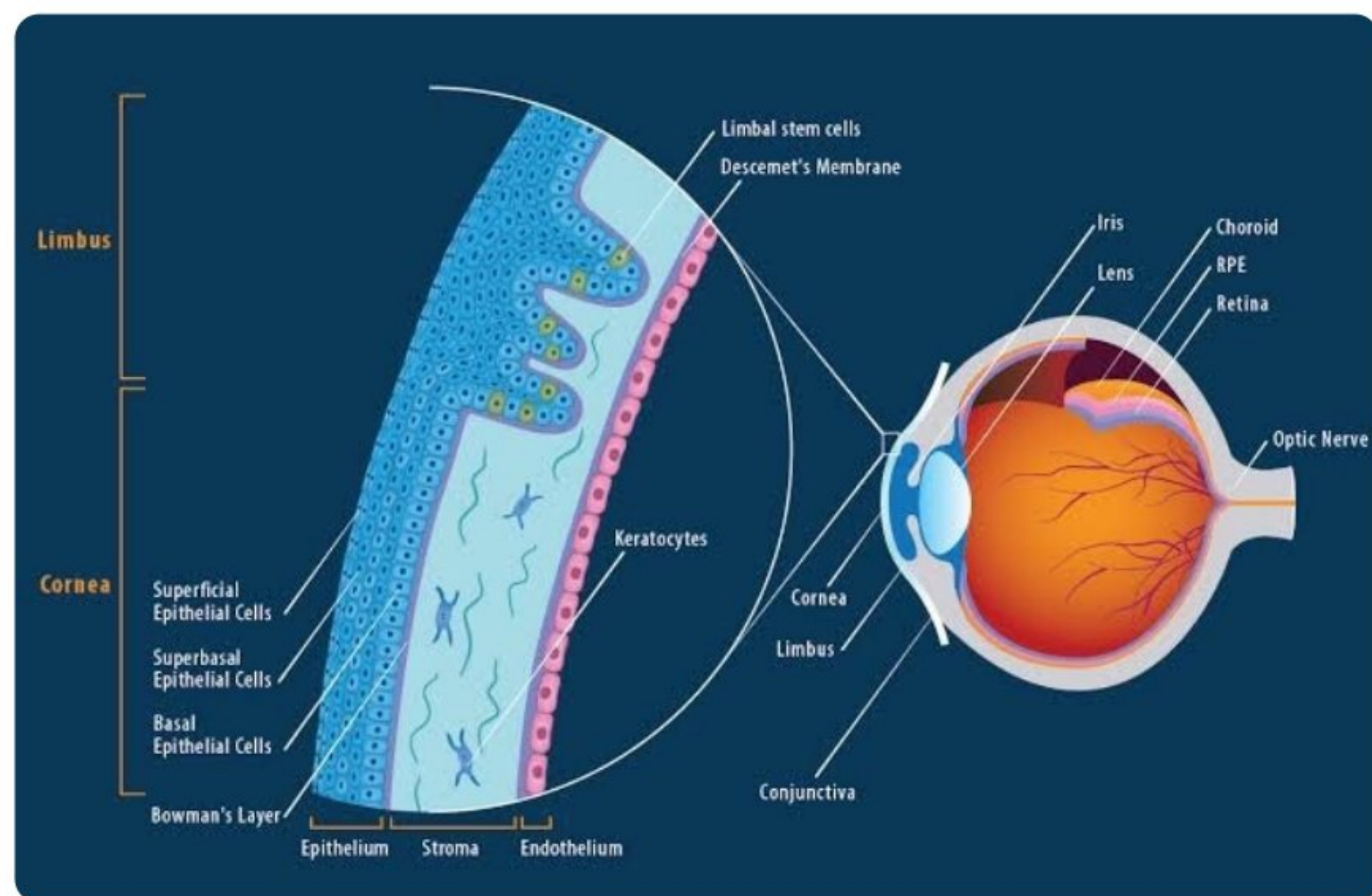
Refractive index of cornea = 1.376

Q: Maximum Refraction occurs at

- a) Lens nucleus
- b) Cornea
- c) Lens cortex
- d) Air-tear film interface

1.00 $\longleftrightarrow$ 1.357

LAYERS OF CORNEA:

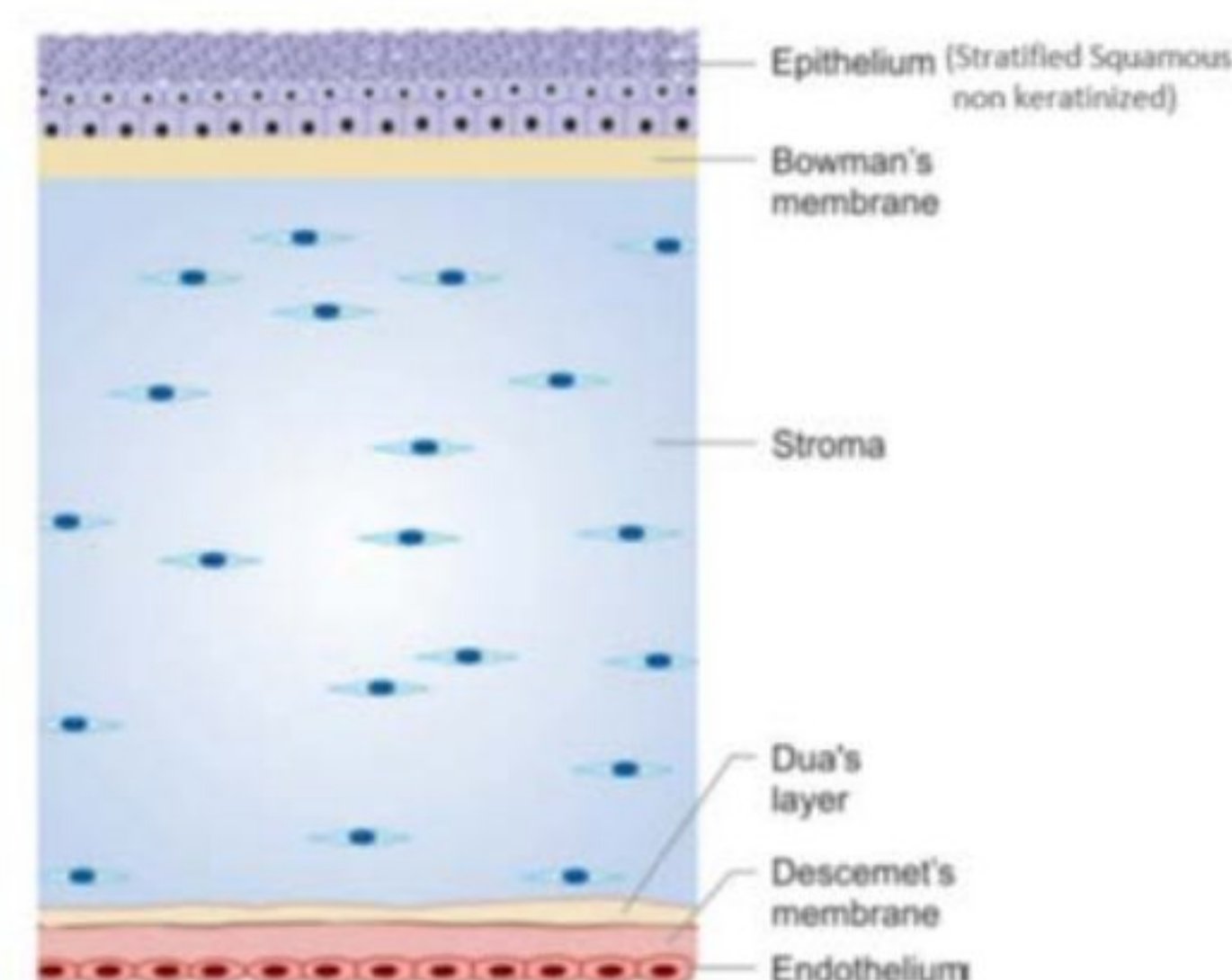


- 1) Epithelium
- 2) Bowman's layer
- 3) Stroma

Q: DUA layer = pre-DM
↳ TOUGHEST LAYER

- 4) Descemet's
- 5) Endothelium

Microscopic Structure of Cornea



Pachymetry $\rightarrow$ measure thickness of cornea
(N) $\Rightarrow$ 500 - 600 μ m

Q: Central cornea thinner than periphery

- Keratometry $\rightarrow$ measure curvature of cornea
 - i) Steep cornea $\Rightarrow$ Keratoconus
 - ii) Flat cornea $\Rightarrow$ Cornea plana

i) Epithelium

- smooth surface $\rightarrow$ d/t non keratinised stratified squamous epithelium
- Epithelium damage $\rightarrow$ heals completely without any scar or damage.

Epithelial Iron Line:

- i) Stocker's Line $\rightarrow$ Pterygium
- ii) Ferry's Line $\rightarrow$ filtering bleb of Trabeculectomy
- iii) Hudson - Stahi Line $\rightarrow$ Old age
- Q: iv) Fleischer ring $\rightarrow$ Keratoconus

FLEISCHER RING

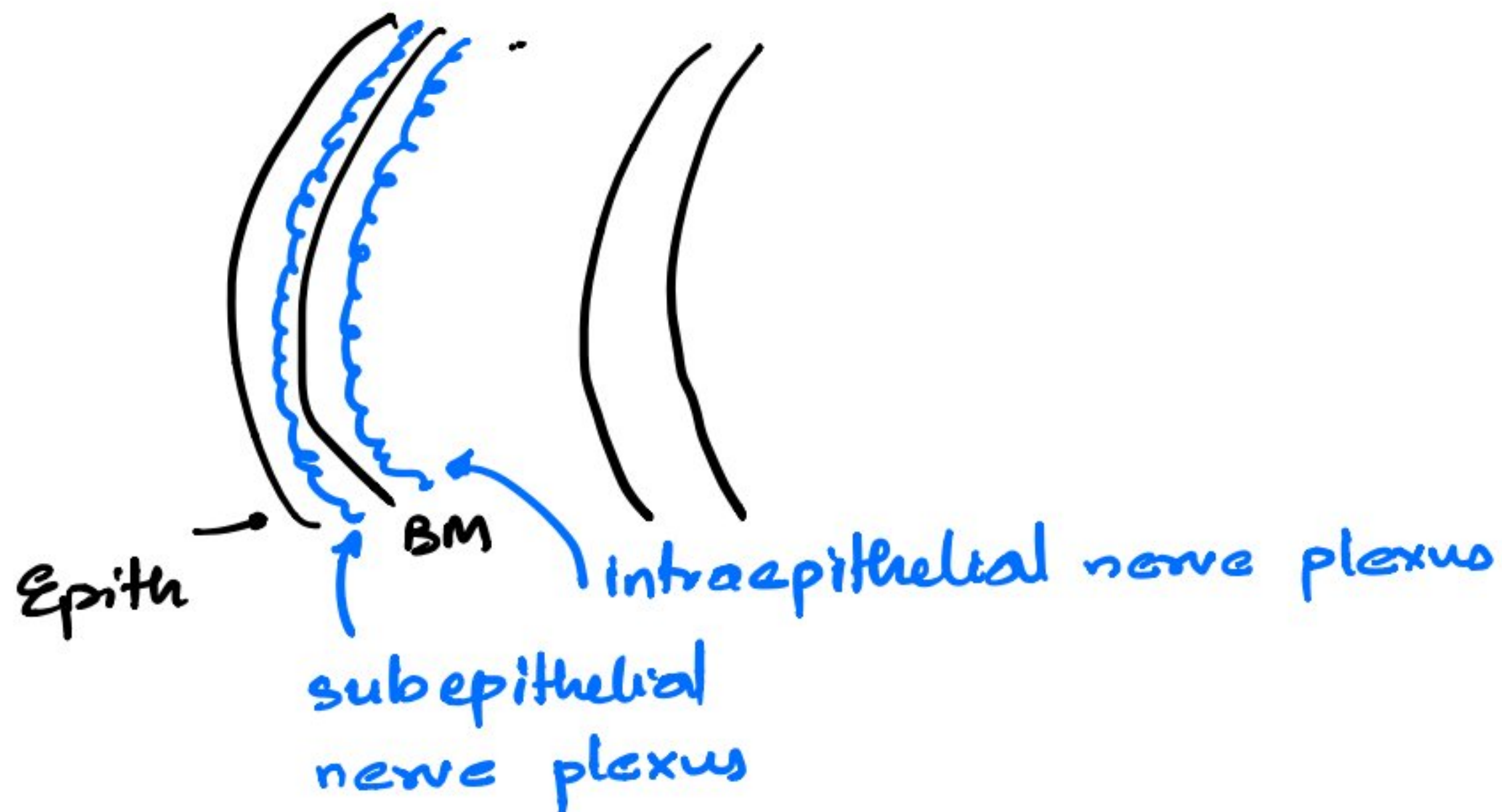
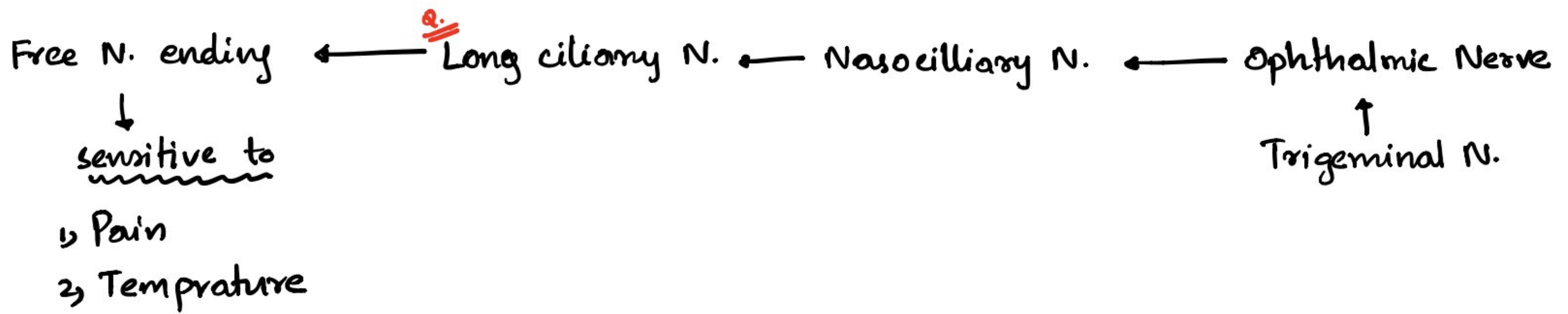
- Epithelium of cornea
- @ base of the Keratoconus



Pachymeter

- 2) Bowman's Layer →
- Acellular
 - Compact layer of collagen
 - Gives integrity of cornea
 - Never regenerates on damage
 - Scar or opacity formed

Corneal Nerves

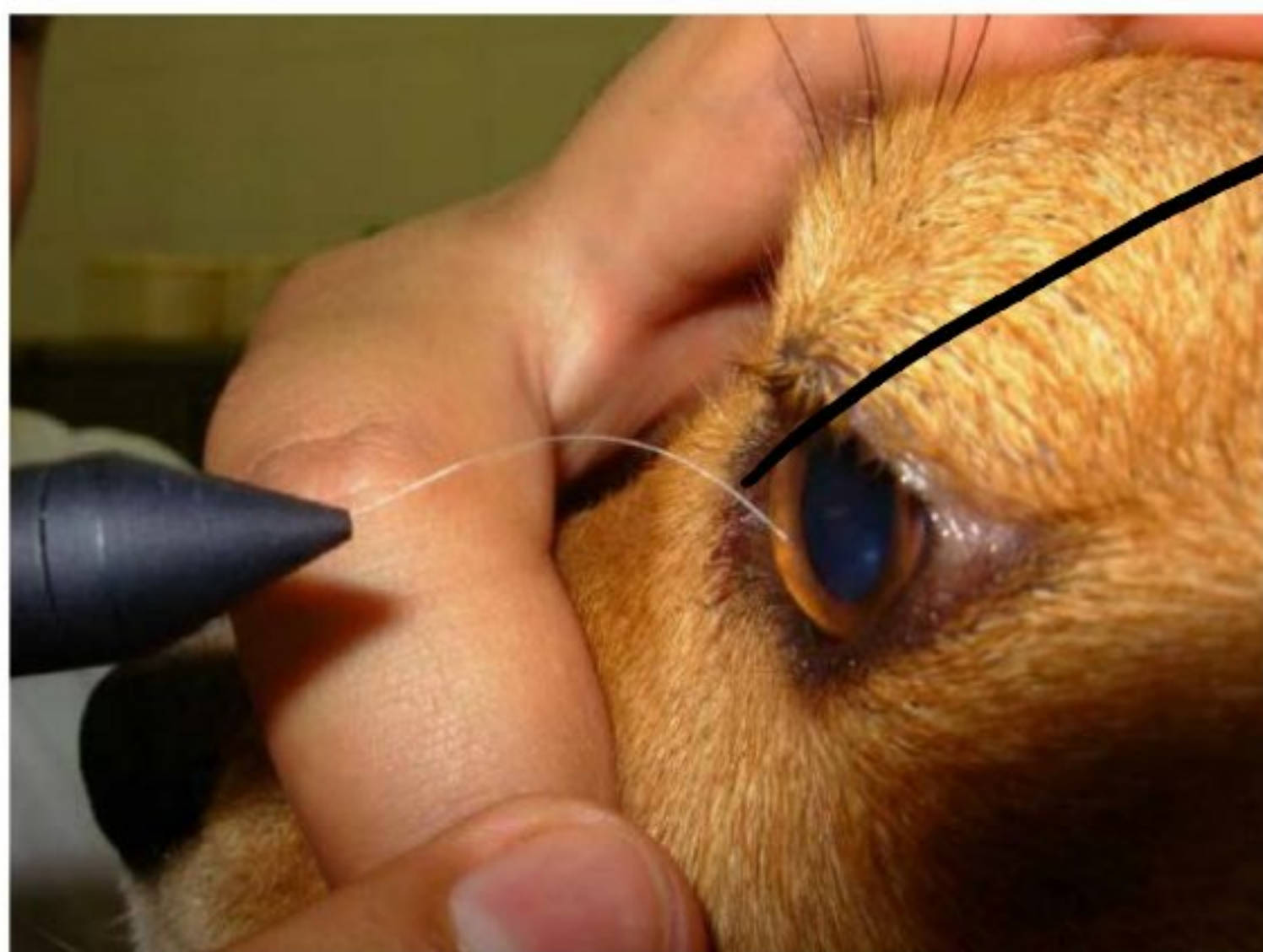


Corneal Reflex : B/L

Afferent : Vth Nerve

Efferent : B/L VIIth N. → orbicularis Oculi

Aesthesiometer → measure corneal sensation



Length of filament

CORNEAL ESTHESIMETRY

Figure 2. Photographic image of corneal esthesiometry in an adult male Beagle. The nylon filament is in contact with the axial cornea.

3, Stroma:

- 90% of corneal thickness
- Bed of proteoglycan → Hydrophilic
- Regularly arranged of Collagen I > III (provide transparency)

Proteoglycans: i, Keratan Sulphate
ii, Dermatan sulphate.

• Stroma kept in relatively dehydrated state becoz if water absorbed by cornea it will disrupts the arrangement of Collagen leading to haziness

4) Descemet's Membrane:

- Secreted by endothelium
- Forms the Basement membrane of endothelium
- **Collagen Type IV** { All Basement membranes in the body have collagen Type IV }

Q: Peripheral termination of DM @ limbus } **Schwalbe's Line / ring**
• Visualised on Gonioscopy.

KAYSER - FLEISCHER RING { K-F ring }

- Cu deposition at DM.

Causes:

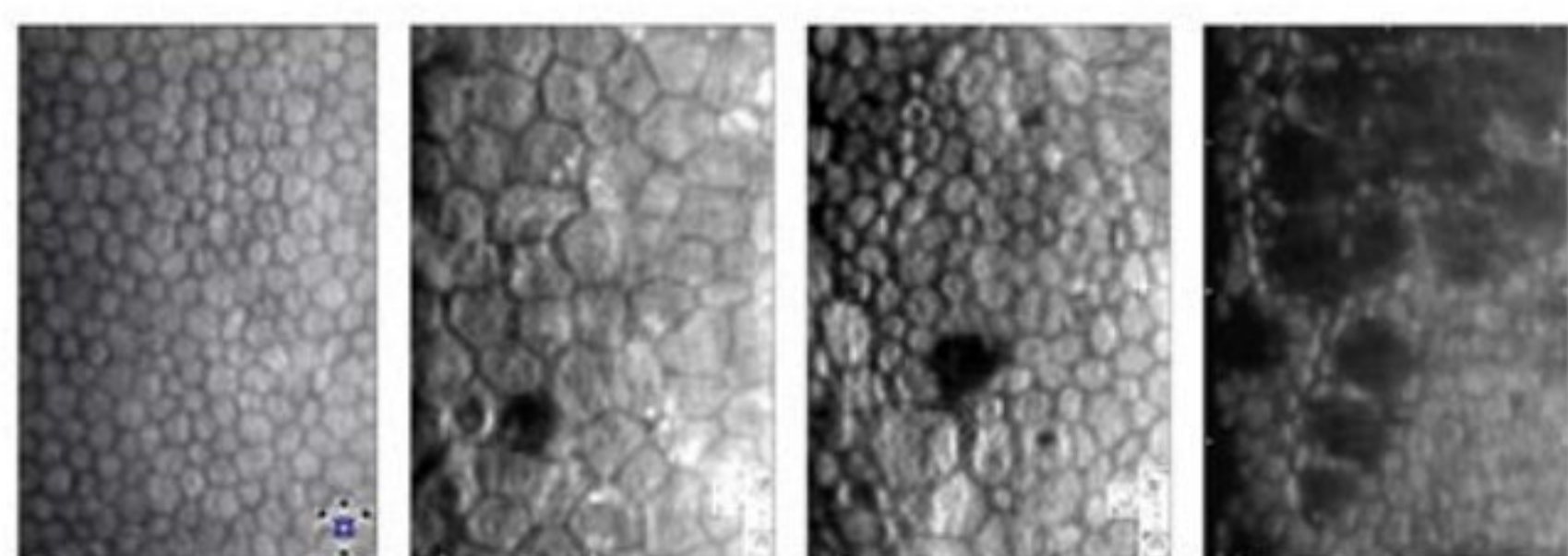
Systemic Cu toxicity → Wilson's d/s { **Reversible** }

Ocular Cu toxicity → Chalcosis

5) Endothelium:

- Single layer of cuboidal
- Visualise the endothelial layer by :-
 - i) Specular Microscopy → **HEXAGONAL PATTERN**
 - ii) Confocal Microscopy.

Cornea Endothelium



Normal Endothelium High Cell Density
Very Low Density High Surgical Risk
Polymegethism EW Contact Lenses
Stage 3 Guttata Normal Cell Count
Konan CellChek® Specular Microscope Imaging

① endothelium count } 3000 - 4000 cells / mm²
@ birth

↓
↓ ses by age

Q: Adult = 2500 - 3000 cells / mm²
Q: Old = 2000 cells / mm²

NEET PG 19

Function of Endothelium

Na⁺ - K⁺ ATPase pump

stroma $\xrightarrow[\text{H}_2\text{O}]{\text{Na}^+ - \text{K}^+ \text{ATPase pump}}$ Aqueous Humour

Relatively dehydration

critical density

↓

< 500 cell / mm²

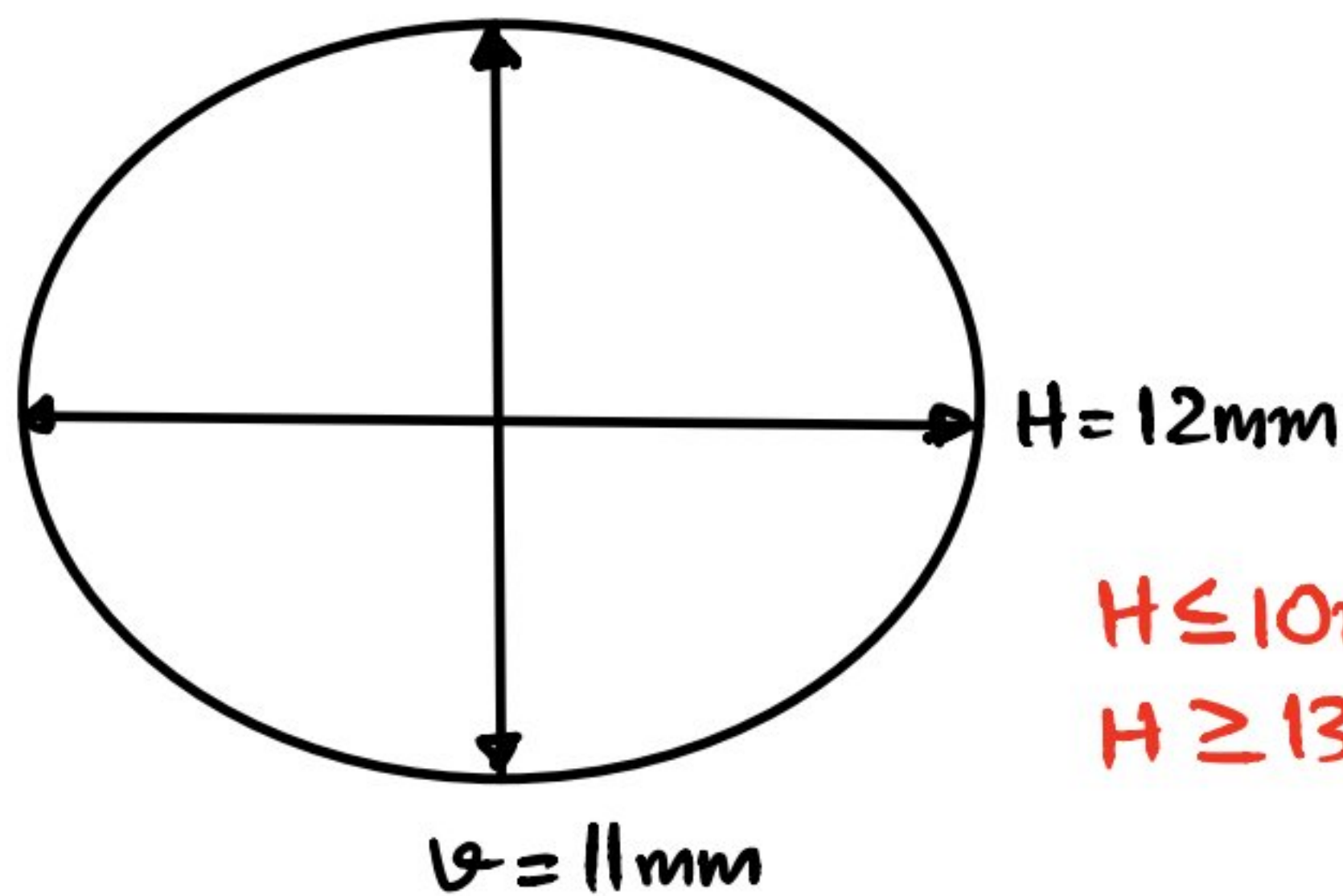
↓

stroma gets hydrated

↓

CORNEAL HAZY

DIMENSION OF CORNEA



$H \leq 10\text{mm}$: microcornea , AD
 $H \geq 13\text{mm}$: Megalocornea , XLR & male?

Grading of Corneal opacities

Nebula : faint opacity
 ↳ most visually handicapping

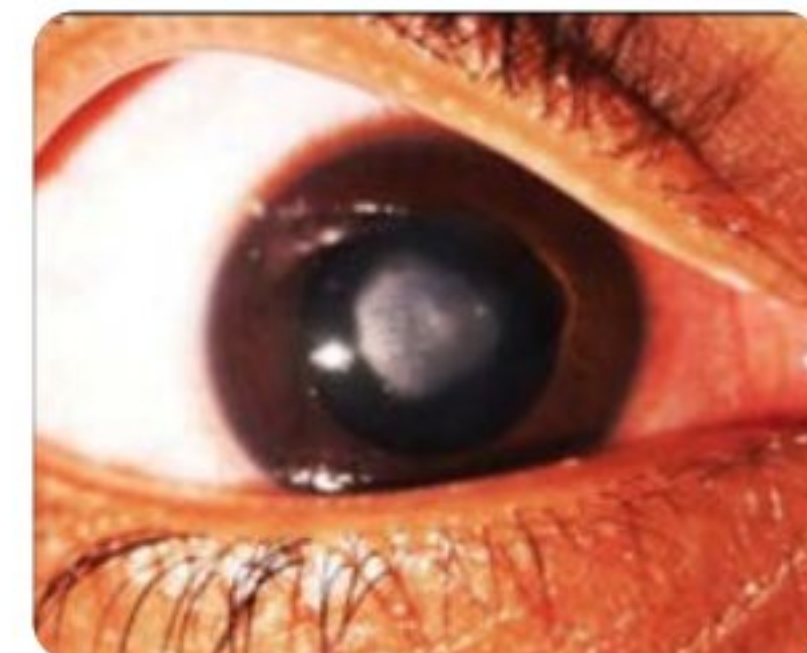
Macular : Moderate opacity

Leucoma : Totally opaque
 (asymptomatic)

⚡ // ⇒ only BM is damaged

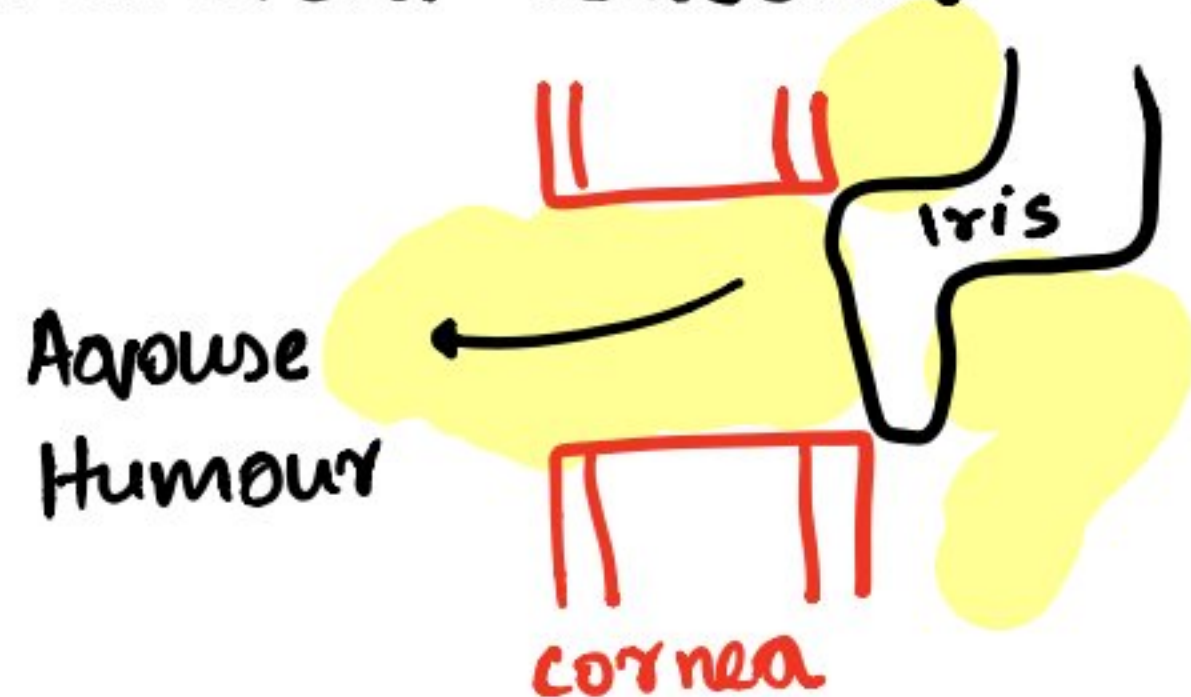
⚡ // ⇒ BM + $< 1/3^{\text{rd}}$ stroma

⚡ // ⇒ BM + $> 1/3^{\text{rd}}$ stroma



Macula

Adherent Leucoma : Complete perforation



LEUCOMA

Factors involved in Corneal Transparency:

- 1) Avascularity
- 2) Unmyelinated^a nerve fibres
- 3) Absence of lymphatics
- 4) Absence of Antigen Presenting cells.
- 5) Non Keratinising nature of epithelium
- 6) Intact Bowman's layer
- 7) Regular arrangement of collagen → stroma
- 8) Relative dehydrated state of stroma
- 9) Very few keratinocytes in the stroma
- 10) Endothelial cells → adequate no. & $>$ critical density?
- 11) Endothelial pump fx $\propto \propto$ IOP
 (10 - 21 mmHg)

Nutrition of Cornea

- Aqueous Humour
- Perilimbal Capillaries

O₂ to Cornea

Air dissolve in tear film.

Metabolism

≠ Aerobic Glycolysis

Corneal Ulcer : Breach / Discontinuity of Epithelium → only epithelium : Abrasion
⊕
Stromal necrosis

Causes :

1) **Bacterial** R.F is corneal epithelial defect

- * F.B
- * Prev. Viral Keratitis
- * Contact lens
- * Dry eye.

PATHOGENS WHICH CAN PENETRATE INTACT CORNEA EPITHELIUM

- NEISSERIA GONORRHOEAE**
- HEMOPHILUS**
- LISTERIA**
- N. MENINGITIDIS**
- C. DIPHTHERIA**

Bacterial Corneal Ulcer

MC organism

India → Str. Pneumonia

West
contact lens } Pseudomonas

Peripheral corneal ulcer → Staph aureus

Pneumococcal Corneal Ulcer :



≠ "Serpiginous" corneal ulcer a.k.a creeping Corneal Ulcer

- active @ one end & healthy from the other end
- rapidly spreading ulcer.



Ulcer Serpens

Pseudomonas Corneal Ulcer:

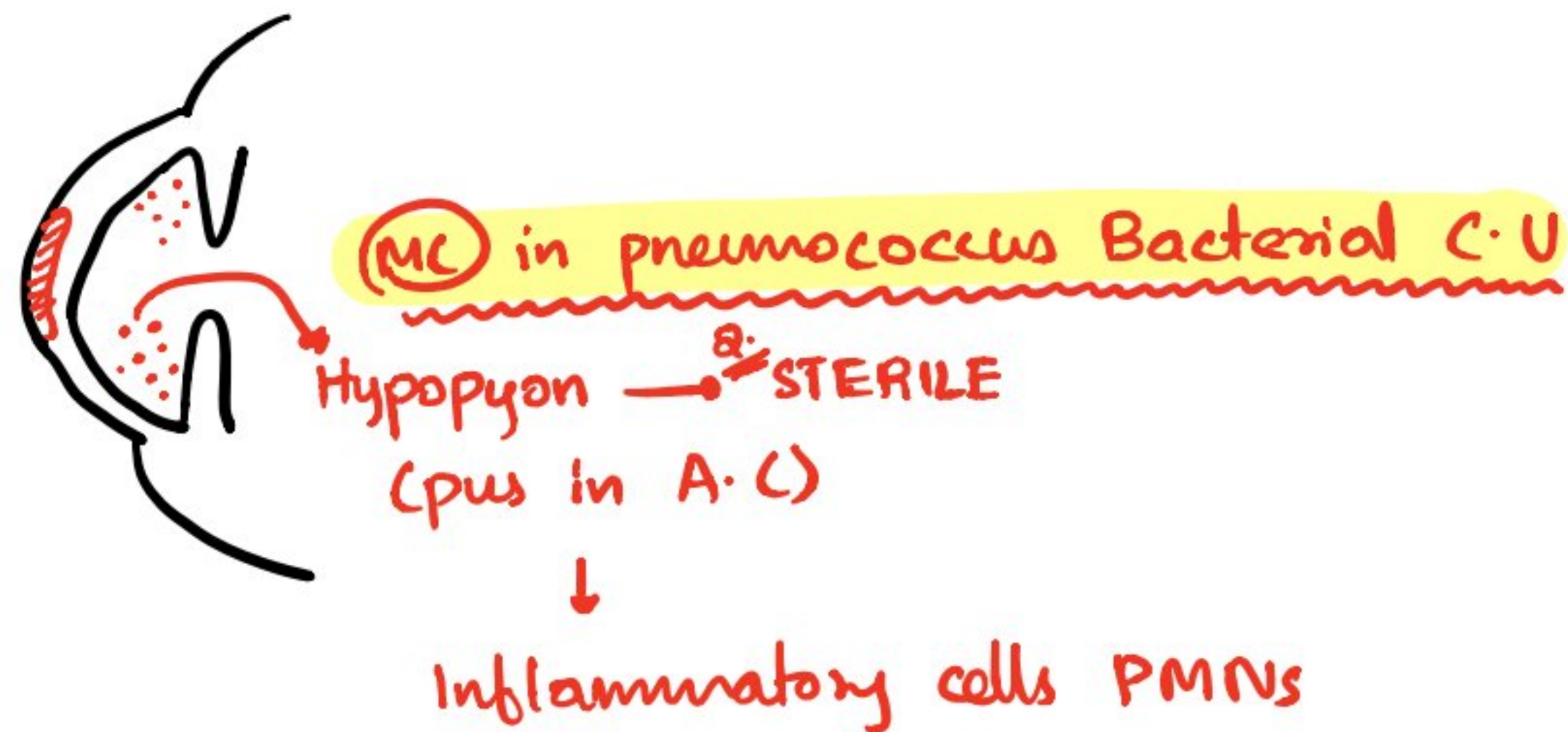
(MC) → Western countries
CL users.

- (MC) gram -ve organism causing C.U
- Greenish discharge
- Perforating ulcers

Pathogenesis:

Adheres damaged → Rapid stromal invasion → Destructive Enzyme release → **PERFORATION within 48 hours**

- Protease
- Lipase
- Elastase
- Exotoxin



INVESTIGATION:

1) Corneal staining: **Flourescent dye = Topical 2%.** (Orange dye)

↓
Fills the **epithelial defect**

↓
observe ↓ Cobalt blue light

↓
Flouresence: Yellowish green

2) Corneal scraping

↳ Staining on slide → Gram stain
↳ C&S

Rx:

Non specific

1) Cycloplegia: Relieve ciliary spasm
[1% Atropine] Pain

- ii) Dark goggles → To reduce Photophobia
- iii) NSAIDs

Specific Rx:	Uncomplicated Bacterial C.U	Mx. of Non healing Ulcer	Impending Perfor.
	1) Topical fortified Antibiotics = 5% Vancomycin {4+ve} [5%] + Cephalosporin {4+ve} [5%]	1) Debridement 2) Chemical cauterization [10-20%] Trichloroacetic acid 3) Conjunctival flapping	i) ↓ IOP ii) Cyanoacrylate {Tissue adhesive glue} iii) Bandage C.L

Fungal C.U {Mycotic keratitis}

• Fungal Infection → Generally associated with trauma to eye with vegetable matter

(MC) fungi in India → **ASPERGILLUS FUMIGATUS**

Thorn, while farming

1) C/F: Symptoms: Patient is **less symptomatic** compared to signs of disease

2) S/S: a) Mild watering, photophobia, redness, pain, mild discharge

b) Fungal ulcer → **DRY LIKE PARCHMENT PAPER**

c) Immune ring or Wessely

d) Feathery edges

e) Satellite lesions in periphery

f) Hypopyon immobile & sterile → contains hyphae

g) No vascularization.

3) Investigation:

a) Smear exam → i) 10% KOH wet preparation

ii) Gomori methamine silver stain

b) Culture → Sabouraud's Glucose agar.

TREATMENT

Antifungal

• Topical ⇒ i) Natamycin } Filamentous
 ii) Amphotericin }

iii) Imidazole } Candida Infection
 iv) Flucystosine }

• Systemic antifungal → severe keratitis & endophthalmitis

Adjuvant / Concurrent

• Cyclopegics → Atropine
 ↳ Scopolamine

• Carbonic anhydrase ⊖ → to prevent complication of 2° glaucoma



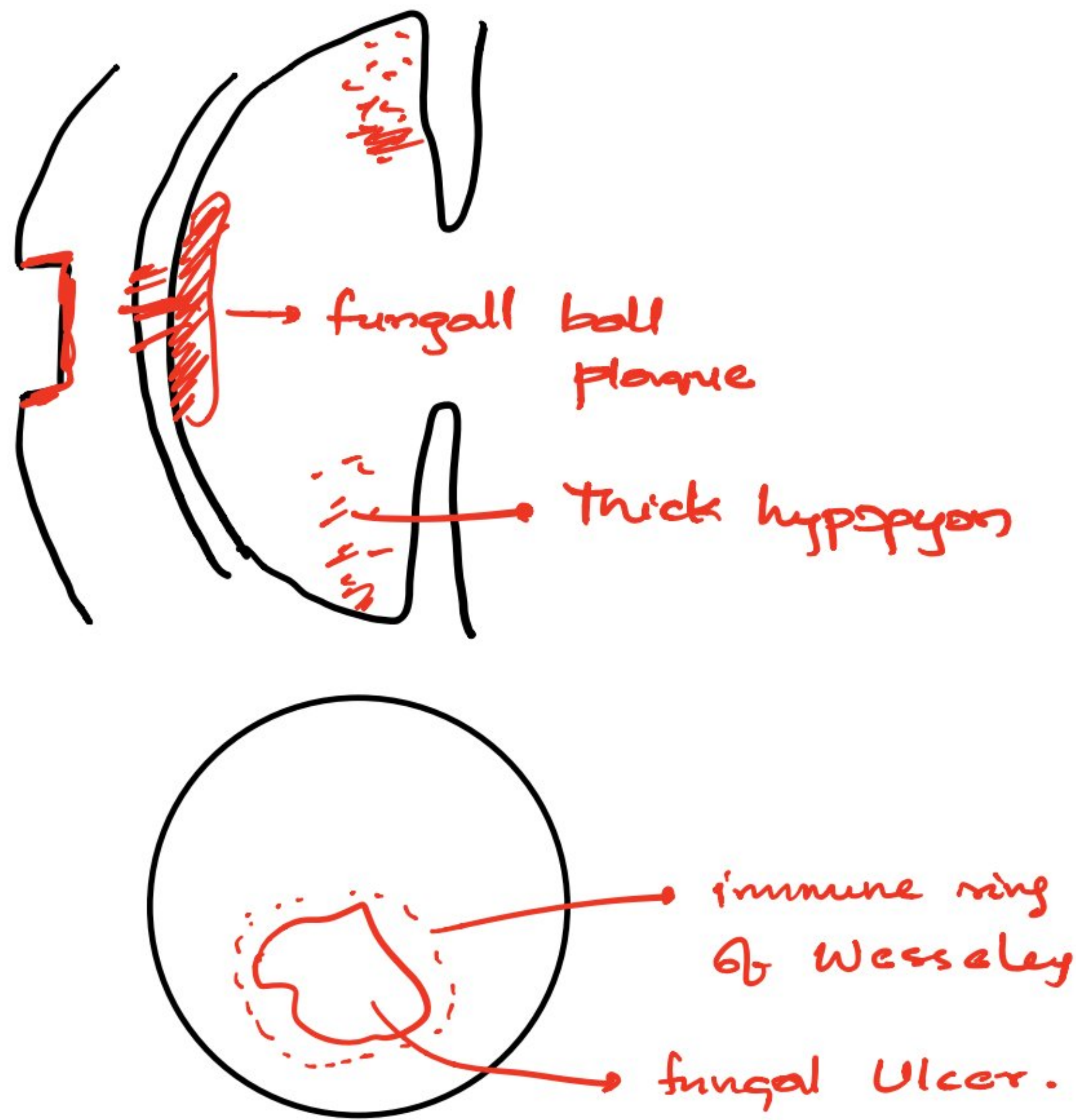
• Dry as leather parchment app.

• It defined feathery margins

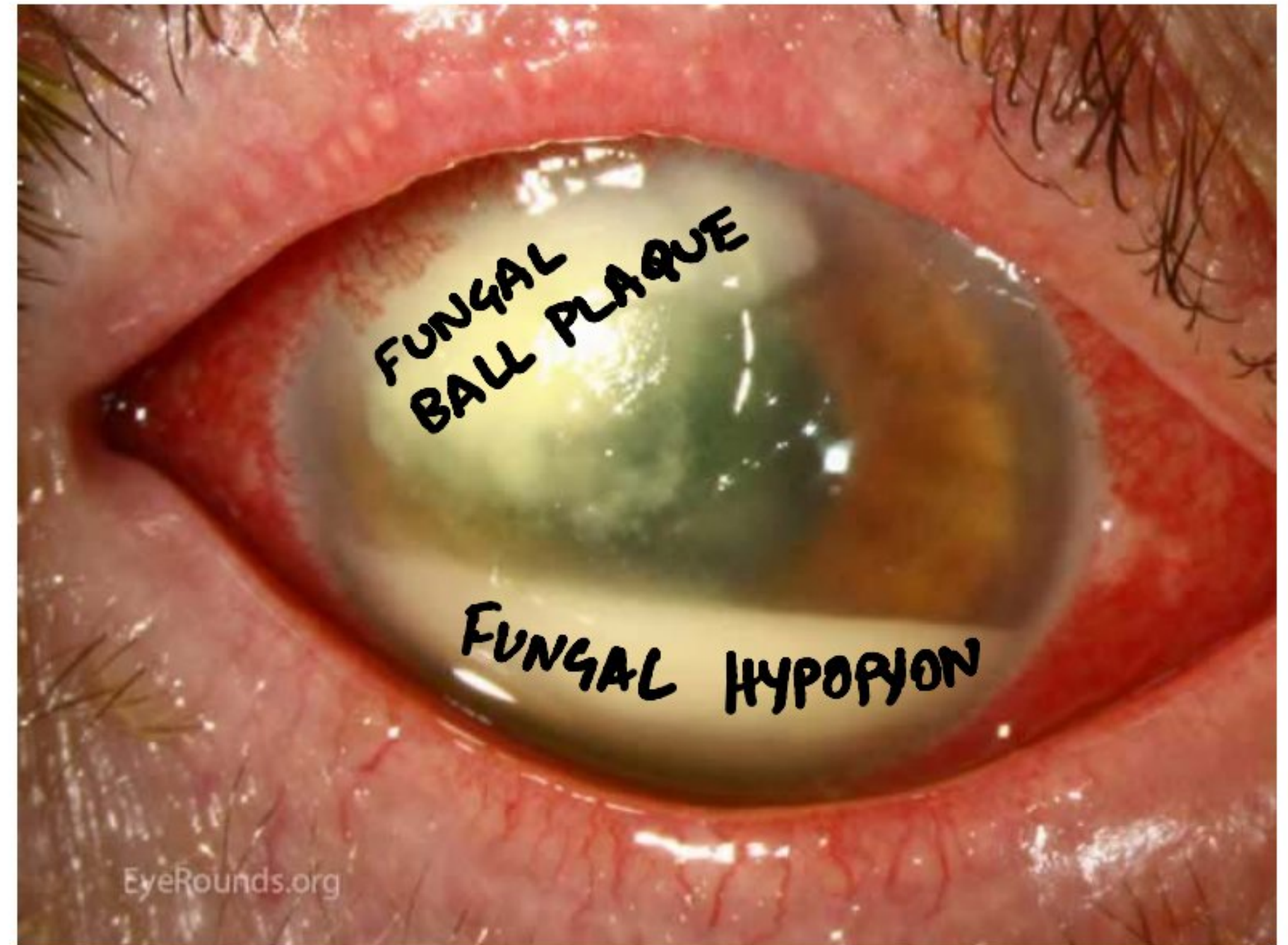
• Sterile immune ring & yellow line of demarcation

• Satellite lesions

• Perforation is rare



- Hyphae → invade intact DM
- Fungal ball → plaque @ Endothelial layer
- Hypopyon → Fixed
Infected
Thick



3) Viral keratitis:

a) Involves the epithelium:

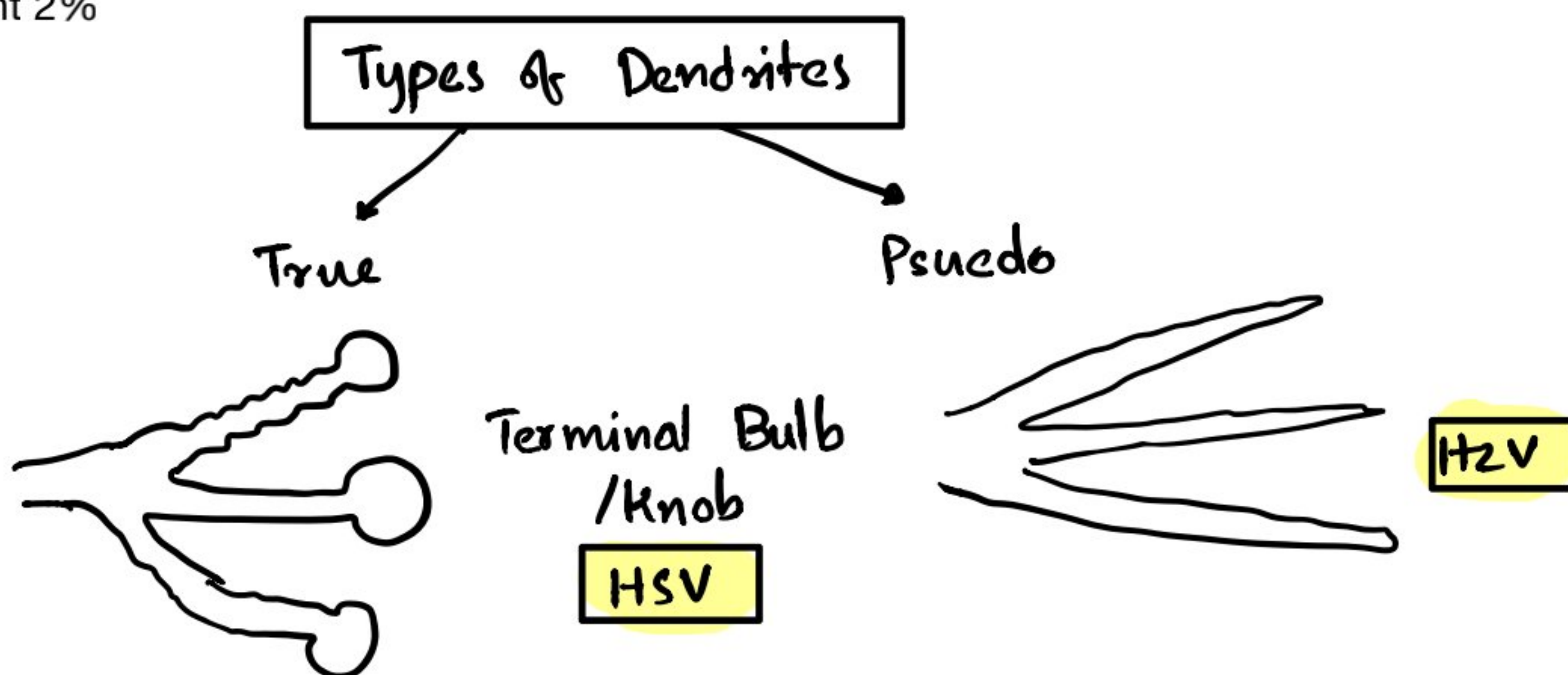
- Dendritic Ulcer



Dendritic Ulcer - staining is fluorescent 2%

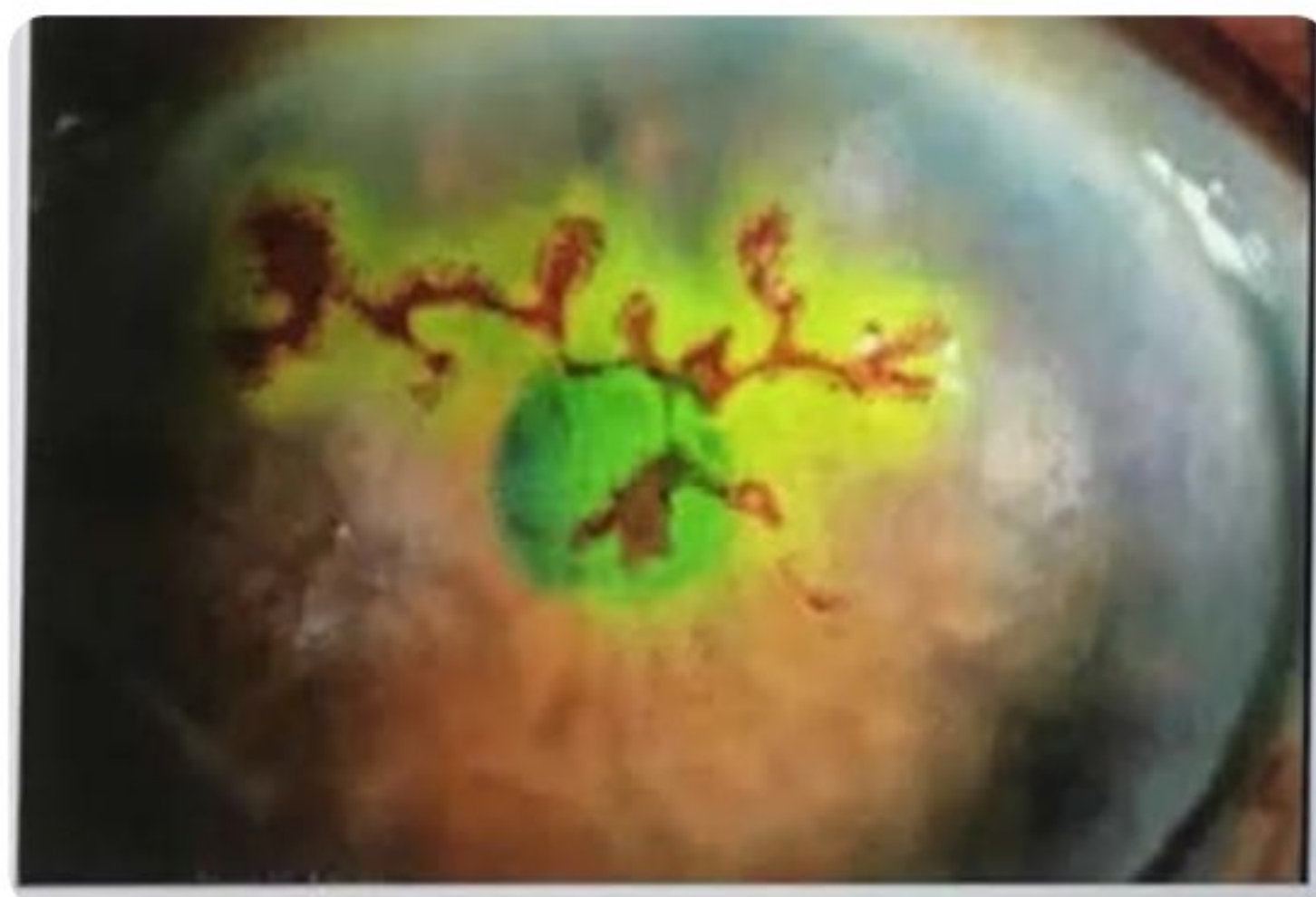
Dendritic ulcer :

- has linear pattern with branches and terminal bulbs at the ends (resembling a tree branches). The vesicles on epithelium unite to form the linear pattern. The bed of the ulcer stains with fluorescein and the diseased cells at the margin takes up rose Bengal stain. corneal sensation is diminished or absent.



Other Stains

- Rose Bengal Staining: 1% topical
Stains the margins of Ulcer $\Rightarrow$ dead epithelial cells



Flourescent
2% Topical
Central stain
Stain the floor of ulcer.

Rose Bengal

- 1% topical
- Peripheral stain
- Stain margin of ulcer

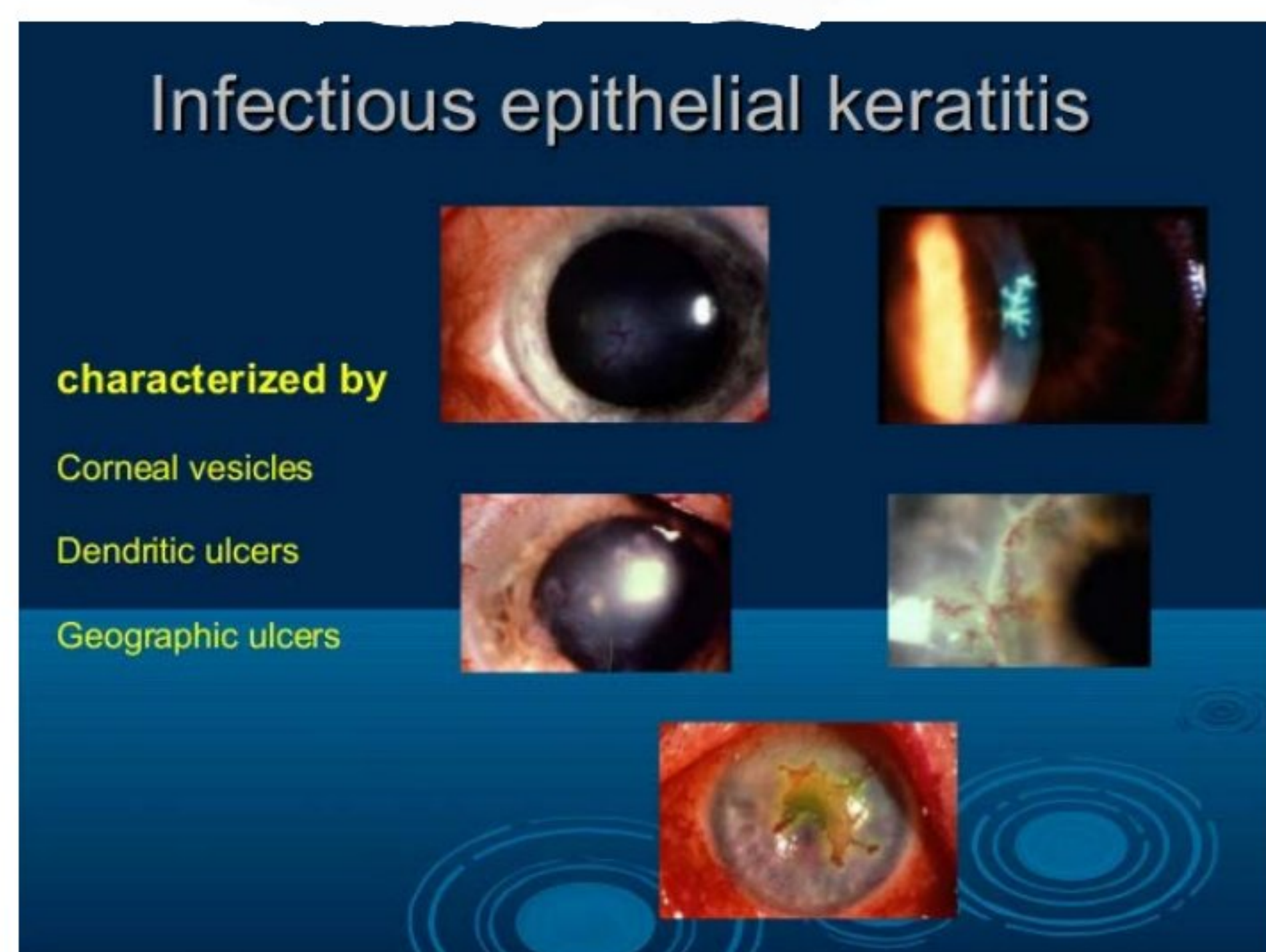
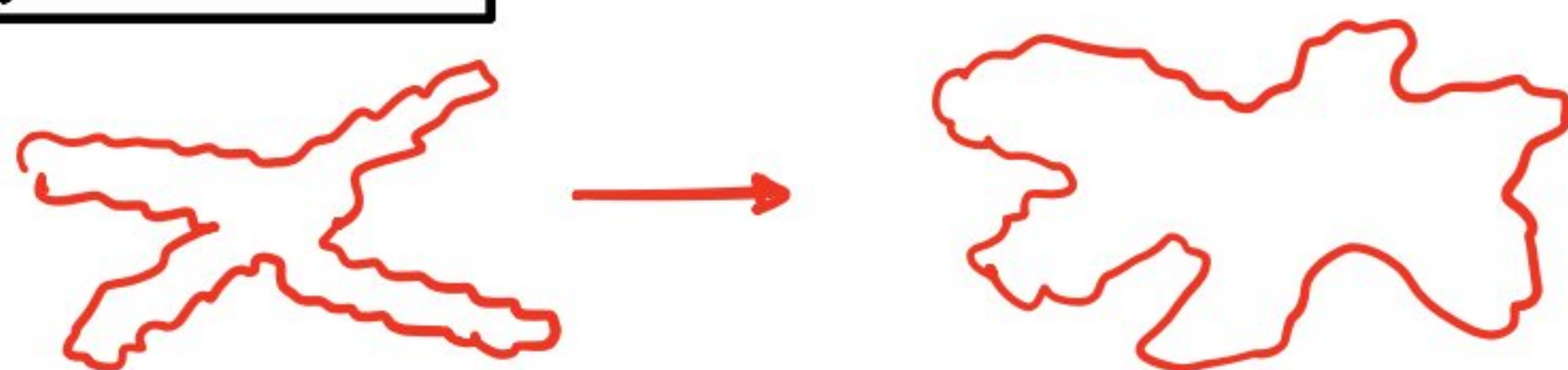
No corneal scraping done {cause its a virus}

Rx:

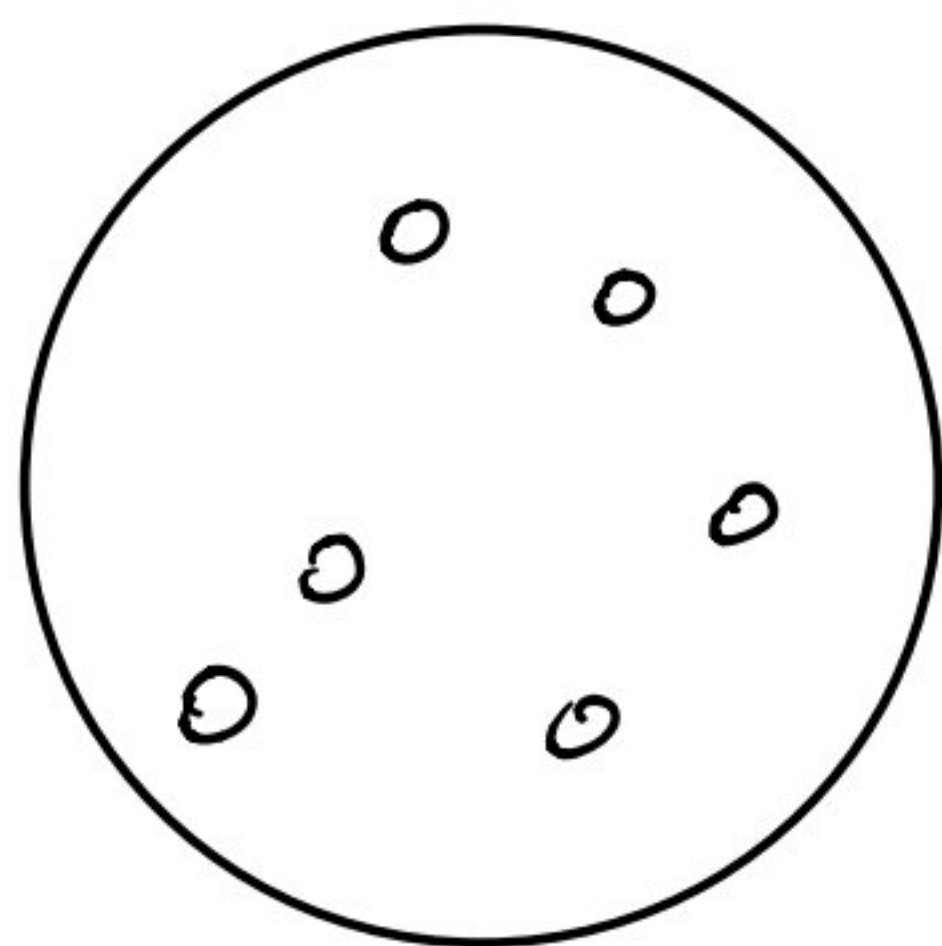
Specific Rx:

Topical antiviral DOC: ACYCLOVIR 3% eye oint 5 times a day

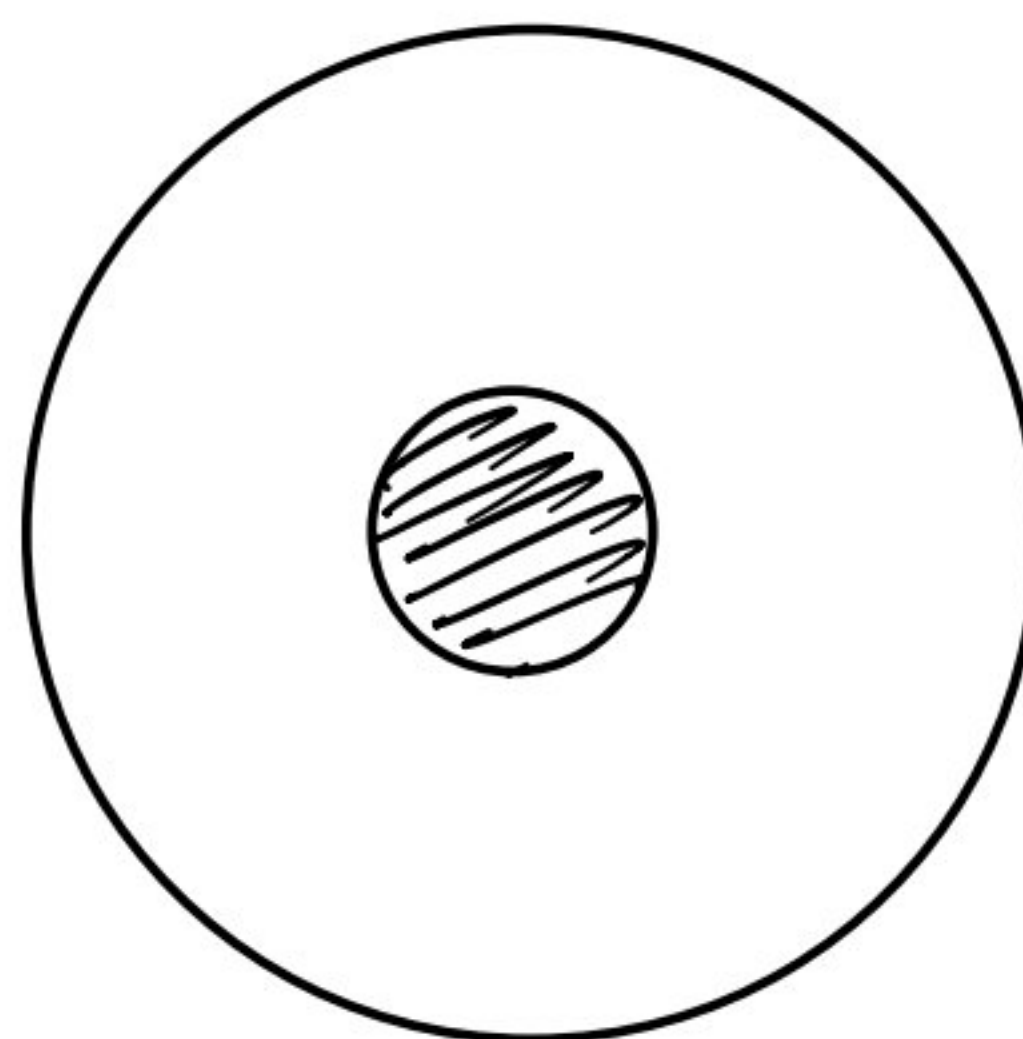
Geographic Ulcer $\Rightarrow$ fusion of the dendrites



b) Stromal Keratitis: • No active viral replication
• Ag-Ab Rn.



b) Nummular Keratitis



Disciform Keratitis

- 1) DM fold $\oplus$
- 2) Keratic precipitates
- 3) Type IV H.S

Mx: Topical Steroid

c) **ENDOTHELITIS** → Trabeculitis

↓
↑sed IOP

Rx: Topical steroid

⊕
↓ IOP

HERPES ZOSTER OPHTHALMICUS

1° infection: Blepharconjunctivitis *recover spontaneously*

↓
virus = @ Vth N

↓
Gasserian Ganglion / *Trigeminal Ganglion*

Latent infection ≈ Trigem =

- fever
- Emotional stress
- heat

2° Reactivation

Nasociliary N.

↓
cornea 

↙ Tip of nose { *Q.T VI* }
{ *via external branch* }

HUTCHINSON'S RULE / SIGN: Clinical predictor of corneal involvement based on presence of lesion @ tip of the nose.

• NOT 100% predictable

Q: (M.C) Nerve involved in H₂O → **frontal N.**

Q: (M.C) Cranial N. involved in H₂O → **IIIrd N.**

HERPES ZOSTER OPHTHALMICUS

Mx: **NO ROLE OF TOPICAL ANTIVIRALS**

Systemic antivirals:

• Tab. Acyclovir 800mg ⇒ 5 times a day **x 2wk**

• Steroids → for neuralgia

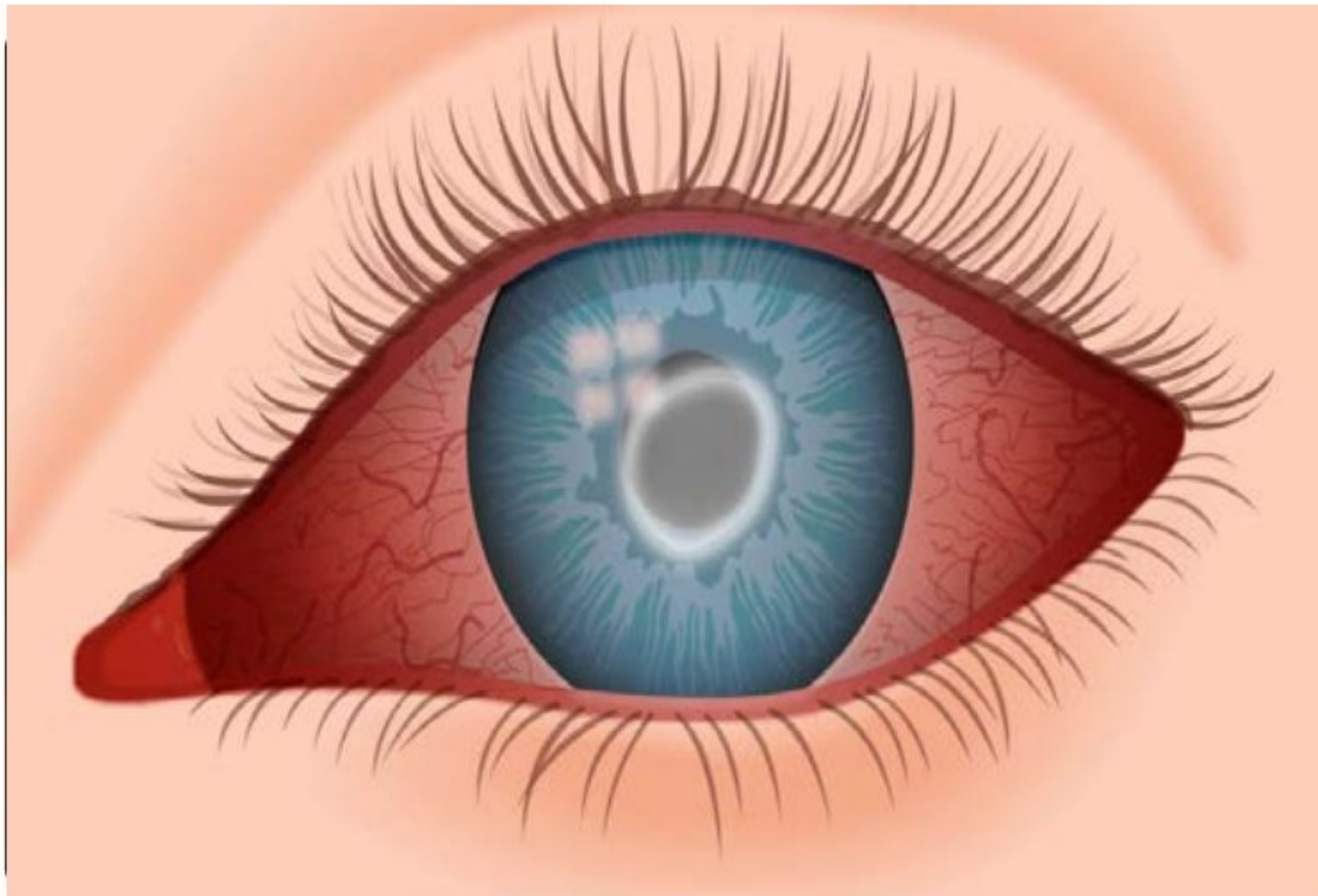
Acanthamoeba Keratitis:

Risk factors:

- i) CL usage → unhygienic practices
 - extended wear CL
 - [⚡] tap H₂O, home made saline for cleaning
 - swimming with CL.

Presentation:

- Symptoms out of proportion to sign ⇒ Excruciating pain
- Hallmark sign: [⚡] Radial perineuritis (cause of pain)



Acanthamoeba Keratitis
Ring Infiltrate - Ring Stromal Abscess



Radial Perineuritis

Investigation:

- Smear Examination →
- a) Calcofluor white: cysts stain bright green
 - b) KOH mount
 - c) Lactophenol blue

[⚡] Culture medium: E.coli enriched Non nutrient agar

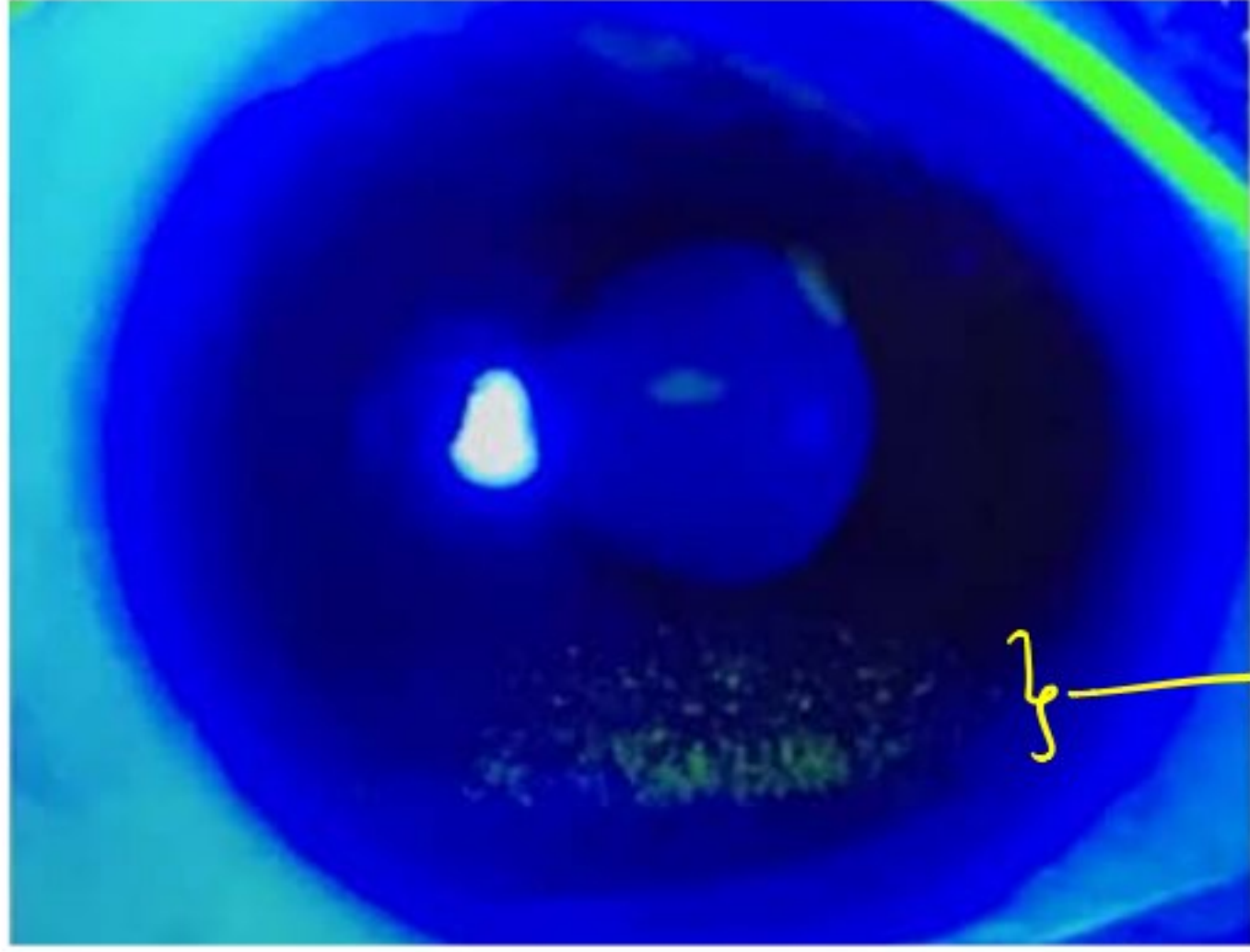
Treatment:

- a) Combination of dipropamide and polyhexamethylene biguanide drops (PHMB)
 - ↳ DDC
- b) Other drugs:
 - i) Propamide isethionate
 - ii) Neomycin
 - iii) Chlorhexidine (0.02%)

Exposure Keratitis:

Causes:

- 1) Lagophthalmos : Incomplete closure of eyelid
In Bell's palsy (VIIth N) → orbicularis oculi & closure of eyelid?
- 2) Proptosis



staining of Inf. 1/3rd of cornea

Neurotrophic Keratitis / Neuropathic Keratitis

- Trigeminal N. affected | **MCC** → Varicella Zoster | Rx: Recombinant human nerve growth factor CENEGERMIN.

MOOREN'S ULCER: {It is a severe inflammatory peripheral ulcerative keratitis}

- non infective
- Autoimmune
- Chronic serpiginous ulcer
- Over hanging edge, ulcer : undermined edge
- Peripheral, superficial ulcer which rarely perforates

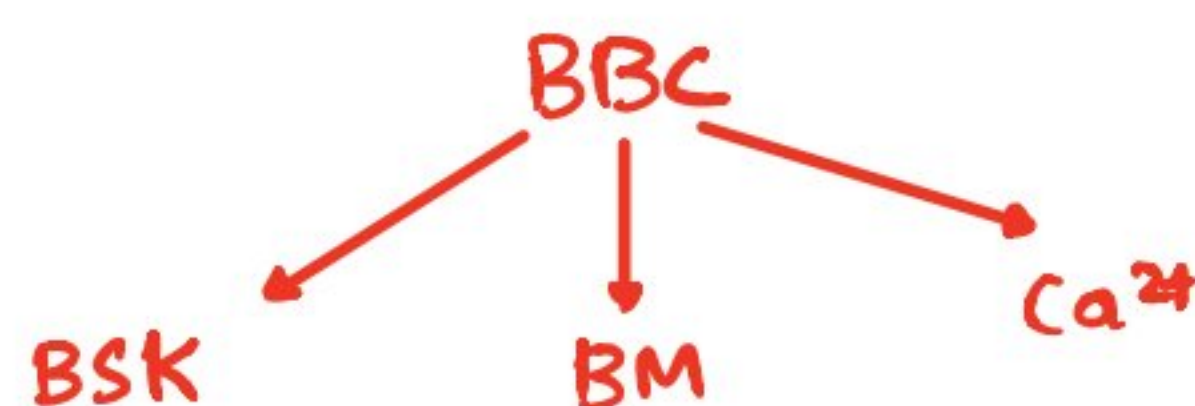
Rx: Steroid (topical)

Peripheral Ulcerative Keratitis (PUK)

MCC: Rheumatoid Arthritis

Corneal Degeneration:

Band shape keratopathy:



- Calcific degeneration
- Present @ interpalpebral
- Clear interval from limbus
- "Swiss-cheese" appearance

MCC: Chronic Iridocyclitis

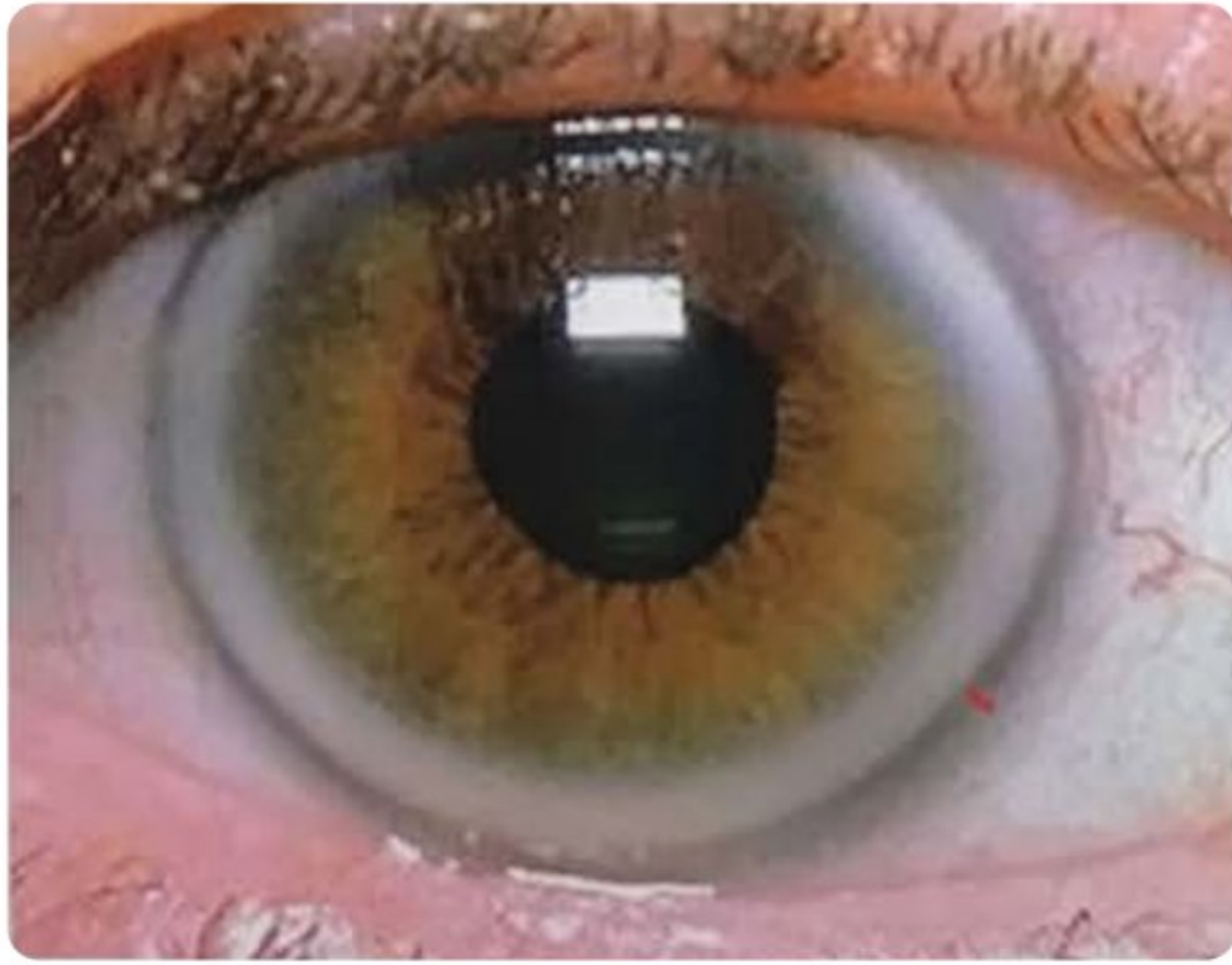
Causes:

- 1) Idiopathic ⇒ **MC**
- 2) Juvenile Rheumatoid arthritis → c/c uveitis
- 3) Hypercalcemia
- 4) Sarcoidosis

Rx: Topical Ca chelators → EDTA

↳ Laser: PTK: Phototherapeutic Keratectomy
↳ **Excimer Laser**

ARCUS SENILIS:



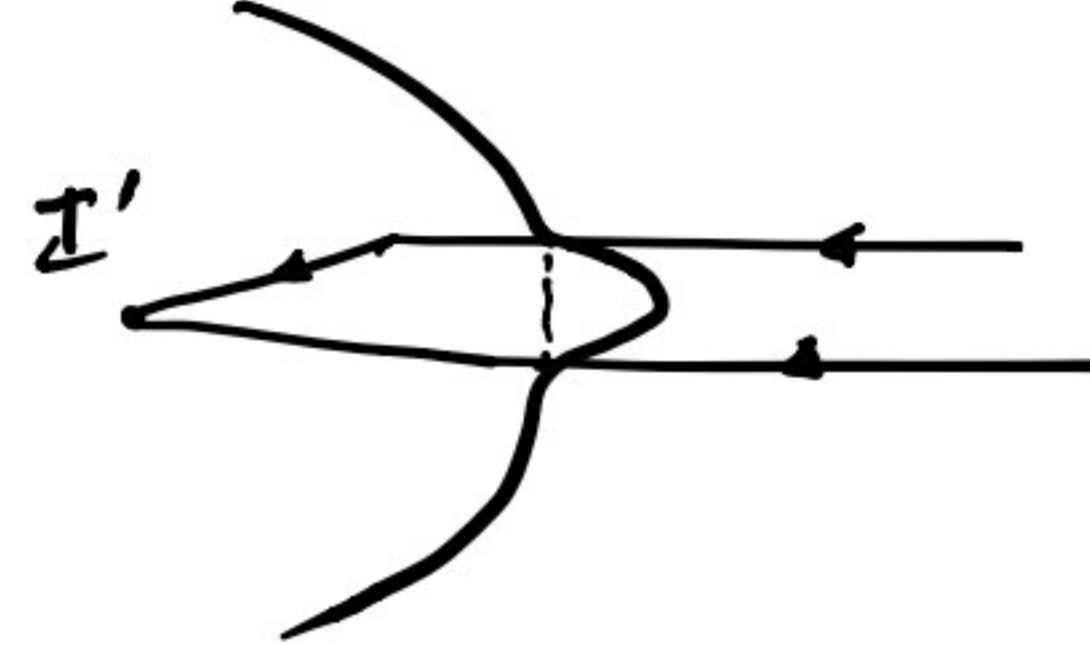
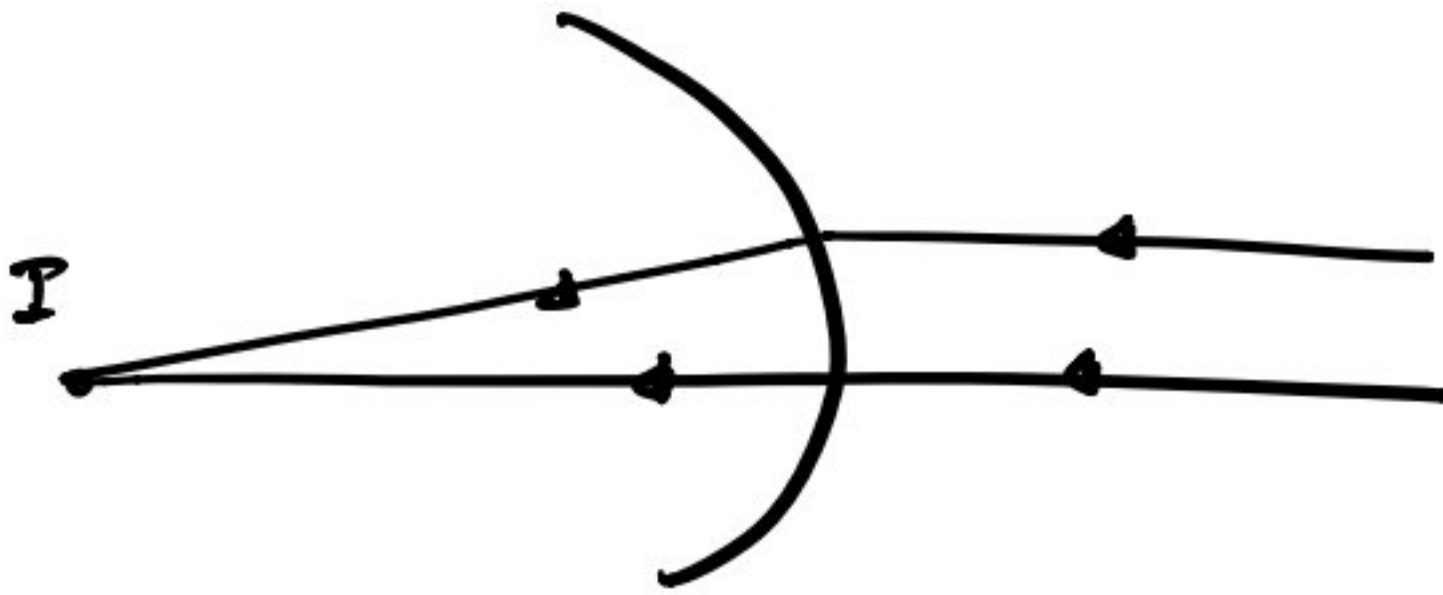
- Physiological
- Lipid deposition in stroma & BM
- Lipid degeneration

Arcus Juvenilis ∴ Young people

- a/w DM
- Familial Hypercholesterolemia

Pseudogerontoxin ⇒ VKC

KERATOCONUS: Non inflammatory degeneration in central cornea resulting in corneal thinning & ectasia



- Refractive Error = Curvatural Myopia
- Irregular astigmatism
- Freq. change in glasses & Progressive Myopia?
- Contact lens intolerance & eventually CL become unfit?

Signs:

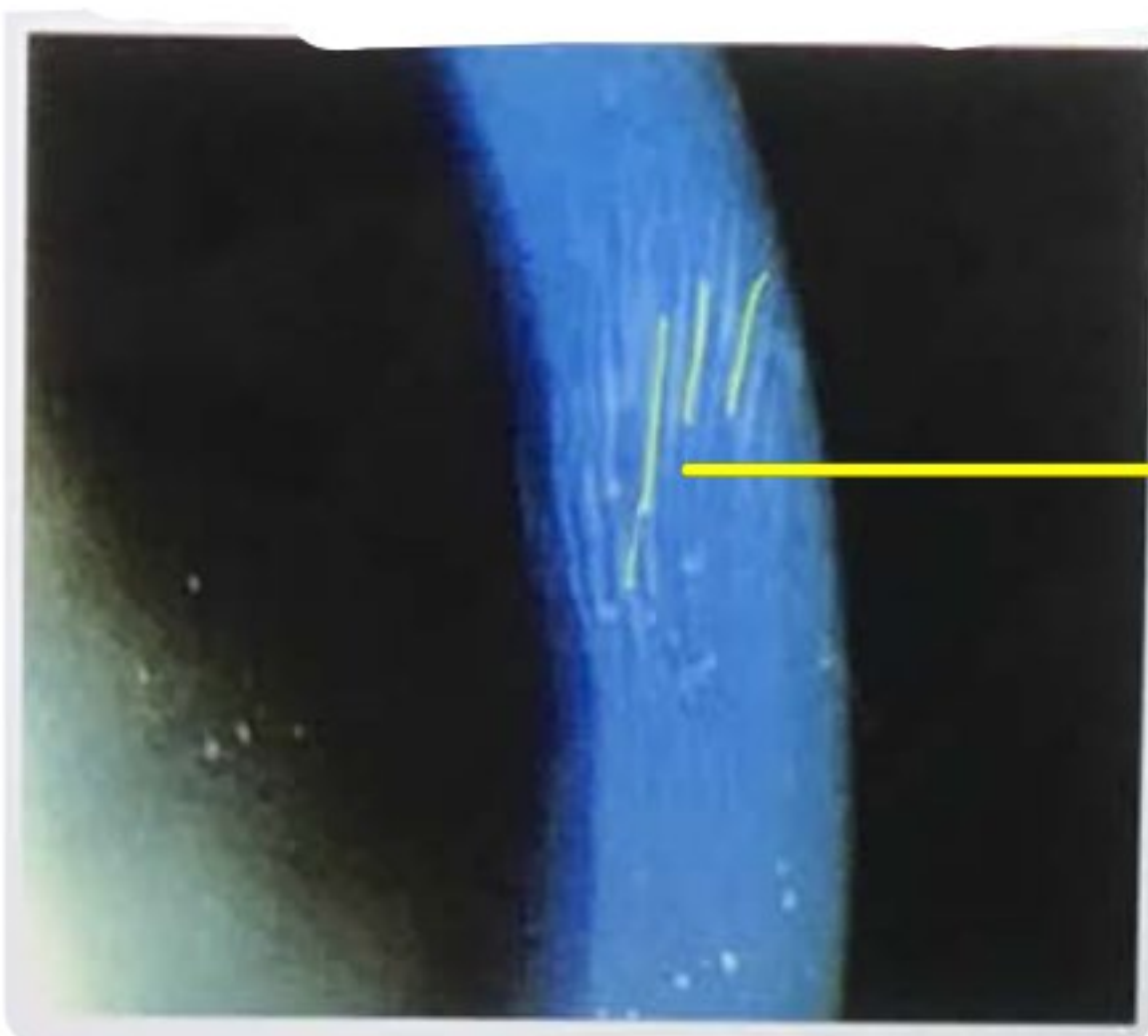
- Scissoring Reflex ← Earliest sign on Retinoscopy

"SCISSORS" MOVEMENT



vertical tears / fold of DM

Haab's striae → Congenital Glaucoma



→ **VOGT'S STRIA**

Haab's Striae-Congenital Glaucoma



Vogt's Striae-Keratoconus



A/c Hydrops → aqueous humours start entering into stroma through the vogt's stria

ACUTE HYDROPS

Apical scarring → collagen fragmented, new collagen is haphazardly arranged



APICAL SCARRING → leads to functional blindness



MUNSON'S sign : V shaped indentation of lower eyelid on looking down

RIZZUTI'S SIGN : Sharply focused beam of light near nasal limbus produces as temporal illumination of cornea



Investigation ::

Video Keratography > Corneal Topography { Test for corneal surface : { Pentacam / orb scan / Placido disc }

Mx ::

- 1, Optical Correction :
 - Cylindrical spectacles (for astigmatism)
 - Rigid gas permeable CL
- solt C/I C/I**



ICRS

Definitive Rx:

- 2, Keratoplasty for apical scarring :
 - ↳ 1, Full - thickness (penetrating)
 - ↳ 2, DALK [deep anterior lamellar K.P]

3) INTACS → Corneal ring in to the stroma.

ICRS = Intra corneal Ring segment { made up of PMMA (polymethyl methacrylate) }

4) C₃R / C₃XL : Corneal collagen cross linking → with Riboflavin

Riboflavin drops on cornea
↓
UV-A

} Photopolymerisation

- arrest progression
- But not reverse the existing d/s.

CORNEAL DYSTROPHIES:

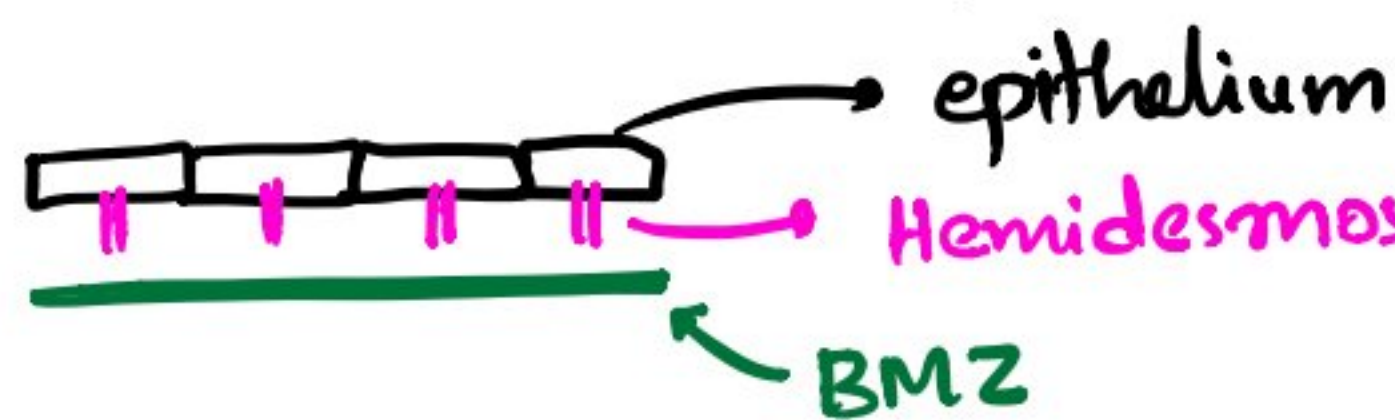
- B/L
- Inherited and a/w deposition of ab(N) materials in specific layers of cornea.

Epithelial dystrophy:

- 1) Cogan's ⇒ (MC)
- 2) Meesman's.

Cogan's dystrophy / EBMD = Epithelial Basement Membrane Dystrophy

- Hemidesmosome defect

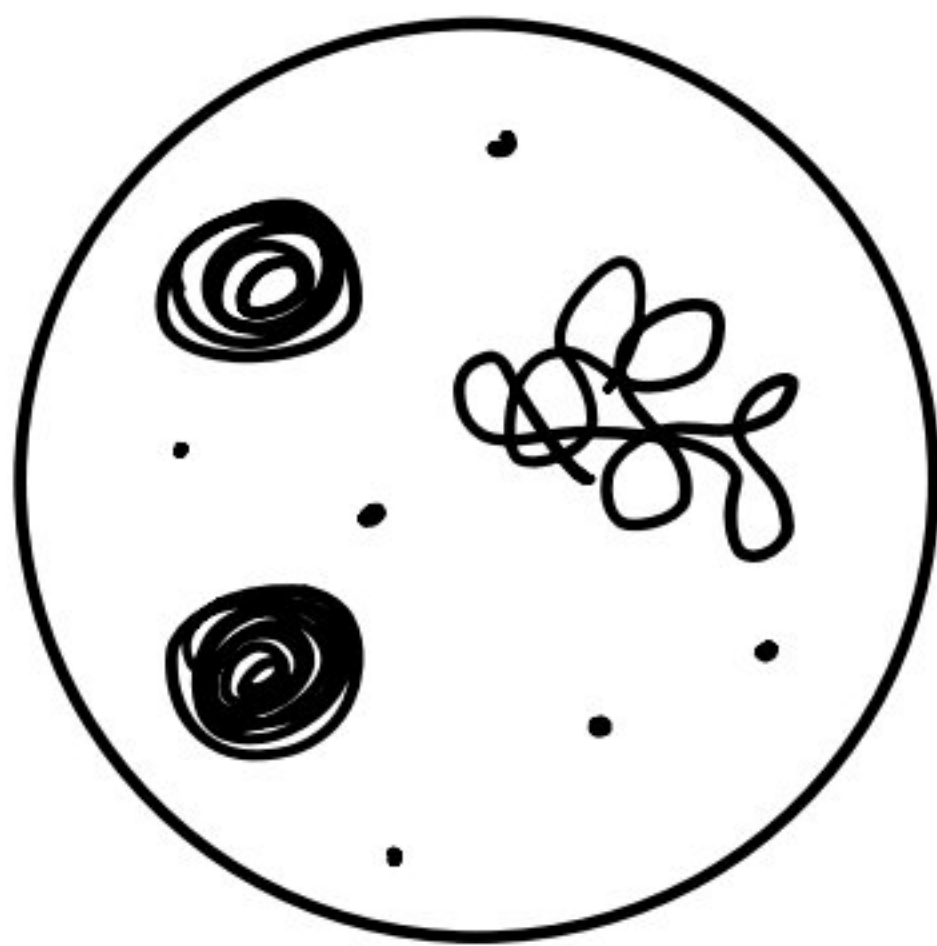


Hemidesmosome: not intact

• ↑ shed epithelial shed off

• Recurrent corneal erosion → VERY PAINFUL

Accumulated Epithelium : Fingerprint / Random map on cornea



Bowman's Dystrophy:

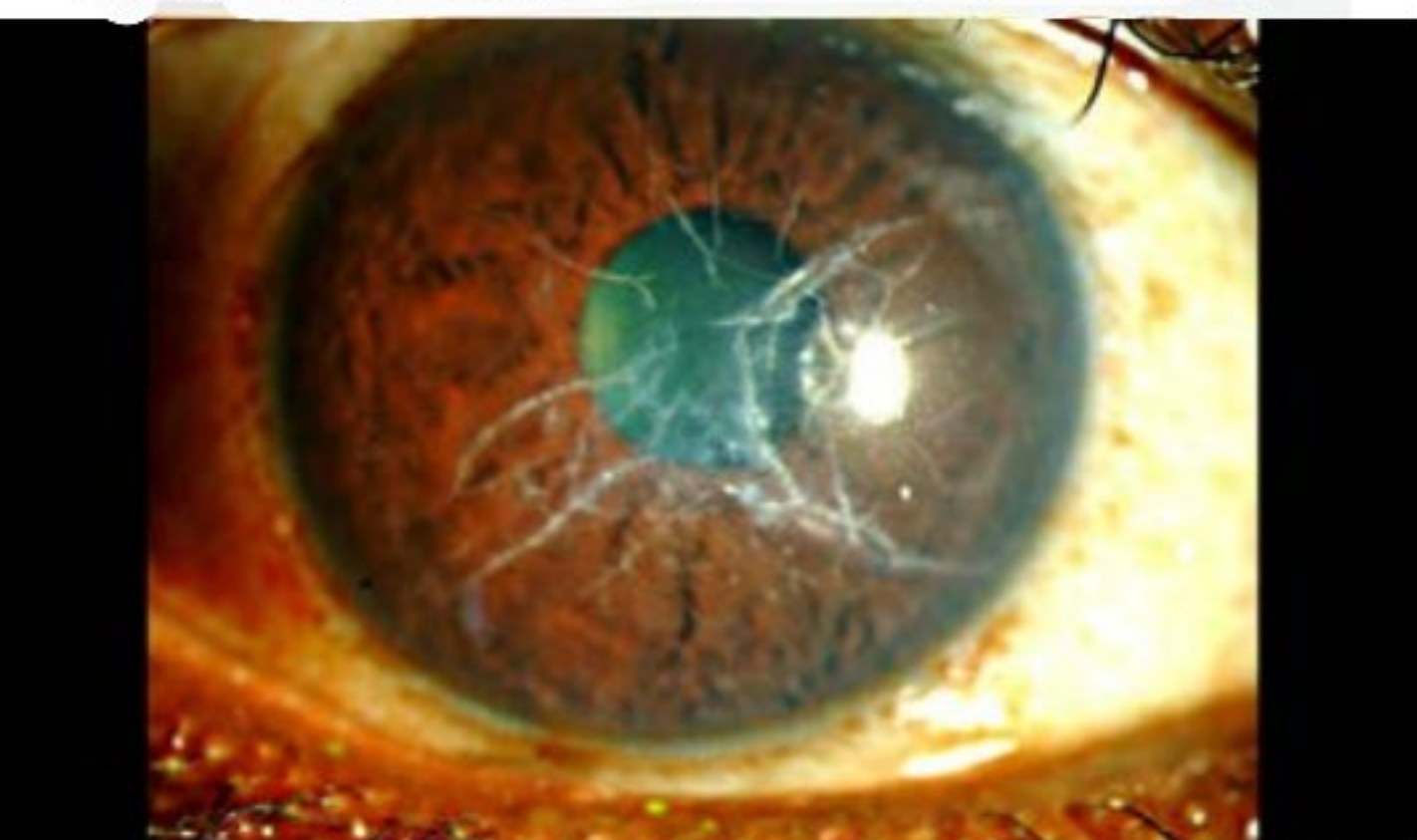
- 1) Reis - Buckler dystrophy
- 2) Thies - Behnke dystrophy

Stromal Dystrophy:

NAME	AB @ deposit	Stain
"Marilyn"	Monroa	Always
① Macular (AD)	Mucopolysaccharide	Alexan Blue
② Granular (AD)	Hyaline	Masson's Trichrome
③ Lattice (AD)	Amlyoid	Congo Red

① Least common ⇒ Macular AR
Most common ⇒ Lattice AD

LATICE CORNEAL DYSTROPHY GRANULAR - HYALINE MASSON'S TRICHROME RED



MACULAR CORNEAL DYS.



Endothelial Dystrophy:

- (MC) → Fuch's end dystrophy
- * Guttate on endothelium
 - * 'Beaten metal' appearance of endothelium

Congenital Cloudy Cornea:



"STUMPED"

- 1) Sclero cornea
- 2) Trauma (forceps delivery) → tears in DM
- 3) Uterine Infection { Syphilis }

Intestinal Keratitis

- 4) Mucopoly sac haroidosis
- 5) Peter's anomaly
- 6) Endothelial dystrophy { congenital Hereditary Endothelial dystrophy }
- 7) Dermoid

② INTERSTITIAL KERATITIS: Non suppurative inflammation of corneal stroma without primary involvement of epithelium or endothelium.

Causes:

a) Syphilis / Leutic IK { congenital >>> acquired syphilis }

b) Herpetic Keratitis { HSV (MC) }

H2V, EBV, mumps, measles etc.

- c) TB, Leprosy
- d) Sarcoidosis
- e) Trypanosomiasis
- f) Malaria
- g) Coogan's Sx.

COGAN'S SYNDROME $\Rightarrow$ C/L B/L non specific IK + vestibulo auditory dysfunction
 { Deafness + Vertigo }

- C/F of IK:
- i) Corneal Vascularization
 - ii) Salmon Patch Appearance
 - iii) Ghost vessel deformity

Rx: Steroids / Penicillin



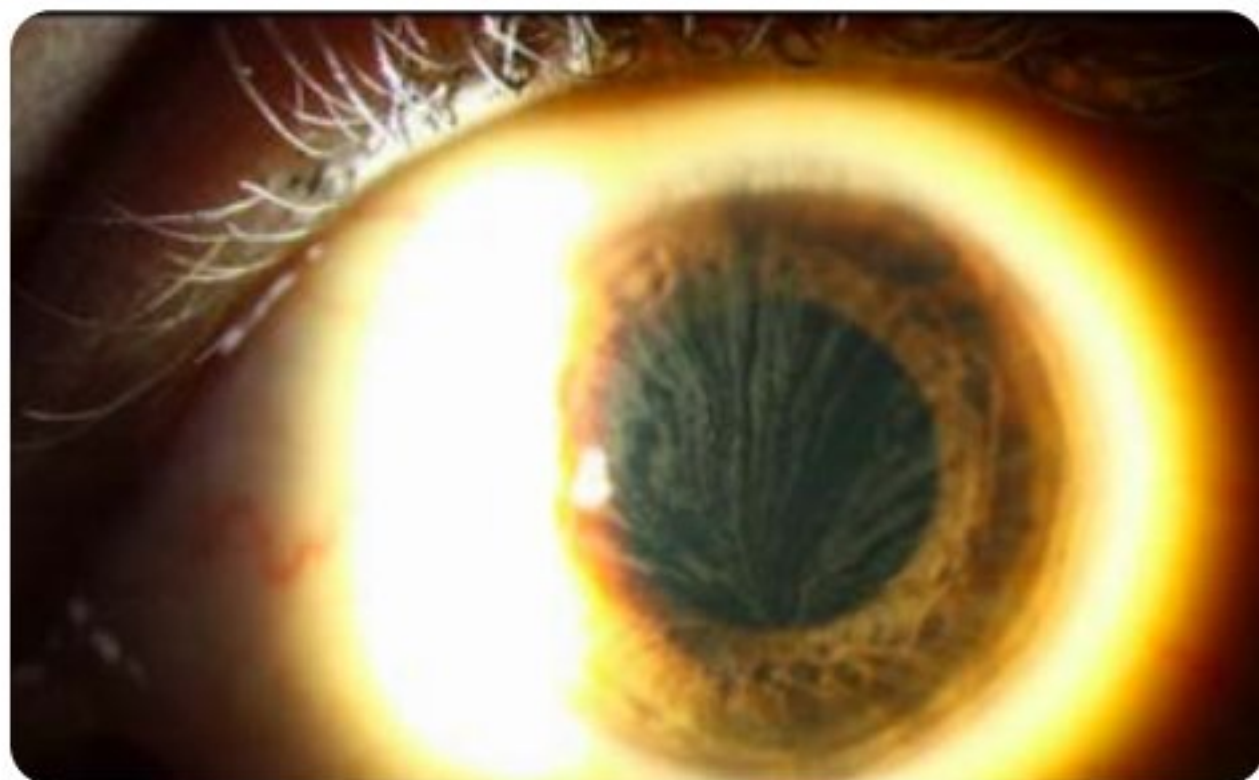
Syphilitic IK

NEET-PC '20

CORNEA VERTICILLATE / VORTEX KERATOPATHY : whorl like opacity in corneal epithelium

Causes:

- 1) Amiodarone: Reversible on stopping the drug
- 2) Tamoxifen
- 3) Indomethacin



CORNEAL TRANSPLANTATION

- National Eye donation fortnight $\Rightarrow$ 25th August to 8th September



- i) Awareness about importance of eye donation
- ii) Motivate people to pledge their eyes for donation after death

Preservation of cornea:

SHORT TERM

- 1) MK medium
- 2) Refrigerated @ 2-8°C
- 3) Cornea to be used within 48-72 hrs (max. 96 hours)

Intermediate

- i) Medium containing Chondroitin Sulphate
- ii) Refrigerated at 2-8°C for 7-14 days
- iii) Mod. Eagle's medium $\rightarrow$ extends to 30 days

Long term

- 1) Cryopreservation in Liq. N₂ / Glycerol
- 2) Upto 1yr
- 3) Disadvantage: Loss of transparency of tissue.

• MC infection after corneal transplant in India → Staph. Epidermidis
developed countries → • Pneumococcus
• Staph. aureus

q. DELLEN: localised thinning of peripheral cornea at the limbus

LABRADOR KERATOPATHY / SPHEROIDAL DEGENERATION / CLIMATIC DROPLET KERATOPLASTY

• Occurs in men whose working outside
cause: Exposure to UV rays

HISTOLOGY: Irregular proteinaceous deposits in the anterior stroma that replace the Bowman's layer.

Prev. GT
Rpt.

LIMBAL STEM CELLS:

- Stem cell marker:
- 1) ABCG2 transporter {universal stem cell marker}
 - 2) Nuclear protein: p63
 - 3) Cell membrane / transmembrane protein :-
 - i) Integrin
 - ii) EGF receptor
 - iii) Transferrin receptor
 - iv) ABCG2 transporter
 - 4) Cytoplasmic proteins :-
 - i) Cytokeratin 19
 - ii, α Enolase.

CONGENITAL HEREDITARY ENDOTHELIAL DYSTROPHY {CHED}

Clinical//

- B/L Congenital Cloudy cornea
- Corneal thickness (x3)
- (N) IOP
- (N) Corneal diameter
- a/w →
 - i) Nystagmus
 - ii) Corneal Plannus
 - iii) Esotropia (inward strabismus)

A 6 month old baby is brought with w/o of B/L hazy cornea since birth. Corneas are thick, but the diameters are (N). Facies was (N). There is associated Nystagmus. Which one of the following regarding this condition is True.

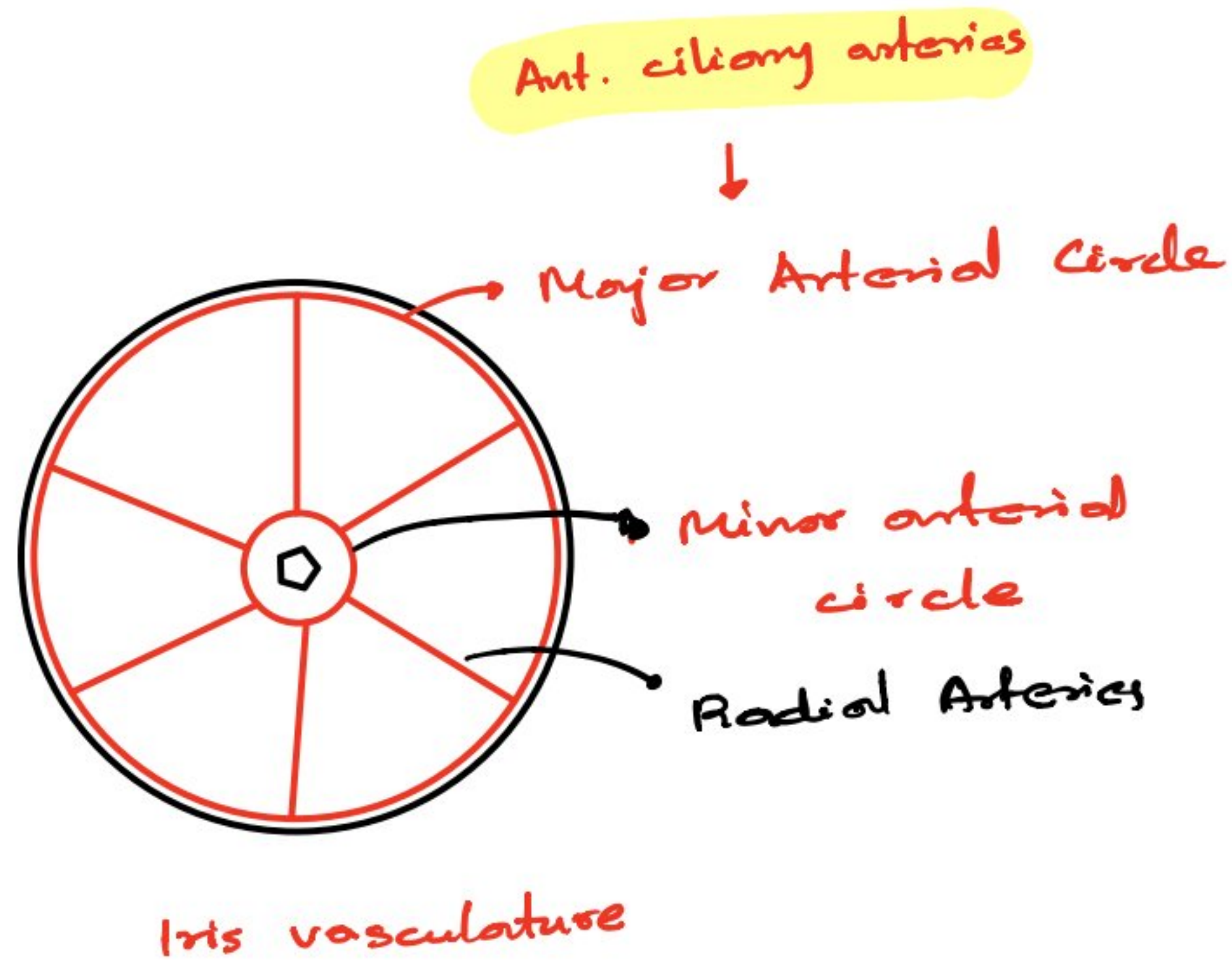
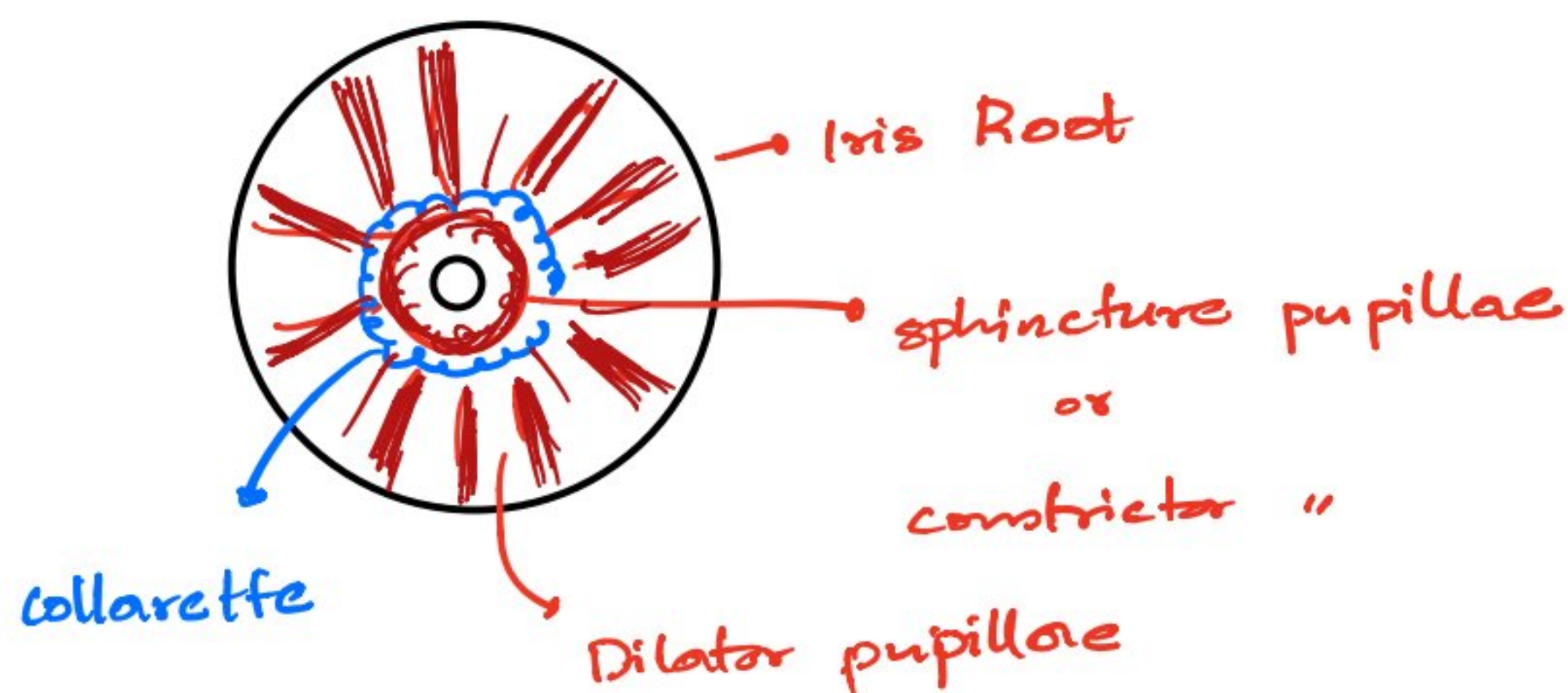
The Descemet's membrane is ab(N) thickened d/t to Ab(N) endothelium

UVEA:. is the vascular middle layer of the eye. It is divided, from front to back

The vocal tract comprises of:

- i) Iris
- ii) Ciliary Body $\begin{cases} \rightarrow \text{Pars Plicata} \\ \rightarrow \text{Pars Plana} \end{cases}$
- iii) Choroid

IRIS:



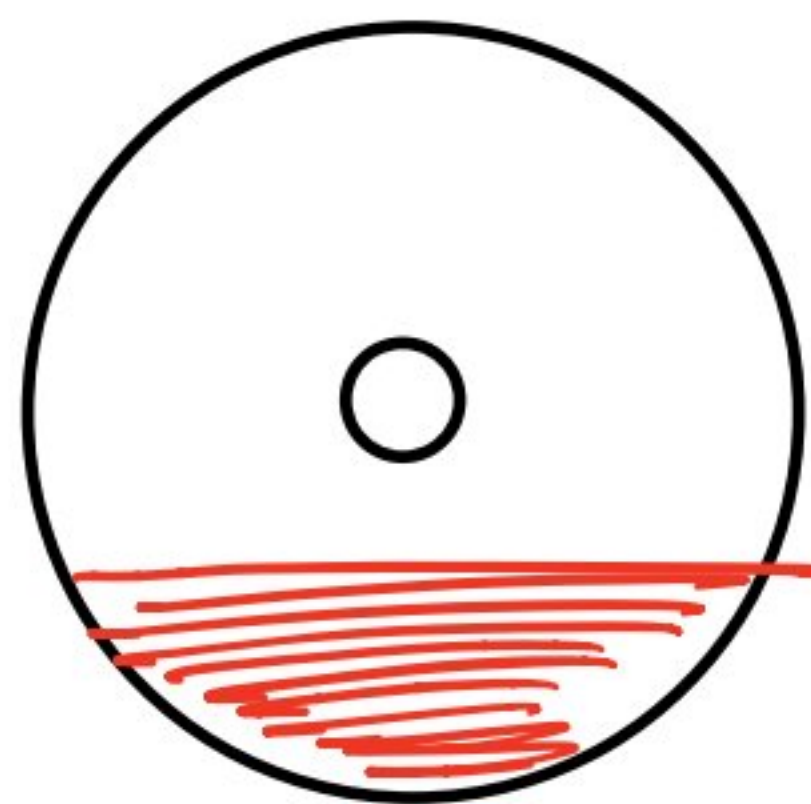
Anterior most part ∴

- a) Central an aperture of 3-4mm called pupil
- b) Iris is thinnest at it root
- c) Entire frill of iris tissue $\rightarrow$ collarette (represents attachment of pupillary membrane)

LAYERS

- i) Anterior Limiting layer: has melanocytes definitive colour of iris depends on this layer
- ii) Stroma (contains vessels, nerves, sphincter pupillae, dilator pupillae muscle)
- iii) Pigmented layer
- iv) Non pigmented layer

Hyphema: Blood in A.C
Source: Major Arterial circle in iris root.



Ciliary Body : Ciliary muscle ← parasympathetic innervation via short ciliary Nerve

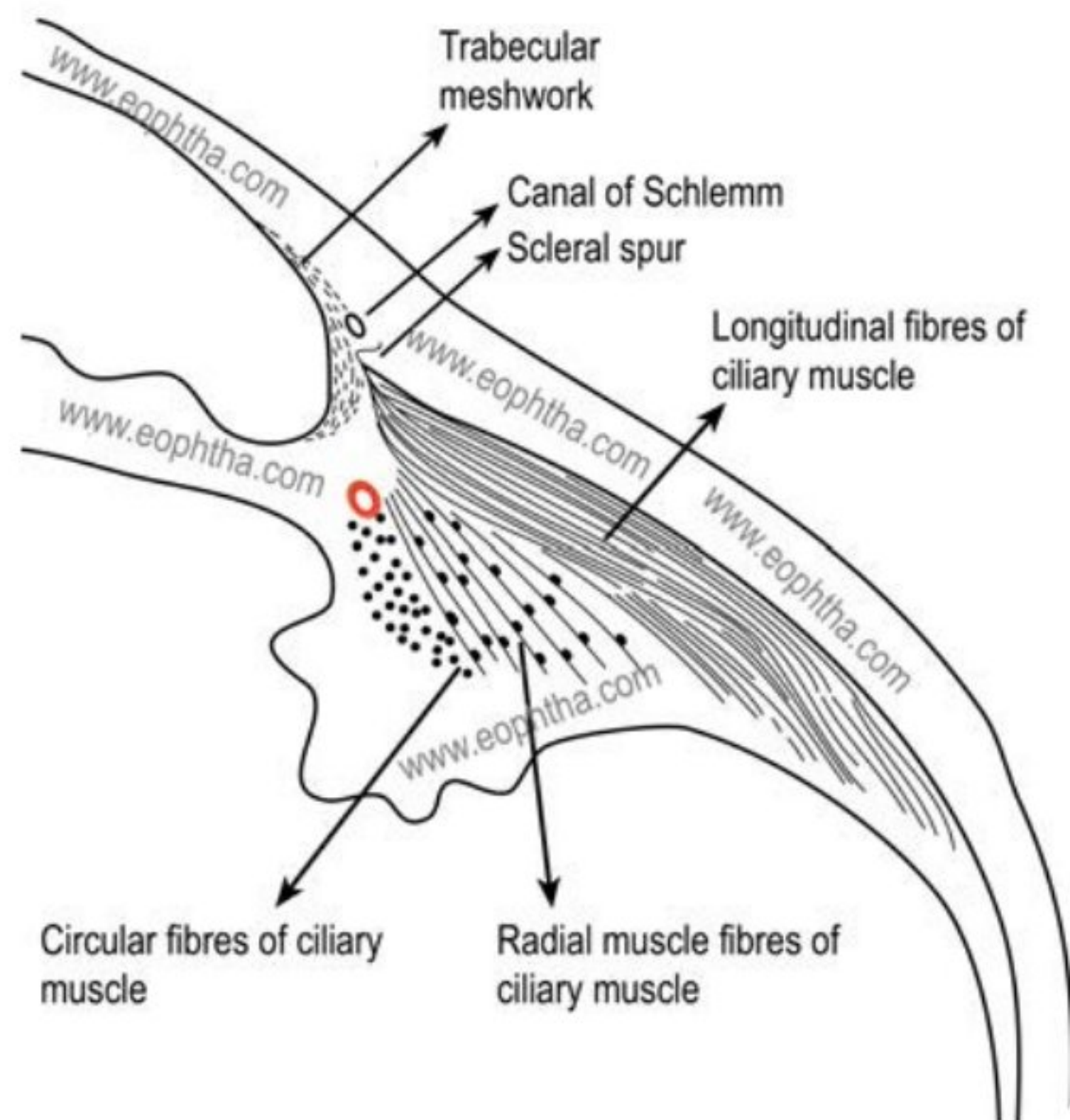
outer → Longitudinal Fibres ↓

middle ↓ Radial / Oblique Fibre

circular ↓ fibres ↓ Accomodation

Trabecular
aqueous outflow

↓
uveal scleral aqueous outflow



Q: B.S of ciliary body:

Long posterior
ciliary artery

Anterior
ciliary artery

Choroid : Highly vascular

Layers - inside out

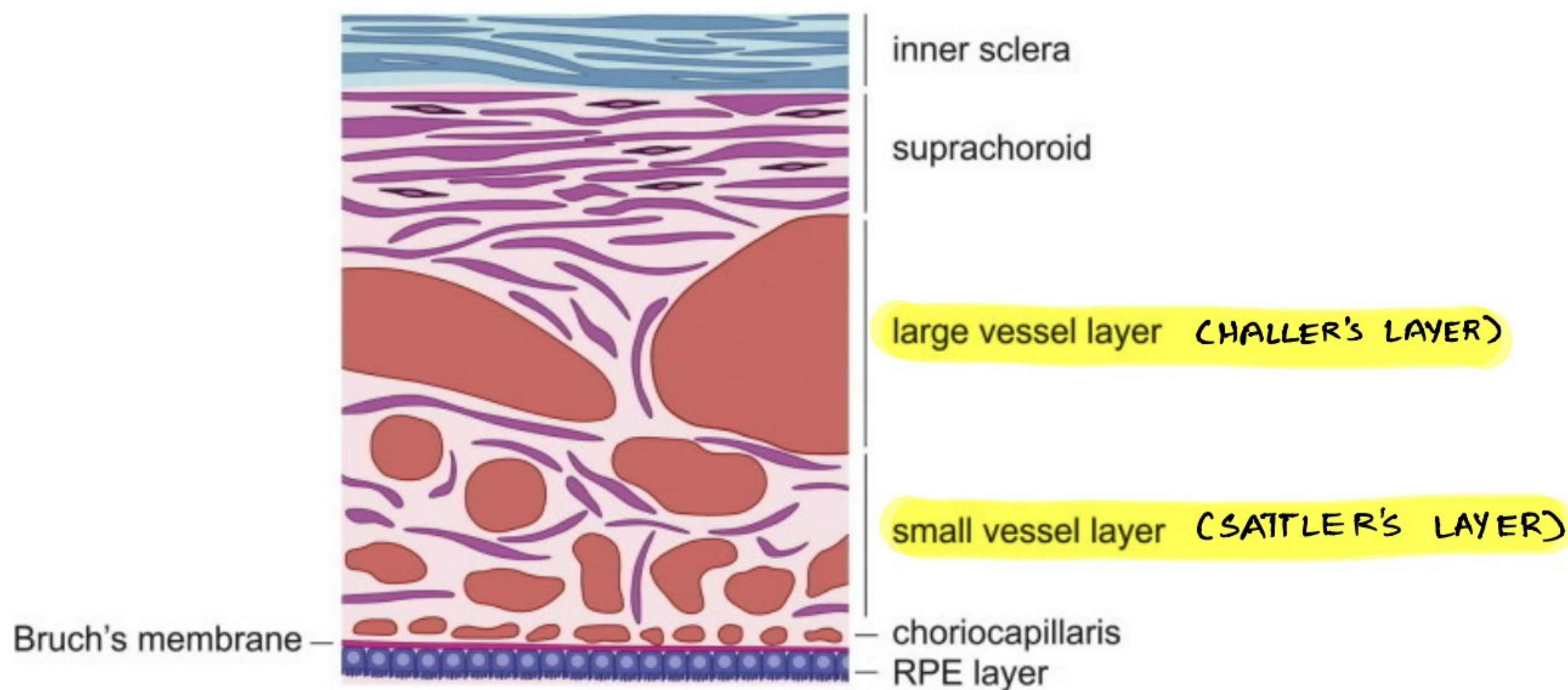
→ BRUCH'S MEMBRANE
CHORIOCAPILLARIES

MEDIUM SIZE VESSELS → SATTLER'S LAYER

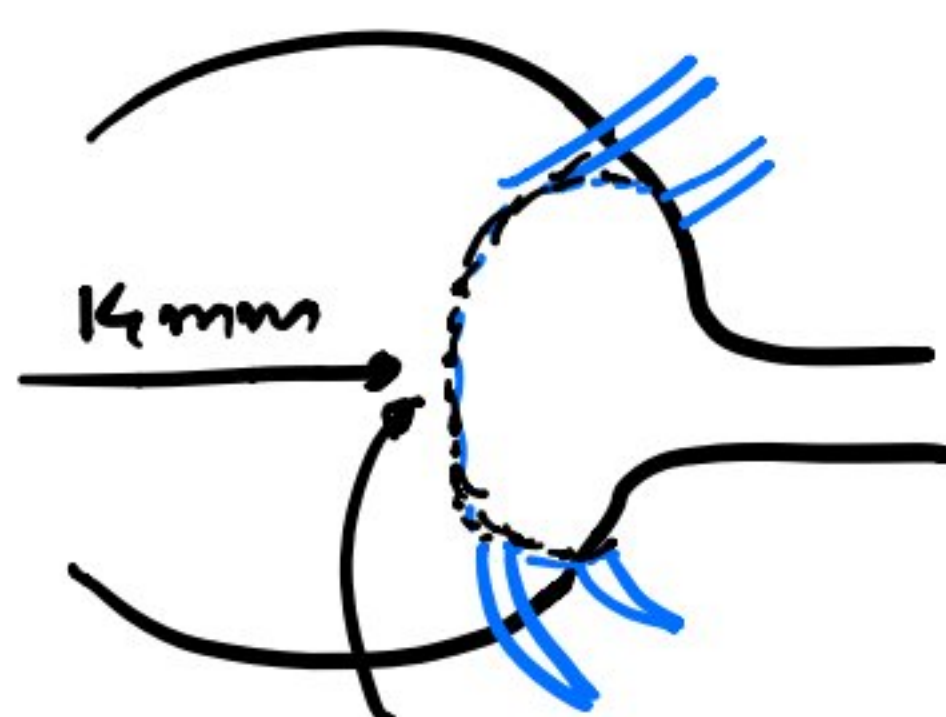
LARGE SIZE VESSELS → HALLER'S LAYER

SUPRACHOROIDAL SPACE

BRUCH'S MEMBRANE {part of choroid} = involves basement membrane of Retinal Pigment Epithelium (RPE)

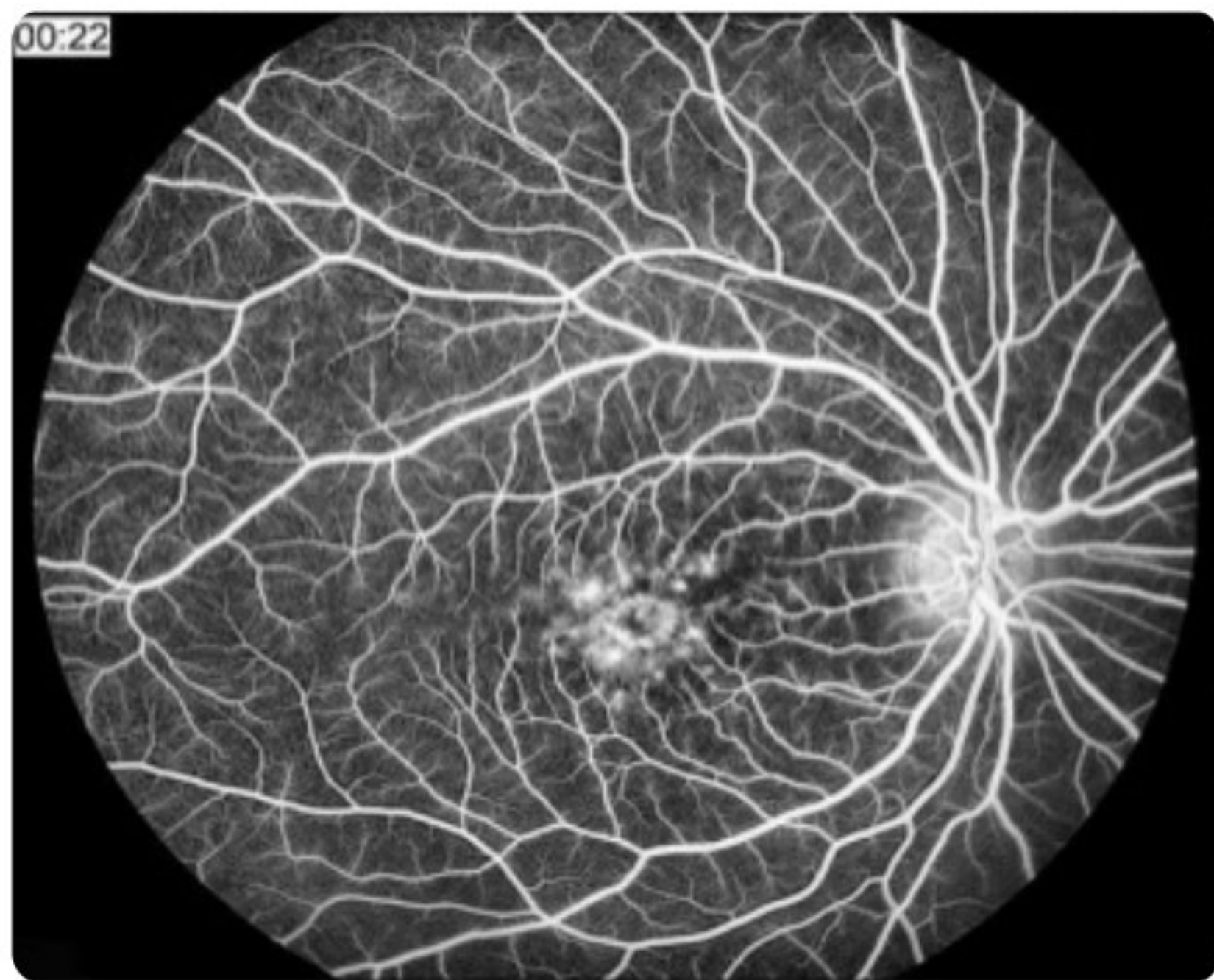


Choroidal veins : VORTEX VEIN → 4 in no.



equator of eyeball : 14mm behind the limbus.

Visualise : Choroidal vasculature $\Rightarrow$ INDOCYANINE Green Angiography



Embryology of Uvea :

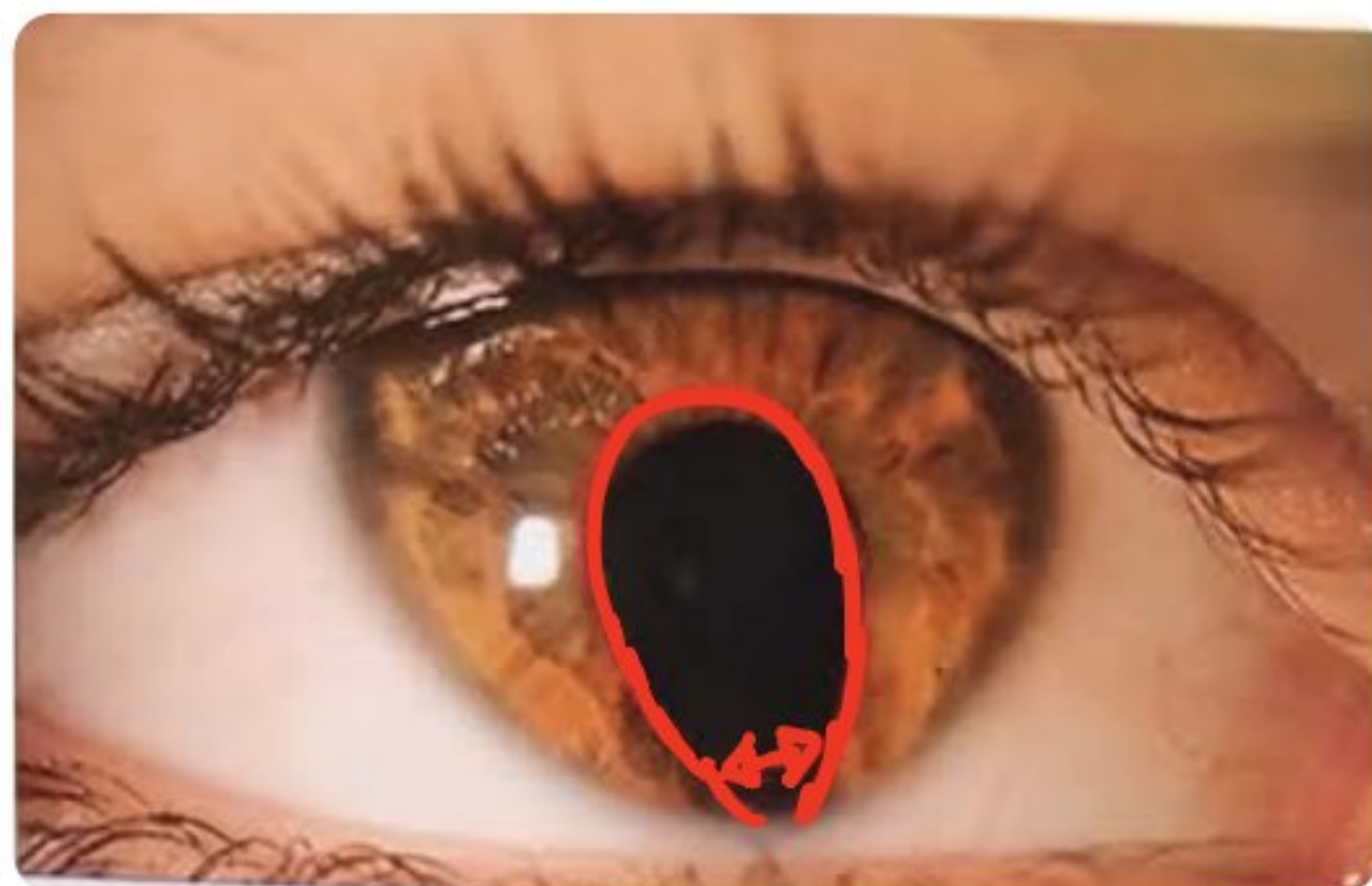
- 1) Choroid $\rightarrow$ Mesoderm
- 2) Iris muscle $\rightarrow$ Neuroectoderm
 {Sphincter & dilator}
- 3) Ciliary muscle $\rightarrow$ Neural Crest Cell

IRIS :

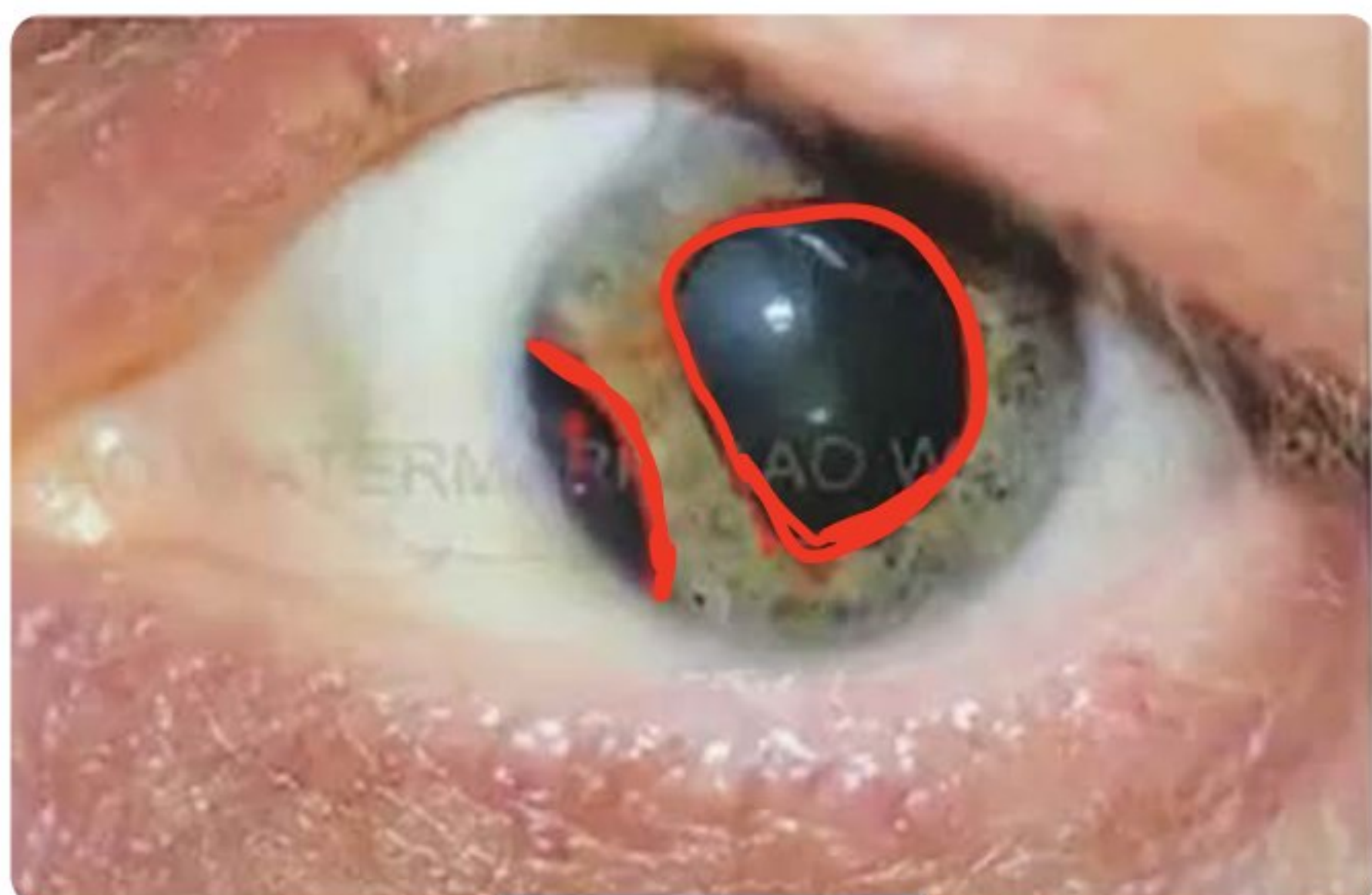
Congenital Aniridia :

- a/w Ectopia Lentis {displacement of lens}
- a/w PAX6 gene mutation

Iris Coloboma : Pear shaped pupil

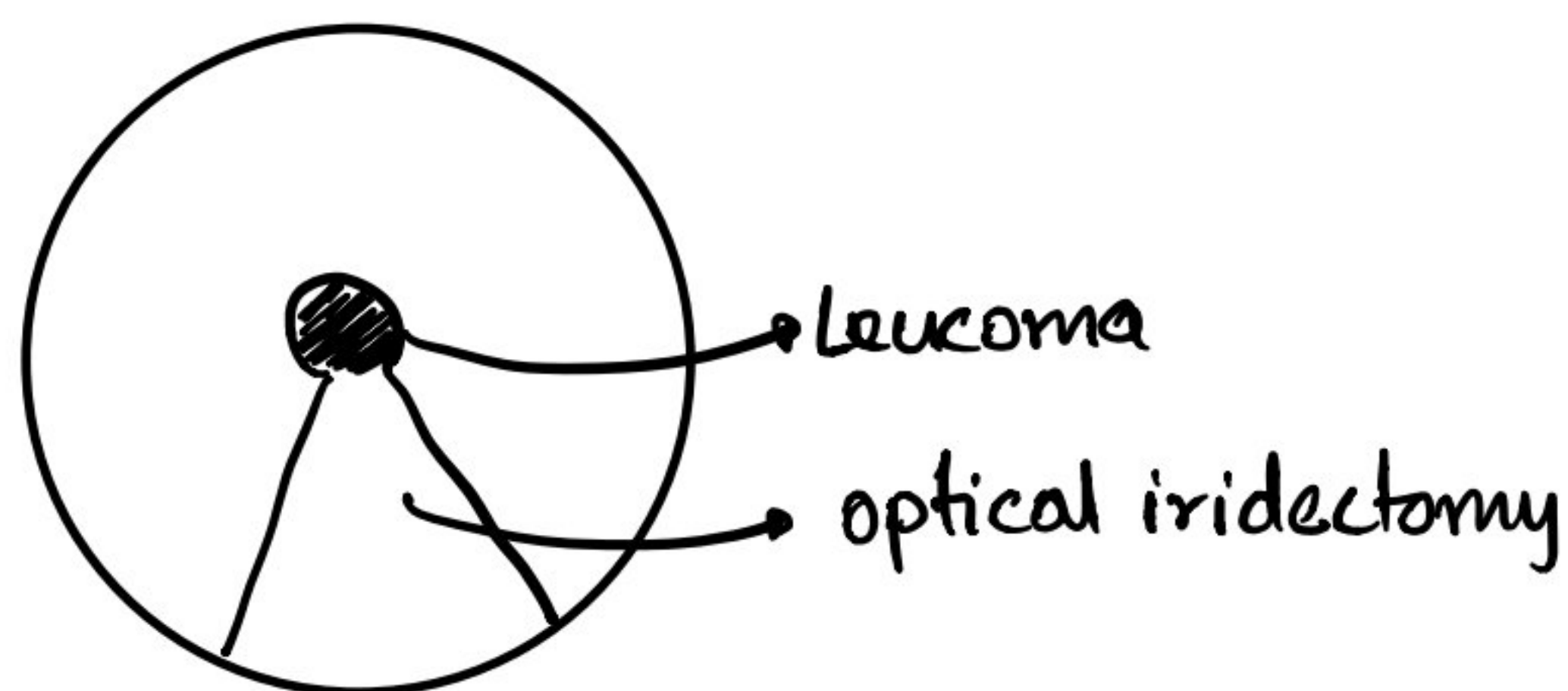


D-Shaped Pupil : d/t IRIDODIALYSIS : separation of iris root from CB attachment
d/t Blunt trauma



Iridodialysis $\rightarrow$ D shaped pupil

Key hole Pupil : Inferior sector iridectomy



Rubeosis Iridis : Neovascularisation of Iris

- d/t VEGF by hypoxia Retina
- 1st NVI @ pupillary border of iris

causes

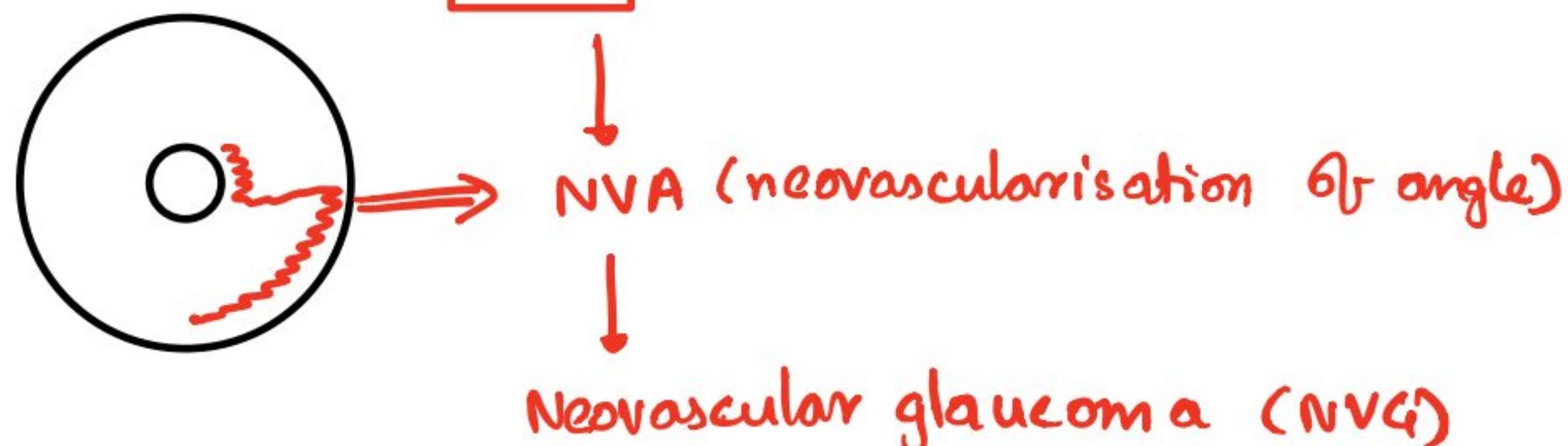
- 1) Proliferative Diabetic Retinopathy → PDR = (MC)
- 2) Ischemic CRVO & central Retinal vein occlusion?
- 3) Retinopathy of prematurity
- 4) Sickle cell retinopathy
- 5) Retinoblastoma.



Rubeosis Iridis

- Rx: 1) Treat the underlying condition
2) Pan - Retinal photocoagulation (PRP)

Complication of NVI



Heterochromia Iris

Hypochromia Iris

- 1) Fuch's Heterochromic Iridocyclitis
- 2) Congenital Horner's Sx

Hyperchromia Iris

- 1) Siderosis
- 2) Rubeosis Iridis
- 3) Prostaglandin Analogue
- 4) Iris Melanoma

Iris Nodule:

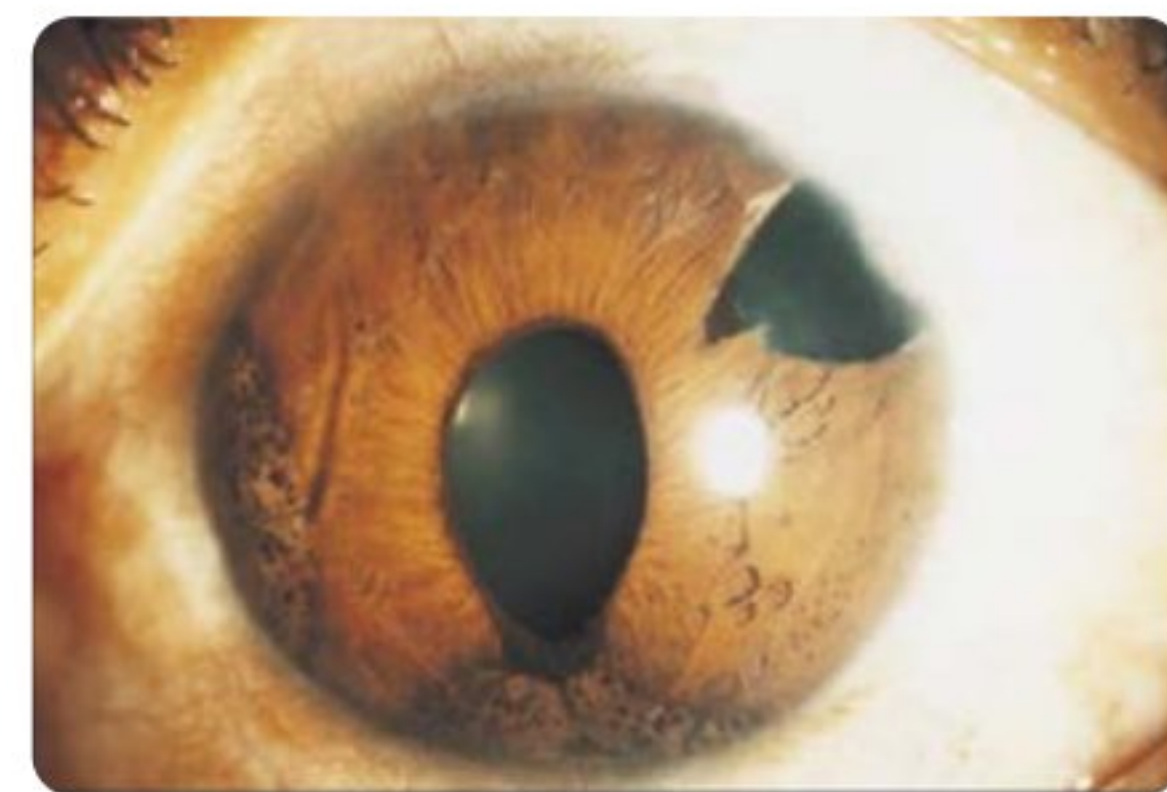
Non inflammatory

- 1) Brushfield spots : Down's syndrome
- 2) Lisch Nodules → NF1
= Iris Hamartoma
- 3) Orangish Iris nodule → Spontaneous Hyphema
: JUVENILE XANTHOGRANULOMA

NEET-PG

- 4) ICE Syndrome { Irido - corneal endothelial syndrome }

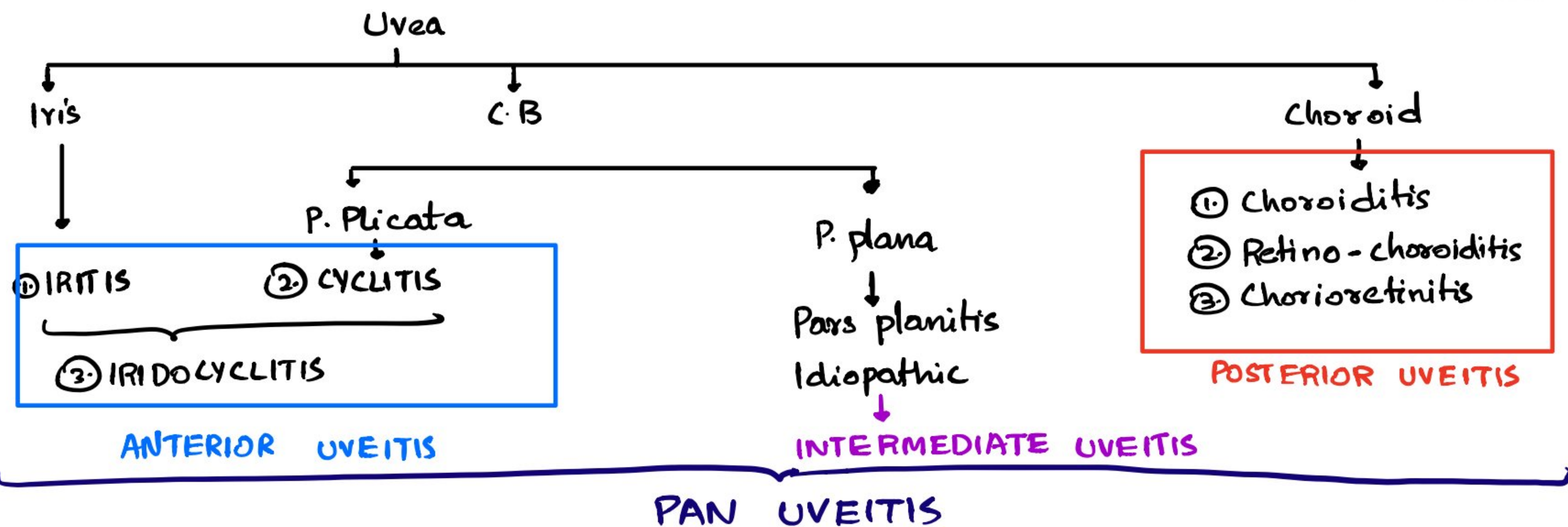
- i) Iris Nevus [Cogan Reese syndrome]
- ii) Chandler's syndrome → Corneal edema + Iris Nodule
- iii) Essential Iris Atrophy
 - ↳ Correctopia: Eccentric position of pupil
 - ↳ Polyopia: Many pupils



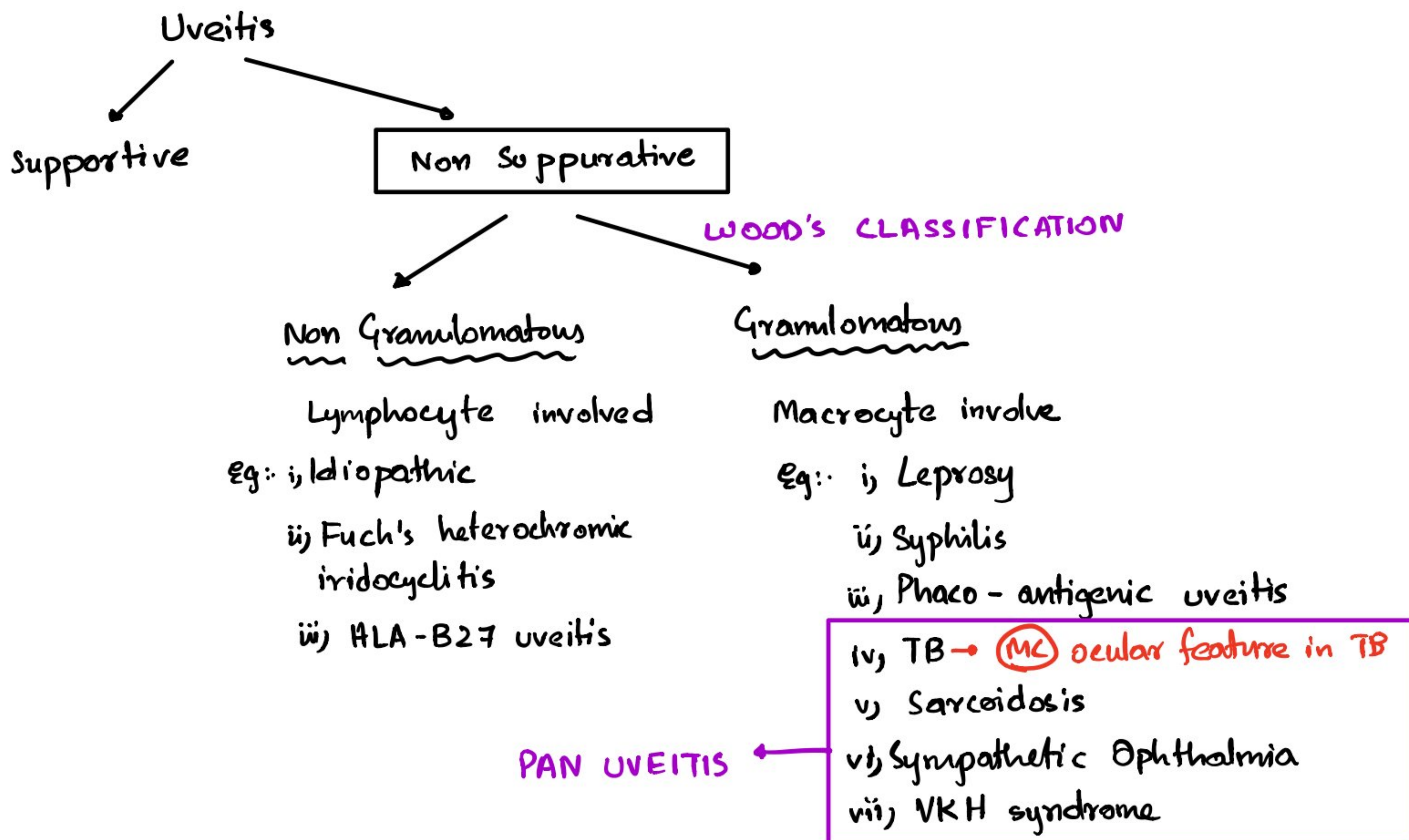
ICE Syndrome

UVEITIS: Inflammation of Uvea

Anatomical classification:



Pathological Classification

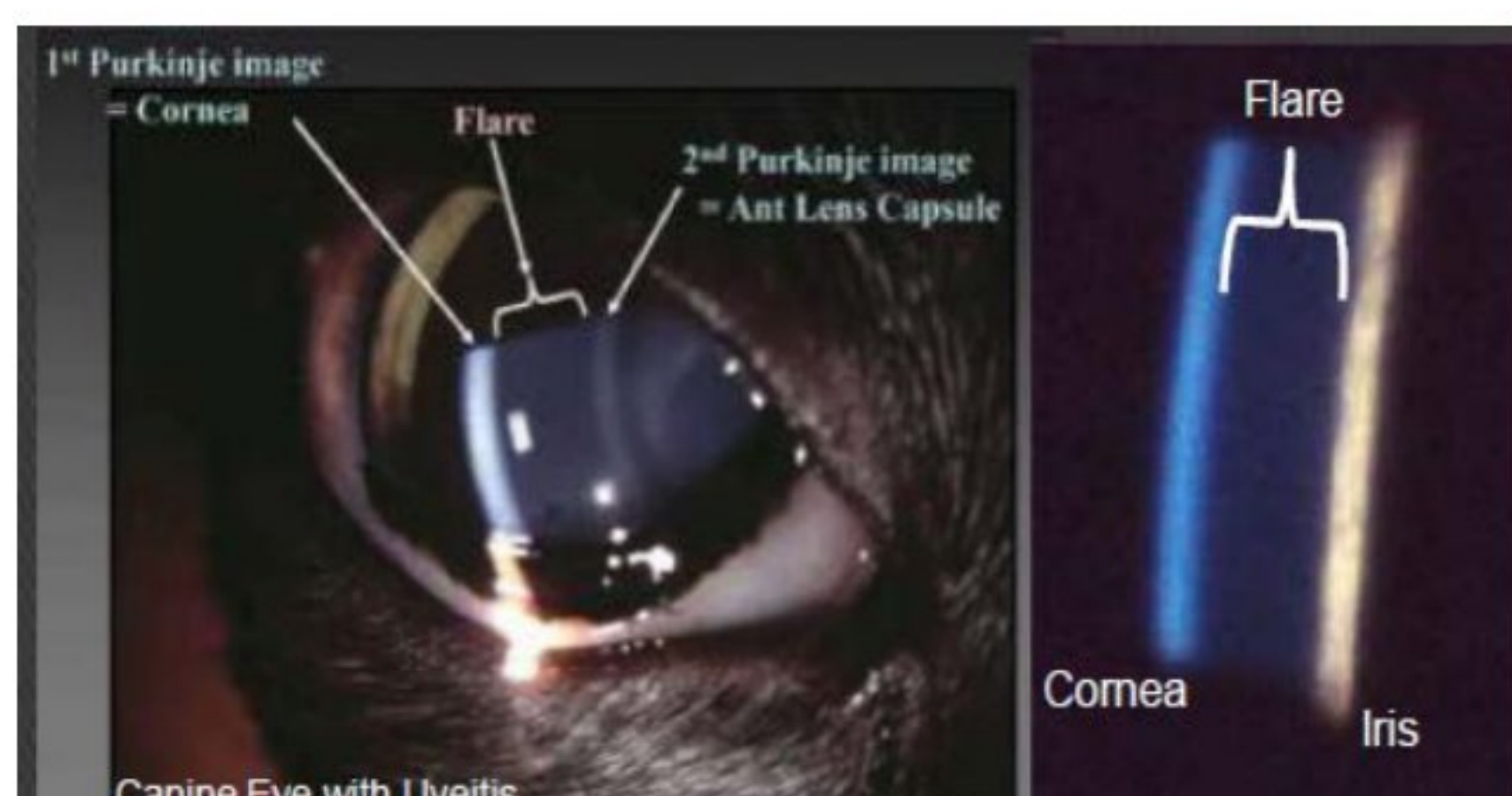


Anterior Uveitis / Iridocyclitis

- Redness of circum corneal
circum ciliary congestion.
 - Spasm of
 - i, Ciliary Muscle spasm → Pain
 - ii, Sphincture muscle " → Miosis
 - Edema - Iris
 - featureless iris
 - Muddy iris
 - Breakdown of Blood aqueous barrier ⇒ leakage of proteins in aqueous
 - ↓
 - Turbid aqueous (↓ se in Vn)
 - FLARE
- RED EYE WITH PAINFUL DECREASE IN VISION**

Tyndall effect ⇒ light passes through proteinaceous aqueous → smoky sensation

1st sign I.C → **Aqueous flare**

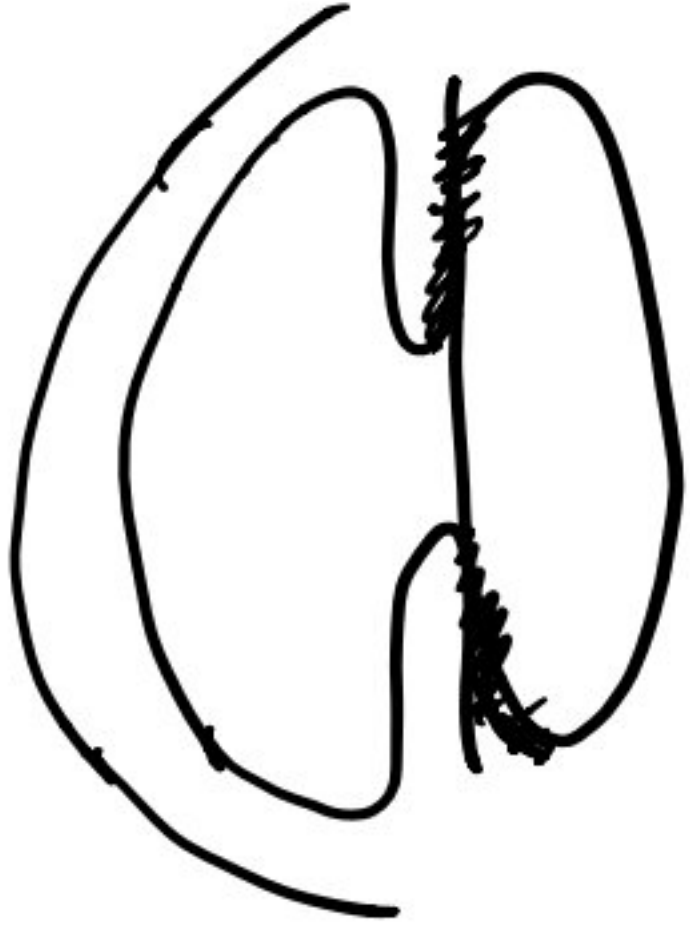


- Iris Adhesion $\Rightarrow$ **SYNECHIAE**
 - ① attached to cornea
 - ② attached to lens

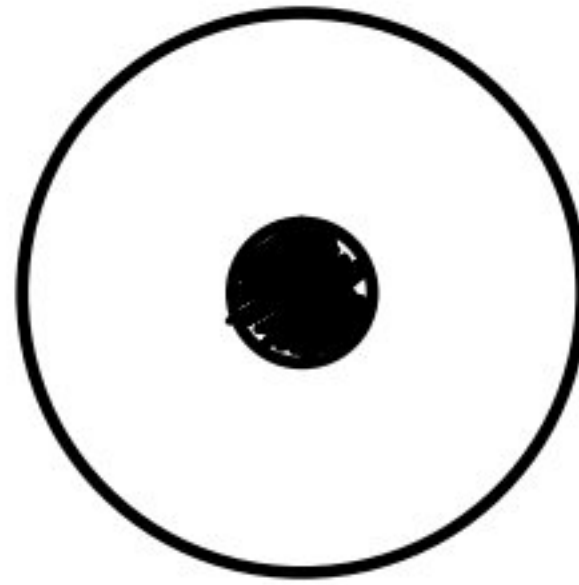
Posterior Synechiae

Types:

1) **Total P.S**



- Entire posterior surface of iris is stuck to lens
- Pupil occluded by exudates $\rightarrow$ **Oclutio pupillae**

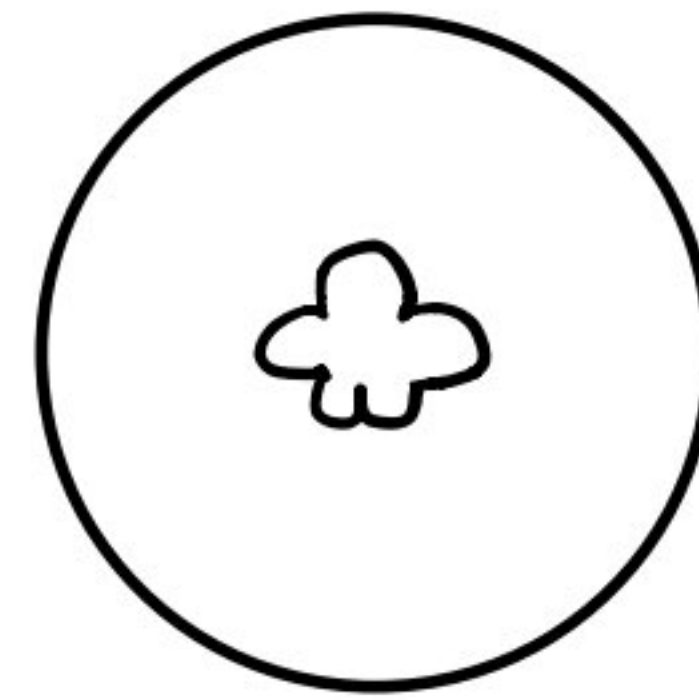


2) **SEGMENTAL P.S**

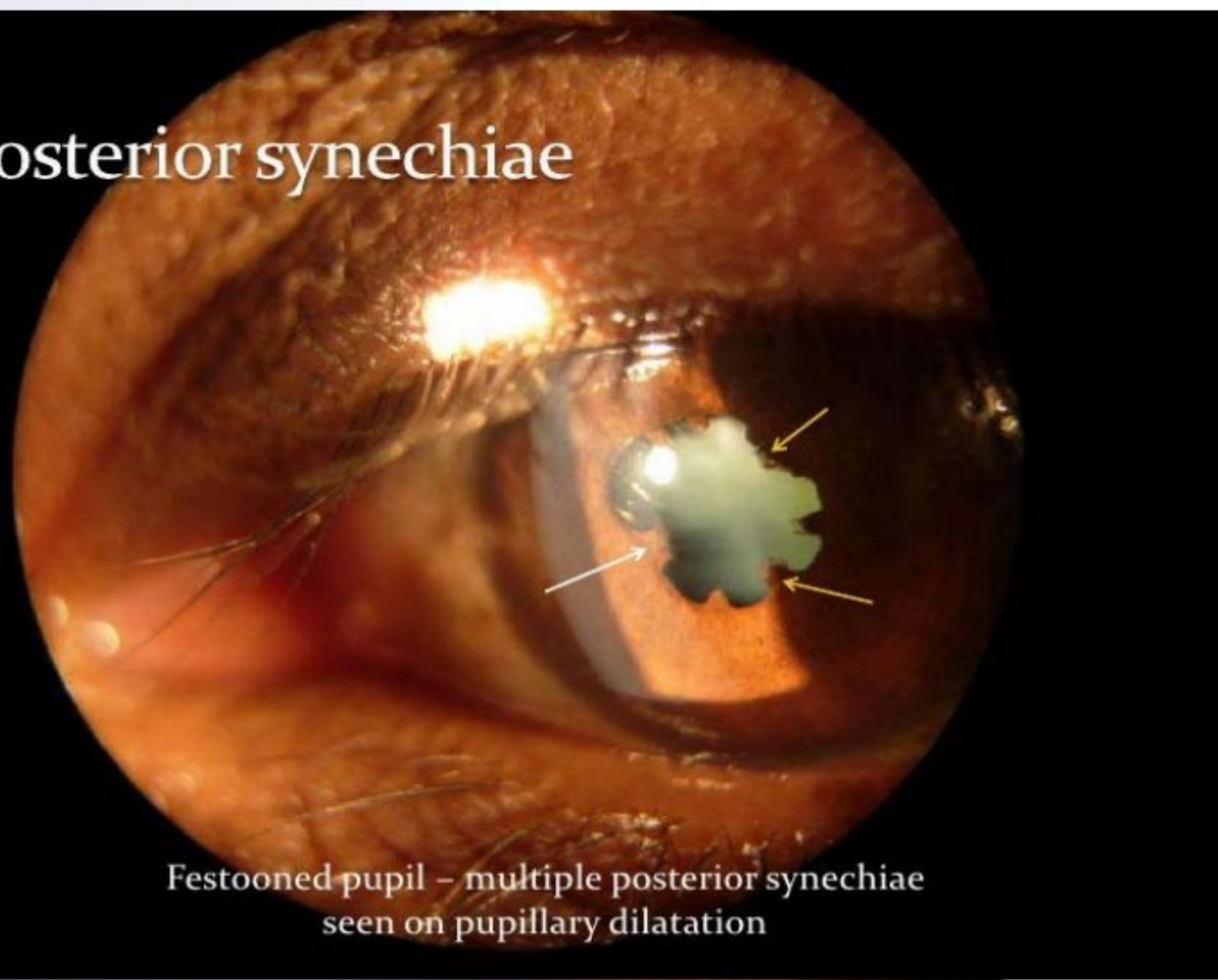
: Irregular pupil
or

Segment of iris is stuck on to the lens

Festooned pupil



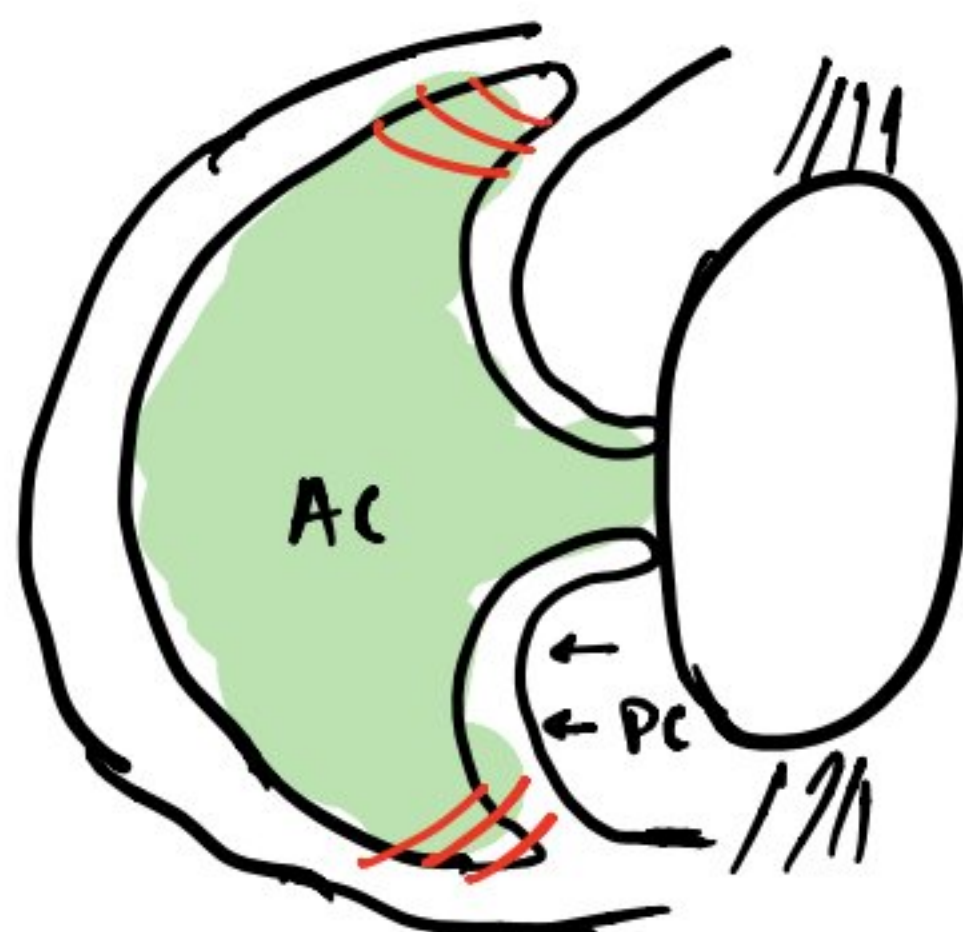
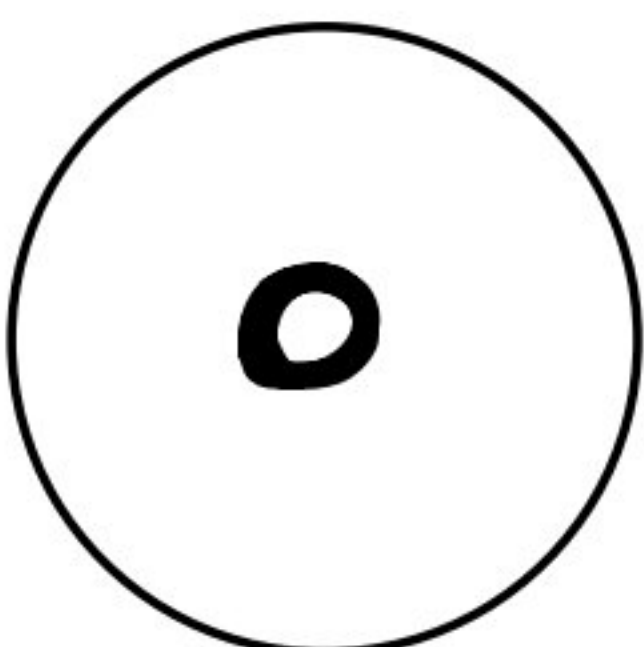
Posterior synechiae



Festooned pupil - multiple posterior synechiae seen on pupillary dilatation

Notespaedia.com
More than just notes...

3) **360° P.S** : Entire pupillary margin is stuck $\Rightarrow$ Ring / Annular synechiae



SECCCLUSIO PUPILLAE
{ P.C. is walled off and cannot communicate with A.C. }

P.C. pressure $\uparrow$ ses

2° Angle closure glaucoma

Peripheral Anterior Synechia

- Forward bowing of Iris "Iris bombe"
- Anterior chamber funnel shaped or irregular

1st sign of Active Iridocyclitis

Aqueous cells : They are inflammatory cells present in A.C / Aqueous flare



Aqueous cells

↓
Block T.M

↓
2° OA Glaucoma

Keratic precipitate : Inflammatory cells stick on endothelium
→ Pathognomonic sign of A/C anterior uveitis

2 types :-

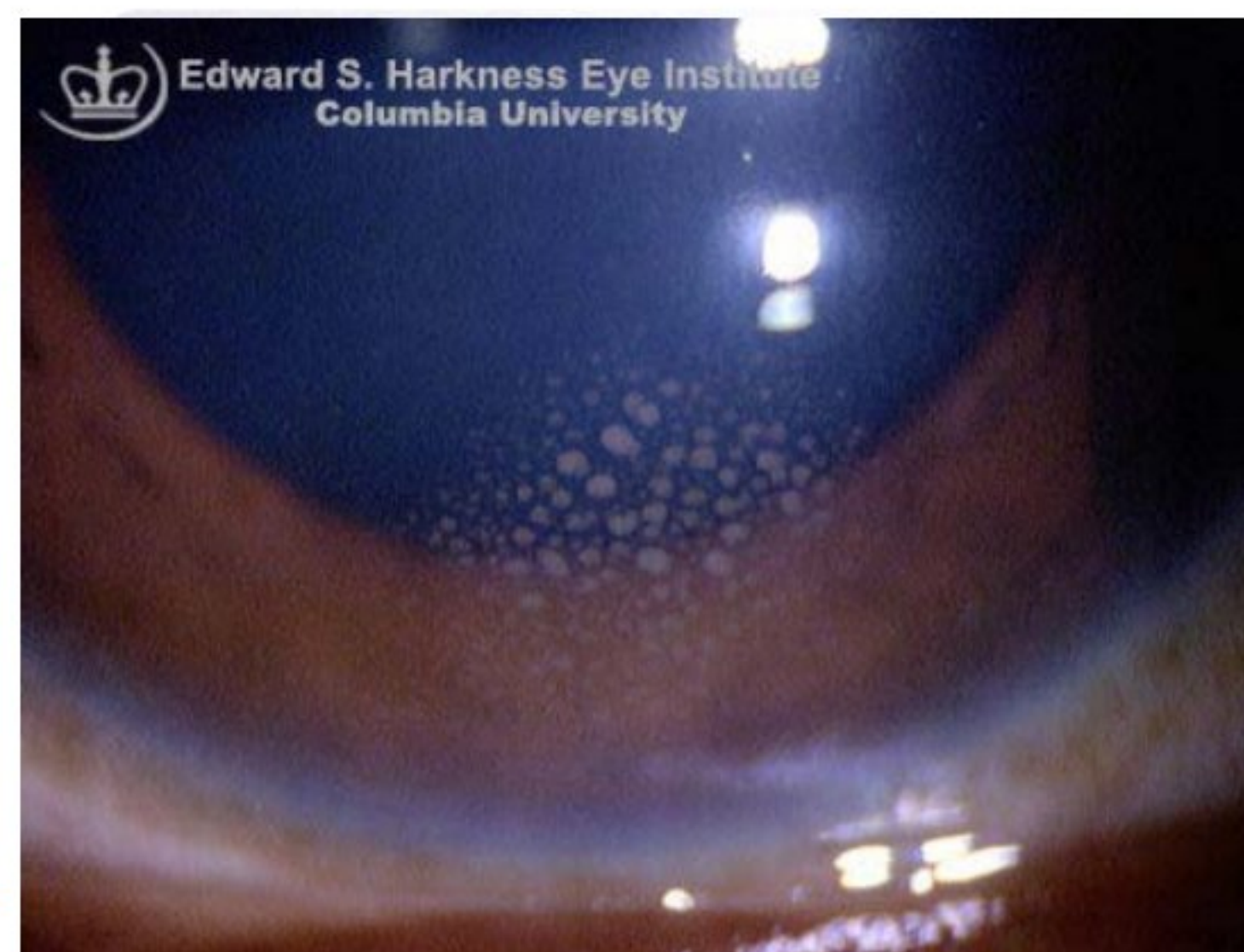
FINE KP

- small sized
- Lymphocytes
- Non granulomatous uveitis

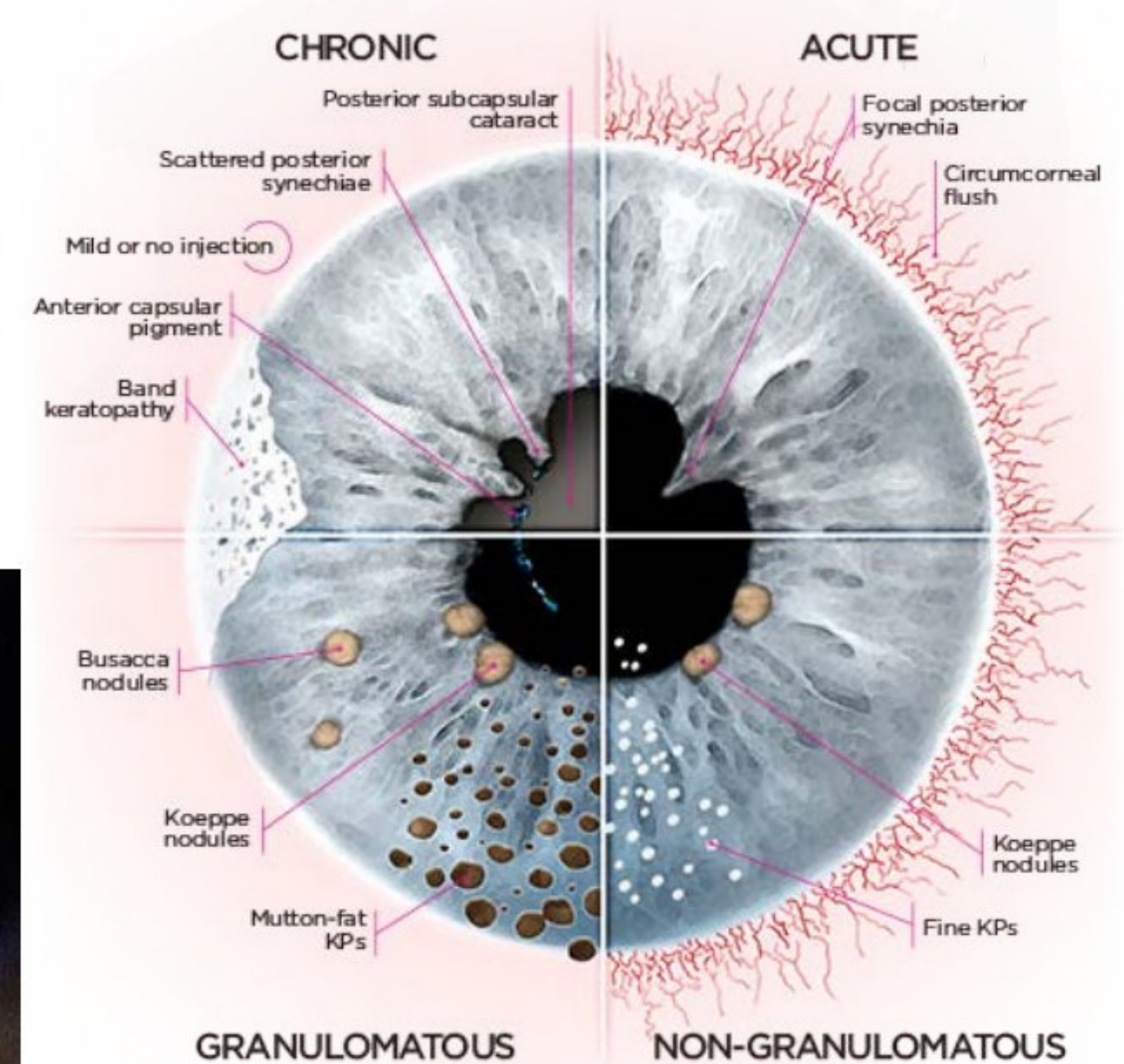
Mutton Fat K.Ps

- large sized
- Macrophages
- Granulomatous uveitis

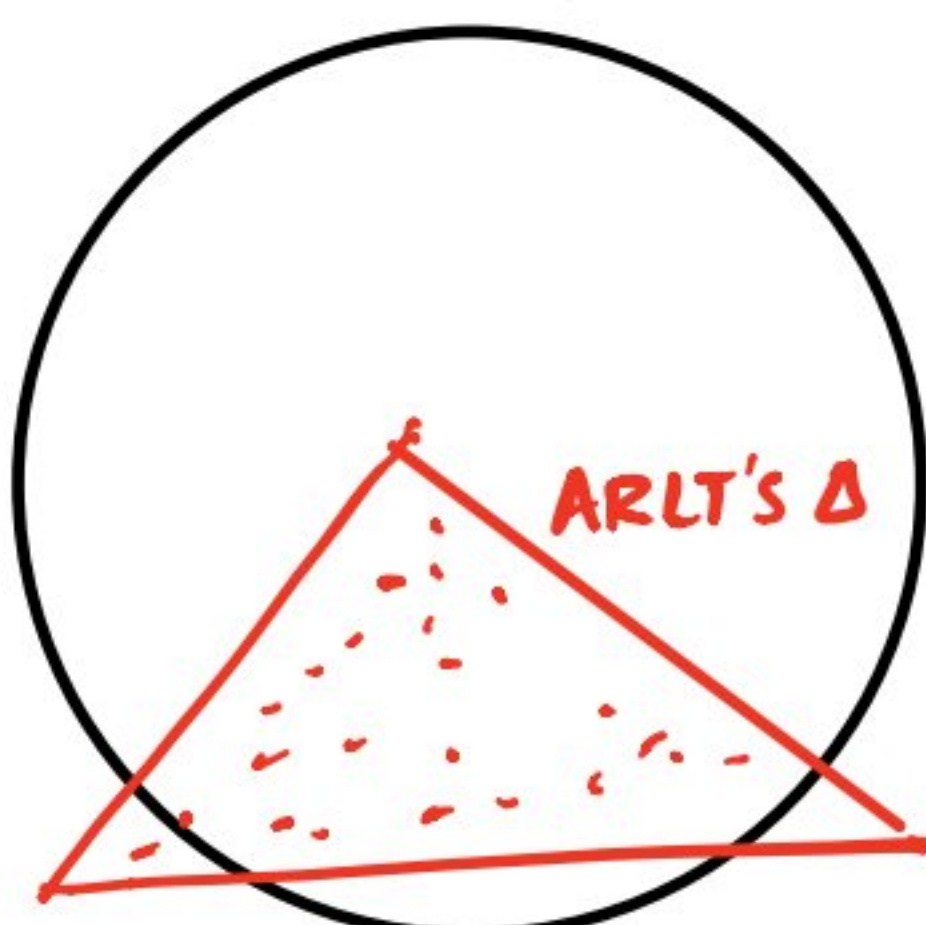
ARLT'S TRIANGLE



Clinical Signs of Anterior Uveitis



ARLT'S Δ : It is the distribution of K.P in the endothelium whose base is down and Apex is up



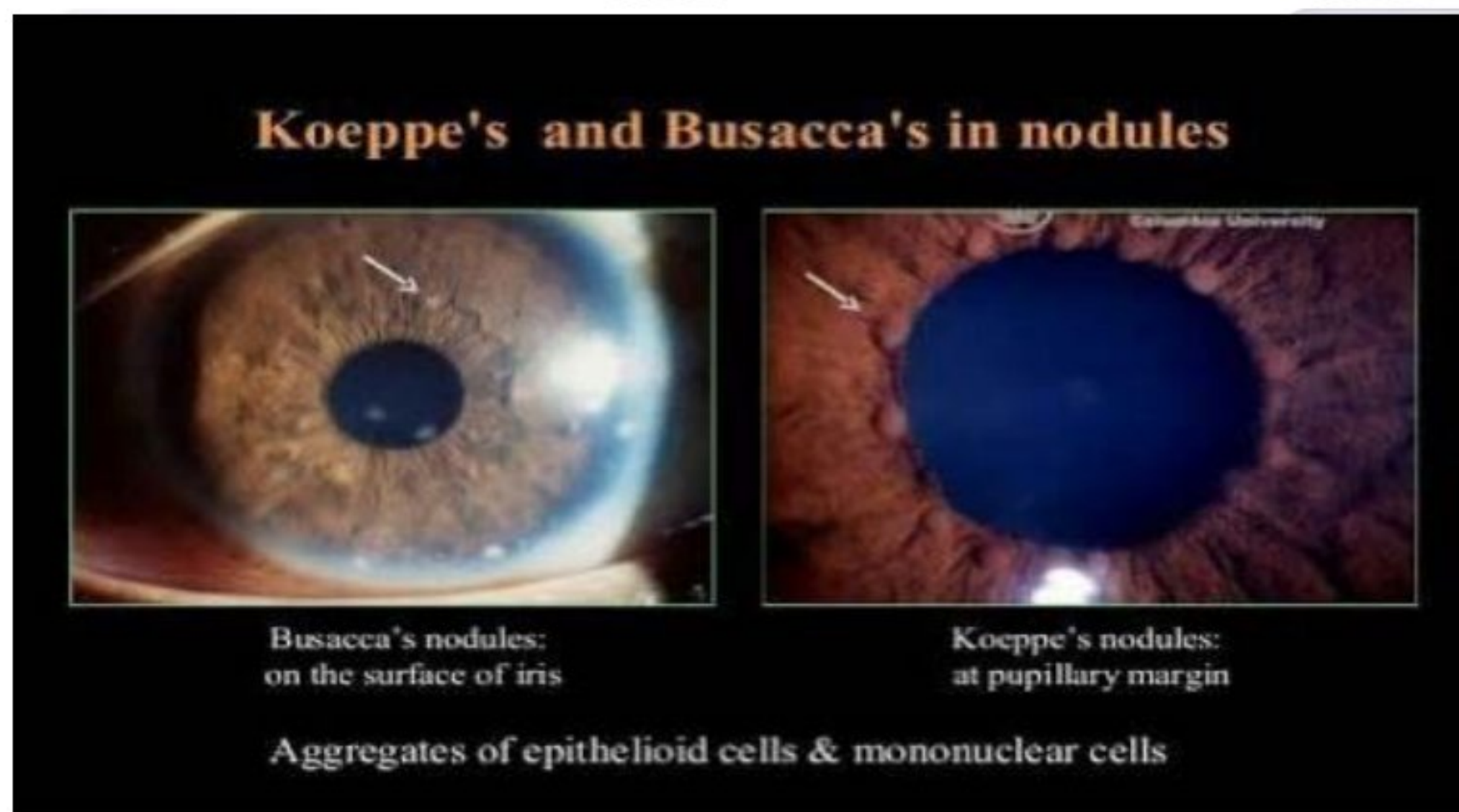
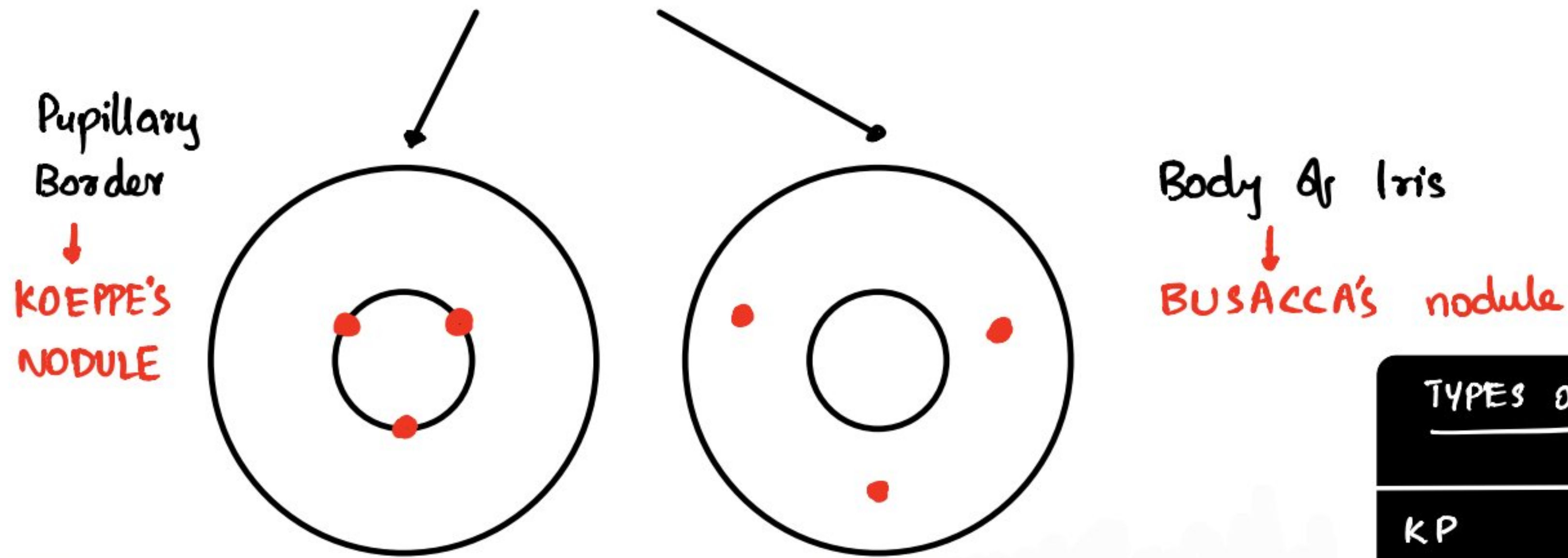
IRIS NODULE ::

- Inflammatory
- Found only in Granulomatous uveitis
- Composed of
 - i) Macrophage
 - ii) Epithelial cells
- In Leprosy → IRIS PEARLS { shiny white in color }



IRIS PEARLS IN LEPROSY

2 types of Iris nodules based on location ::



TYPES OF KP'S :

KP	Cells	S/i
1, Mutton fat KP	Macrophage + epithelioid	Granulomatous U.V
2, Granular / Fine KP	Lymphocytes	Non granulomatous U.V
3, Stellate / Star shaped KP	Endothelial dusting	Fuchs heterochromic
4, Old KPs	Healed UV	
5, Red KPs		A/c H'magic U.V

Complication of IC's ::

- 1) Secondary Glaucoma (most common)
- 2) Cyclitic membrane
- 3) Complicated Cataract
- 4) Pseudoglioma due to vitreous exudation
- 5) Cystoid macular edema → common cause of visual impairment
- 6) Pthisis & atrophic bulbi
- 7) Tractional retinal detachment
- 8) Band Shaped Keratopathy :: More common in children with JRA

Rx ::

TOL : Topical steroids + Topical cyclopegia (for pain)

- 1) Atropine → longest acting
- 2) Homatropine

3) Cyclopentolate

4) Tropicamide → shortest acting

All cyclopegics are mydriatics

ciliary spasm ⇒ pain
↓ cyclopegics
ciliary paralysis
↓
relieves pain

Role of cyclopegic in IC's:

- 1) Pain relief
- 2) Breaks, thin newly formed post. synechia → By Mydriasis
- 3) Prevents formation of posterior synechiae
- 4) Brings in more blood supply: more antibodies

Types of IC's:

1) HLA associate uveitis → non granulomatous UV

(MC) → **HLA B27**

- 1) Psoriatic Arthritis
- 2) Reactive Arthritis
- 3) Reiter's Syndrome
- 4) Ankylosing Spondylitis

HLA B5/B1: Behcet's d/s ⇒
• B/L non granulomatous pan uveitis
• B/L fleeting, transient mobile hypopyon.

HLA-A29: Birdshot Chorioidopathy
90% → HLA A29+ve | **≡ Strongest HLA association to ocular d/s**

HLA-DR4: VKH Syndrome

FUCH'S HETEROCHROMIC IRIDOCYCLITIS: U/L condition

- Anterior Uveitis
- Non granulomatous U.V
- Stellate KP
- Diffuse distribution, **No Artl's Δ**
- No posterior Synechia formed
- No role for cyclopegic
- Poor response to steroids
- Hypochromia Iris

AMSLER'S SIGN

During paracentesis (draining of Aqueous)
↓ rupture → HYPHEMA

Fine villiform vessels in the Angle

→ a/w Rubella infection

POSNER - SCHLOSSMAN Sx : a.k.a glaucoma cyclitic crisis / Hypertensive uveitis

- C/c low grade anterior uveitis
- Trabeculitis → Trabecular sclerosis
- IOP ↑↑ : 2° OAG → Intra ocular muscle ischemia

• Painful white eye

• Pupil dilated → action of dilator pupillae

a/w Toxoplasma infection

Inflammatory Glaucoma:

Mx of ↑sed IOP: **NEVER** give Pilocarpine : worsen the inflammation

PG analogues : are inflammatory mediators

Q: What is the DOC for Antr. uveitis → Topical steroids

Q: Management of uveitis & ↑sed IOP → Topical steroids

Q: Mx of ↑sed IOP in uveitis : Antiglaucoma drugs except PGs / pilocarpine

Q: Steroids cause 2° OAG.

Pediatric Uveitis:

• MCC of anterior uveitis in children → Juvenile Idiopathic Arthritis (JIA)

↳ ANA +ve oligo-articular subtype (≤ 4 jts)

C/c ante uveitis

Positive ANA req.
periodic slit
lamp examination

JIA : (MC) complicated cataract in children

Q: (MCC) Band shaped keratopathy in children

Intermediate Uveitis : Idiopathic pars planitis

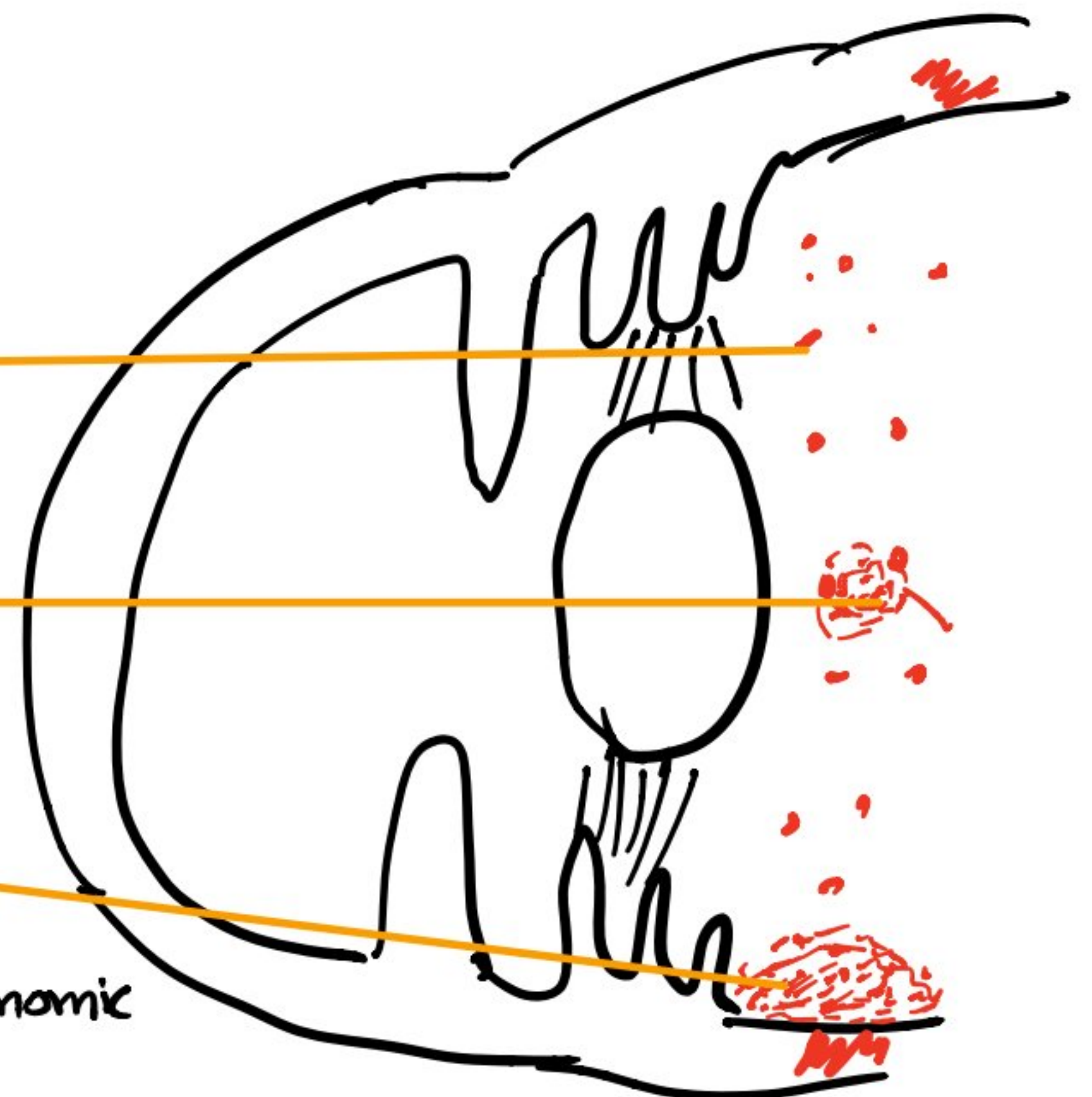
- C/F :
- i) Pain d/t ciliary spasm ± Redness
 - ii) Floater
 - iii) ↓se in vision

FLOATERS
d/t vitreous cells

snowball & clumped vitreous
cells

vitreous cell settling **SNOW BANKING**

@ pars plana ⇒ hallmark / pathognomic
feature of IU



- Inflammatory edema of macula / macular edema
↓
↓se in vision

Mx: Inv. = Indirect Ophthalmoscopy

Px: Steroid : T.O.C

Route : Posterior Subtenon injection

KAPLAN'S STEP LADDER APPROACH

I) PST inj. of TRIAMCINOLONE ACETONIDE

↓ do not respond

II) STEROIDS → • Systemic

or

• Intra vitreal injection of steroid

↓ do not respond

III) CYCLODESTRUCTIVE procedure { destroy pars plana }

1) Laser : Cyclophoto coagulation DIODE LASER

2) Cryotherapy : Ischemic Necrosis

b pars plicata destroyed accidentally $\xrightarrow{4t}$ phtisis bulbi

SKIP STEP III $\longrightarrow$ IV

IV) Vitrectomy : Vitreous → Bank / sponge of inflammatory cells.

Posterior Uveitis:

Choroiditis

Retino-choroiditis
(1°)

Chorioretinitis (1°)



CF: 1) ↓se in vision which is **painless**
⊕ **positive scotoma**
Floaters and **no redness**

1) Toxoplasmosis: H/o of contact with cats

• Toxoplasmosis Gondii

• MCC of posterior uveitis

Metamorphosia d/t macular edema

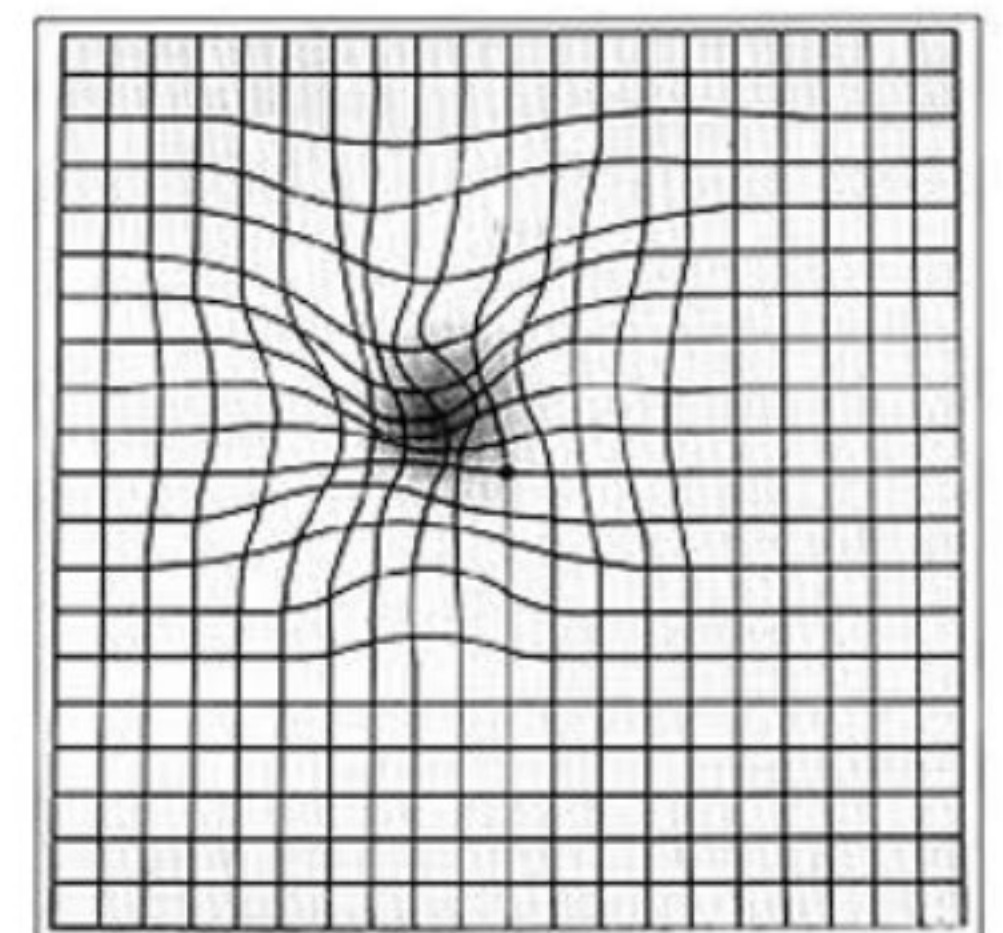
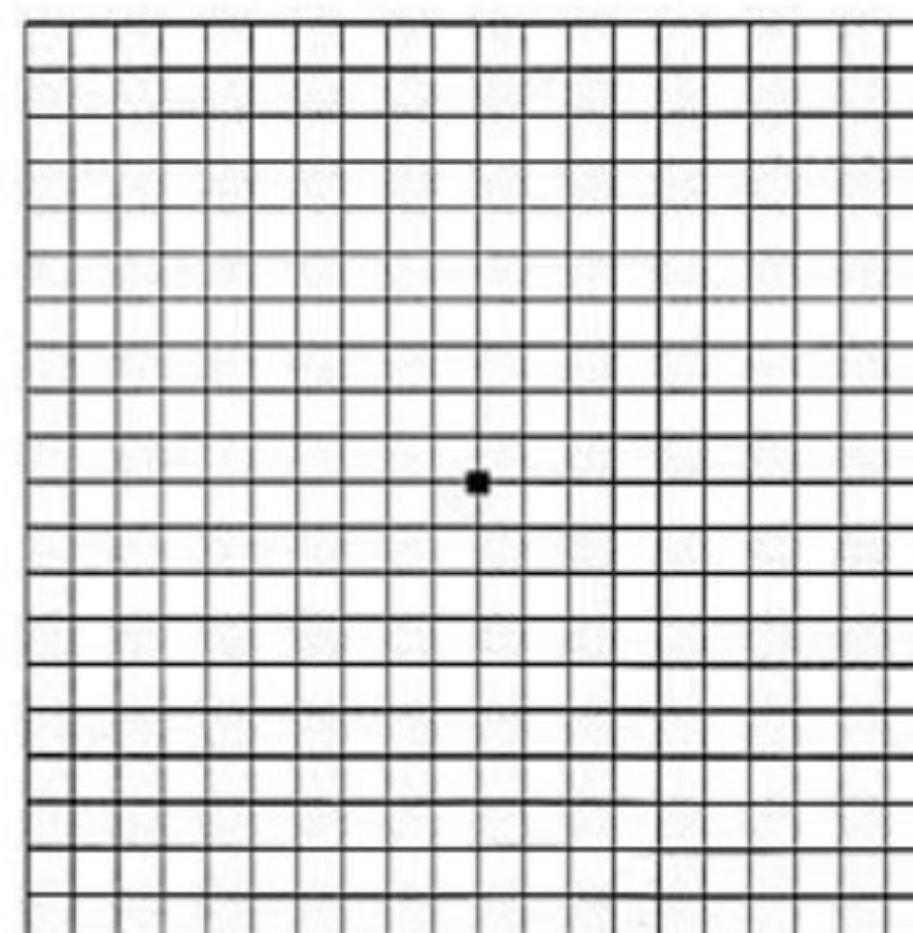


Image courtesy of Nat. Eye Institute, Nat. Institutes of Health

Metamorphosia